Specialisation Project 2 MTE7102

Enhancing User Confidence in Report Timeliness and Data Accuracy through Data Quality and Availability Checks

Course of study	Master of Science MSc in Engineering: Profile Data Science
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► School of Engineering and Computer Science
► Master of Science in Engineering

Abstract

This project addresses the next mandatory steps bridging from a first project on the topic towards the master's thesis. The problem addressed is the identification of a suitable tool with which the automation of the current data quality checks can be performed. During the course of this work, another main goal is to provide the business with tools that can assist them with the usage of new data sources. The Power BI report for the daily checks of the message processing data did exactly that, employing improved visualisations developed by the project team. All this was done while applying the DevOps principles at Swiss Post and implementing user feedback received while building the report for stakeholders.

As a next step, the search for a DQM tool was addressed by following the team during the setup of requirement lists and the evaluation of tools, including demonstrations of such tools. However, as the decision on the tool had not yet been taken by the end of this project, the way forward was defined together with the DWH LS OPS solution architect in a meeting. The monitoring that was set up to check the uptime of jobs such as ETL jobs was selected for testing in its second application, namely running checks on data. This tool was able to run counts on queries executed through Amazon Athena. However, the function had not yet been properly tested. Therefore, the project team carried out tests to verify whether this tool could be applied to the use case defined for the master's thesis. This use case was the automation of the data quality checks performed daily, which were part of the previous research.

With this information, the next step was to test this possible data quality tool, which was developed during the AWS Modernisation and was part of the full BizDevOps team until the end of May. Until then, knowledge transfer sessions were performed. After these transfers, a test was derived based on the full use case of the master's thesis, reducing its complexity and amount of data to perform only one count of one specific sorting centre for one event rather than for every sorting centre.

Finally, it was possible to evaluate the given solution as suitable for the master's thesis. This included not only the setup of a test case but also the setup of the required infrastructure. Thus, everything required for the master's thesis has been tested successfully. The details of this research, the additions to the solution landscape, and the implementation are part of the documentation that follows.

Contents

1. Introduction

Since mankind first began making plans, they were based on the available description of the environment. This approach continued into modern times, although the nature of the predictions has changed. People no longer plan which animal to hunt or when to produce stone tools, but instead how many people the postal service should hire for the Christmas season. These decisions require data, and in order to make the right decisions, the data must be available in the right time, in the right context, for the right people and naturally be correct. This makes data one of the most important assets in a company [The Economist 2017]. Data allow companies to gain an edge over competitors, analyse clients' wishes, or improve internal processes. This is where the relevance of this project emerges.

Swiss Post requires data availability and data quality in its reporting for decision making. This project analyses the current state of a specific data area of Swiss Post and provides steps toward improvements. This chapter familiarises the reader with the topic. It presents the background of the project at Swiss Post and delves into the motivation behind it. It then further explains the business relevance, demonstrating why it is both important and beneficial for Swiss Post to invest in the topic. Next, the problem statement is addressed, explaining what this project aims to solve. After the problem statement, the scope and limitations of the project are outlined more clearly. Finally, a brief overview of the expected outcomes of the project is provided. This chapter concludes with a preview of the document's structure.

1.1. Background

Swiss Post is Switzerland's national service provider, active not only in mail and parcel logistics but also in passenger transport (PostAuto), financial services (PostFinance), and various digital and e-government solutions. Its workforce of approximately 45'000 employees and nationwide infrastructure move millions of items and people every day, supported by highly automated sorting centres and extensive data flows [Post 2024]. In such a diversified organisation, analytical reporting is essential for strategic steering, operational planning, and maintaining service quality across business units. The organisational structure of Swiss Post is shown in the following diagram.

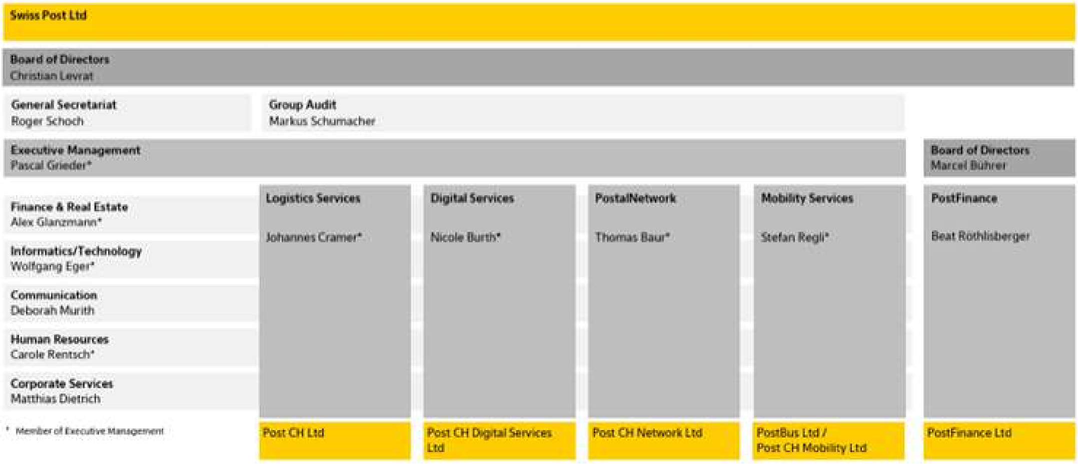


Figure 1.1.: Organisational structure of Swiss Post (online version)

The full mail piece lifecycle

Within this large organisation, the project takes place within a specialised team supporting the entire lifecycle of logistics for mail and parcel delivery. In German, this lifecycle covers every step from A to Z, ASTVZ, which stands for:

- **A**nnahme: Receiving, where parcels and mail are accepted and registered by Swiss Post to be sent through its network
- **S**ortierung: Sorting, the process during which received mail pieces are separated by destination and routed to the next process step, either further sorting or delivery
- **T**ransport: Transport includes the movement between sorting plants and delivery sectors
- **V**erzollung: Customs clearance handles international parcels and letters, including the required customs checks and documentation
- **Z**ustellung: Delivery is the final process step for every item; it is delivered by the staff of Swiss Post

However, this process is more easily visualised using the future vision of Swiss Post. In particular, the version tailored for the IT domain is well suited to describing the project environment. Starting from the top left with the A or V step, followed by S, T, and Z, the process progresses towards the bottom right.

Figure 1.2.: Future way of work illustrating the full postal delivery workflow from receiving an item to delivery.

DWH LS OPS

The project team supports the **D**ata **W**are**h**ouse of **L**ogistic **S**ervices **Op**eration**s** (DWH LS OPS). This **Busi**ness **Dev**elopment **Op**eration**s** (BizDevOps) team is located within the organisational structure of the Logistics Services vertical and the Informatics/Technology horizontal. It represents the single source of truth for analytical reporting at Swiss Post. In contrast to operational reporting, which focuses on real-time guidance for operational processes, analytical reporting focuses on aggregated insights for evaluation and planning. Examples of analytical reporting include working time, process time, and aggregations of sorted items. The stakeholders of the team distribute these reports to various task groups within postal services, such as sorting and delivery.

There are two reasons why a deeper understanding of analytical reporting is currently very important. First, the infrastructure for analytical reporting is transitioning from on-premise systems to the cloud. Second, in addition to the migration of reporting, the infrastructure at all Swiss Post sorting center locations for parcels is undergoing fundamental changes. These changes may include complete modifications of data pipelines, which can make comparisons with previous figures impossible. Therefore, at this critical point in time, it is of utmost importance to maintain users' trust in the data during this change process. The following figure illustrates the shift from the on-premise setup on the right-hand side to the target architecture using cloud hardware on the left-hand side.

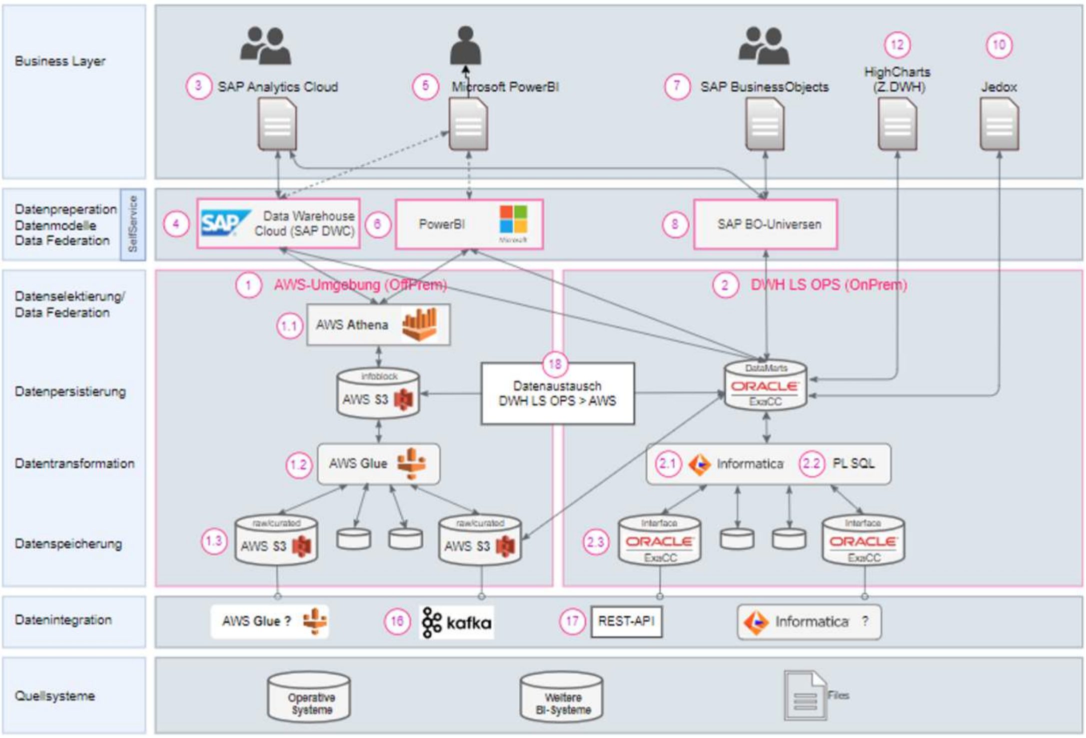


Figure 1.3.: The transition from an on-premise setup to a cloud-based target architecture.

Swiss Post sorting centers are cyber machines

Swiss Post has built and operate a number of massive sorting centers located across Switzerland. These are far more than just mechanical warehouses or factories. Rather, these sorting centers are massive "cyber facilities" that include high-tech hardware but also high-tech IT systems to keep them running.

While the team handles use cases across the entire postal delivery workflow, this project initially focuses on supporting sorting data flows. Sorting at Swiss Post is the process of consolidating received items in centralised plants, to which letters and parcels are sent from regions across Switzerland. One such plant is located in Härkingen:

Figure 1.4.: Aerial view of the Swiss Post sorting center in Härkingen.

Mail pieces are handled by workers and placed onto the automated sorting hardware of Swiss Post. This hardware enables the handling of up to 10'000 parcels per hour [Post 2023]. The system scans the received items and transports them through the plant via conveyor belts to their correct destinations, where they are discharged through chutes. These chutes are shown in the second image, including staff placing the parcels into roll boxes for further processing.

Figure 1.5.: Interior view of the automated sorting system in a Swiss Post sorting center.

Figure 1.6.: Discharge chutes inside a Swiss Post sorting center, where parcels are released to their designated zones. Then put into rollboxes to get them to next processing steps.

1.2. Motivation

Data quality and **data availability** are central to the successful operations of any company. Even a company working in low-tech or non-IT industries [Pas 2025] has basic but essential data needs, for example, information on who is working where and the addresses of clients and staff. Not having this data in the right quality or available to the right people at the right time will cause problems whenever the data needs to be used.

In the context of ongoing changes to the Swiss Post reporting infrastructure, keeping track of data availability and data quality is challenging. These infrastructure-related changes comprise the construction of new reports, one-to-one migrations, and the re-engineering of existing reports. Since the topic of data quality in the Swiss Post operations area could be built from scratch, as if starting on a greenfield, it became an ideal subject for a master's thesis.

Furthermore, the topic was not of the highest priority for Swiss Post, but it was still considered a valuable idea to pursue. The medium-priority nature of this project also allowed the topic to be divided into multiple steps, such as working from a PoC to an MVP setup. Additionally, the medium priority regarding feature development in this area allowed for well-grounded research on the topic. All these aspects combined made it a suitable project leading to a master's thesis. Furthermore, due to the lack of automated data accuracy and data quality checks in the DWH LS OPS environment at the time of defining this project, this topic is highly relevant to the team responsible for both the operational and analytical reports that help guide the company. If the data is incorrect, wrong decisions are made; and if the data is unavailable, management cannot properly lead the staff. Thus, the topic is of high importance and business critical: data quality and availability help Swiss Post continue to deliver reliable services on a daily basis and enable IT to provide better assistance to staff throughout the postal process.

Figure 1.7.: Main customer touch point, delivery of a parcel

1.3. Problem Statement

The two key problems addressed in this project are, first, that data quality in this area of Swiss Post has so far been handled in an ad hoc manner, without formal definitions, frameworks, or an agreed set of KPIs; and second, that data quality is handled manually by the business at a high level rather than being automated by the IT department. Both of these problems lead to issues not being detected early and to longer reaction times. As a further consequence, delays in data processing are not identified and targeted for systematic resolution, which in turn reduces management confidence in the reports. All of this prevents continuous improvement, as the current checks are neither documented nor stored in a way that allows follow-up analysis or structured refinement. This project researches tools with which the next step towards automation can be performed and also tests their applicability for the research in the upcoming master's thesis.

1.4. Scope and boundaries

This research focuses on the data availability and data quality challenges of the DWH LS OPS BizDevOps ASTV teams at Swiss Post described in section 1.1. To simplify the scope of this work, the first step of data retrieval, such as how data is stored from Kafka topics, is excluded. The scope begins with the curated and reporting layers in the ETL process of the retrieved data. The aim of this project is to specifically provide solutions for the mentioned teams and not to offer a general solution or framework for the entire Swiss Post.

1.5. Expected outcome

The high-level goal of this master's thesis pre-project is to demonstrate that automated data quality checks provide additional safety and can be calibrated to the needs of the business. Automated data quality checks refer to checks executed daily or upon data load, which are always carried out by systems without the need for manual interaction. More specifically, the project focuses on establishing the foundations for this goal by analysing the current situation and developing a proof of concept (PoC). It will carefully document and explain the steps taken toward the proof of concept and, in this way, make the decisions made understandable. The success criterion of this project is to provide a PoC, or steps toward a PoC, that help reduce the costs of manual checks.

1.6. Structure of the document

An important note for the reader is that this document is the middle part of a three staged research project. As some of the research and solution design of the first research project is relevant to understanding this one, the content has been kept. This document is intended to guide the reader through the project in meaningful steps, explaining how the project is set up and executed, and then leading into the research and conclusions. It is structured in the following way. First, the document begins with the project management chapter, which explains the methods used during the project. It provides reasoning for tool choices where applicable and lists the tools and teams with which collaboration took place. Next, the research chapter describes the problem in more detail. It clearly defines the specific and measurable goals of the project and presents the research sources used. This allows the reader to understand the decisions made and establishes the basis for the solution design. The subsequent solution design chapter establishes the relevant terms and definitions valid throughout the project. It documents how the problem statement is addressed and provides the architectural groundwork for the implementation. Building on top of the solution design is the implementation chapter, which presents the implementation of this design. It contains the work carried out by the project team and outlines any issues encountered. After the implementation, the target analysis chapter compares the expected outcome with the achieved outcome, assessing how many goals were reached and listing their fulfilment status. The document concludes with the critical review and lessons learned, summarising what went well and what did not, and offering specific suggestions on which methods to retain or replace. This will be valuable for the master's thesis, being the final part of this three-part research. Finally, additional materials are provided in the appendix.

2. Project management

The project management chapter summarises the environment of this project and how it is conducted. First, the teams in which the project staff is engaged are listed, allowing the reader to see the relevant surrounding teams. Second, the different methods for effort and task tracking are introduced, so that the project and progress management methods applied are understood, together with their reasoning. Third, the toolsets used by the different teams are described, which allows the variety of tools and their usage to be grasped. Fourth, time tracking is introduced and explained for transparency, while the tracked time is provided in the Annex. Finally, the risk management section describes the risks handled and the measures taken to counteract them.

2.1. Team constellation

This section explains the team environments in which the project takes place. This serves to provide an understanding of the direct project team working on this project and other teams at Swiss Post that contribute to the goals of the project.

Relevant Swiss Post teams

At Swiss Post, three teams are active and related to this project. In what follows we elucidate their tasks at Swiss Post, how they work together, and their contributions to this project. These teams are:

- BizDevOps team DWH LS OPS
- AWS Modernisation team
- Data Quality Initiative (DQM team)

DWH LS OPS team

The team **D**ata **W**are**h**ouse of **L**ogistic **S**ervices **Op**eration**s** (DWH LS OPS), contains the main project resource and the author of this document is situated, namely Roland Roccaro, Business Analyst in the MPP team. The key mission of this team is to deliver accurate reports guiding decision making in the postal services of Swiss Post. The DWH LS OPS team is responsible for developing and operating the Swiss Post data warehouse DWH for logistics services, which is the business unit that handles all postal services. The DWH is used for analytical reporting, which includes reports not used in direct daily operations but used for various forms of reporting, such as productivity analyses based on volumes and working time, informed long-term decision-making, as well as daily handovers and near-term planning.

Figure 2.1.: Picture of DWH LS OPS team

Within the Swiss Post this team is also referred as a **Busi**ness **Dev**elopment **Op**eration**s** (BizDevOps) team, because it contains all required personnel, to create, deploy and maintain reporting solutions. This starts with business analysts to evaluate existing reports and their usage, and to document and specify them for further updates or re-engineering. This continues through the implementation, which is carried out first by data engineers and then by business intelligence (BI) engineers. Finally, the business side is involved in testing these reports and also serves as the main source when it comes to questions of how or what to implement. Due to the size of this team and the different responsibilities, the DWH LS OPS team consists of three separate sub-teams. These teams are ordered along the process of postal services as introduced in the full mail piece lifecycle being: Receiving, Sorting, Transport, Customs clearance (German acronym is ASTV), **D**elivery **&** **F**inance (D & F), and finally the planning focused Parcel Volume Planning and Forecasting (german acronym is MPP).

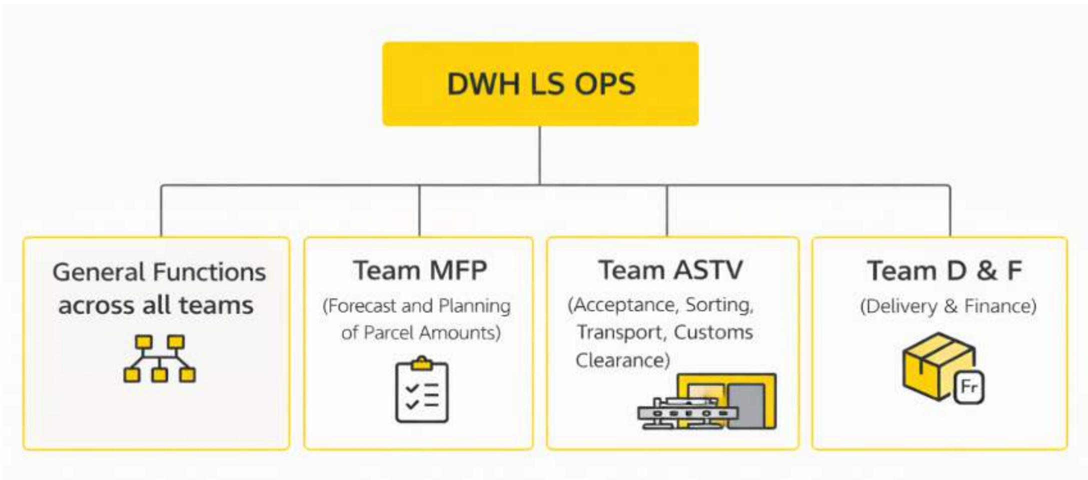


Figure 2.2.: Team and sub-team organisation

AWS modernisation team

Swiss Post is currently in a process the modernisation of the AWS cloud environment, which the Swiss Post itself describes as follows:

Swiss Post is currently pursuing a strategic modernisation of its AWS-based data lake with the objective of improv-

ing developer efficiency, increasing operational reliability, and establishing a modern, data-driven architecture. According to internal documentation, the initiative is aligned with defined key performance indicators and aims for completion by the end of the first quarter of 2026 [Swiss Post 2025].

This modernisation is performed with the assistance of consultants as part of improving AWS usage in DWH LS OPS. The improvements are detailed in 15 KPIs across topics such as AWS Glue job optimisation and deployment pipeline improvements. With the assistance of the consultants, best practices are applied to the development of ETL, as well as general improvements to deployment pipelines. Most importantly for this project, the initiative also includes DQM improvements. As a reference, a screenshot of the Confluence landing page describing the initiative is shown below:

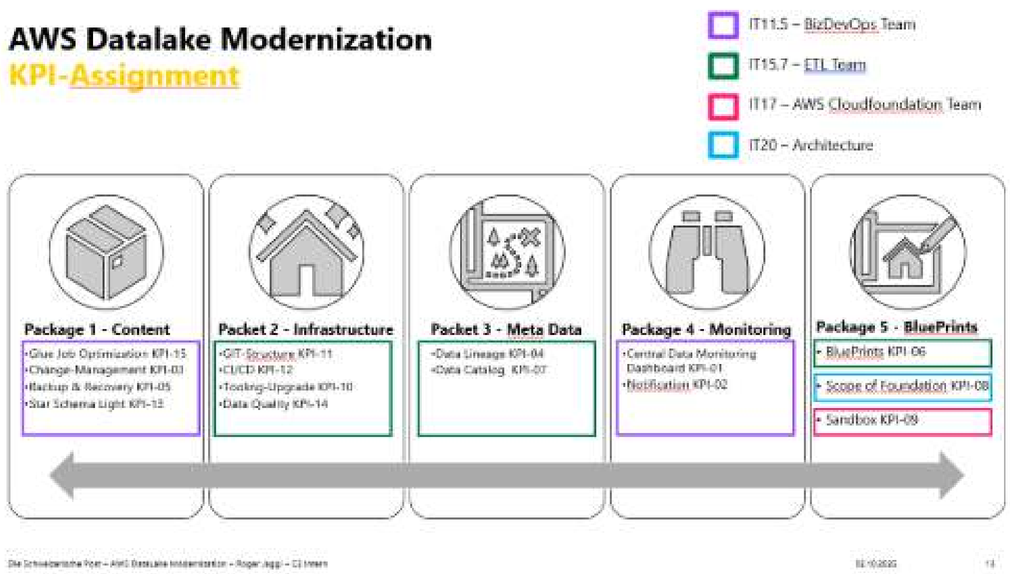


Figure 2.3.: The start page of the AWS modernisation, listing KPI and latest news

This team consists mainly of consultants with long-term experience in assisting companies in becoming proficient with AWS. They support data engineers at Swiss Post by improving existing code and reviewing it for potential enhancements. This is relevant for Swiss Post, as the development of the cloud infrastructure started without full architectural support and therefore has room for improvement.

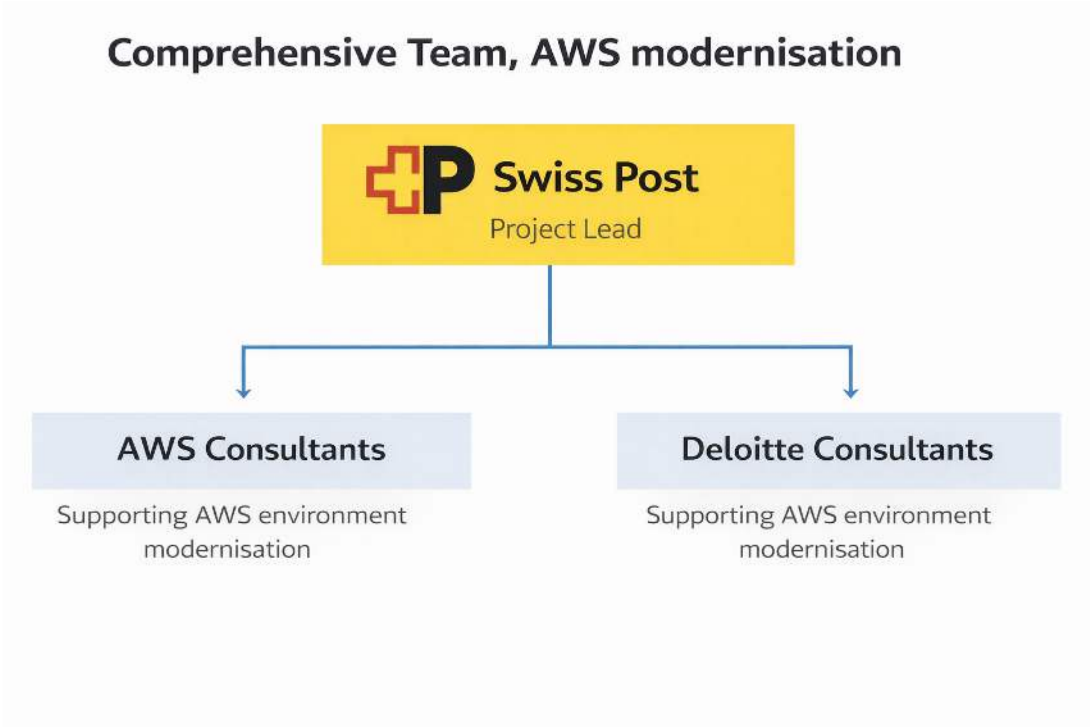
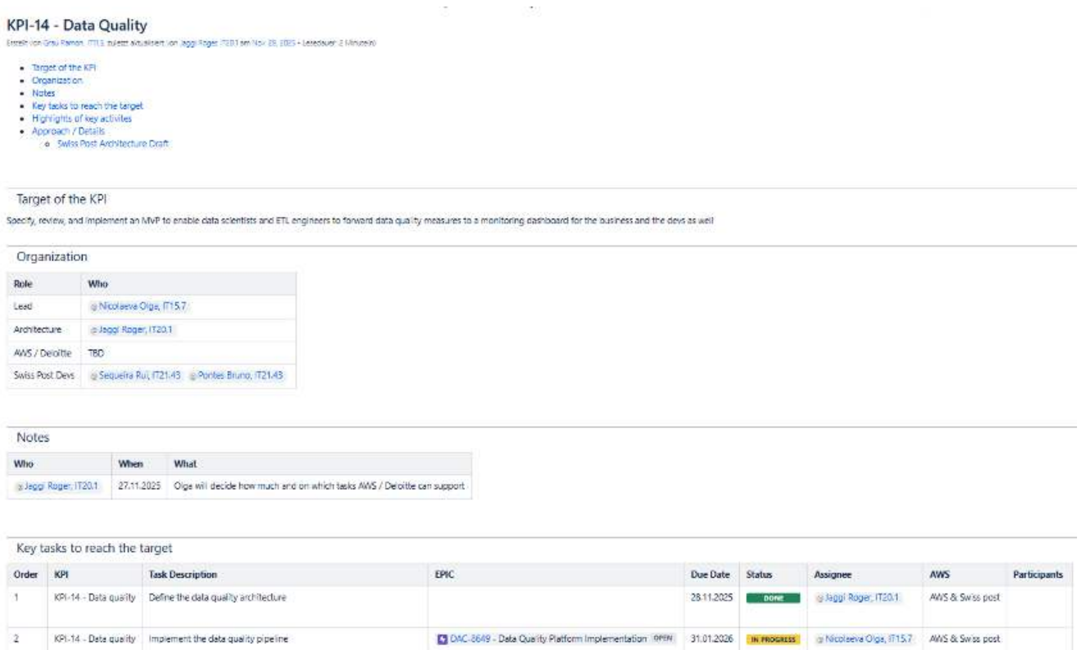


Figure 2.4.: The team setup of the AWS modernisation

DQM team

The third team relevant to the project is the DQM team. The team serves KPI 14 Data Quality within the AWS modernisation. Their goal is described on Confluence as follows: specify, review, and implement an MVP to enable data scientists and ETL engineers to forward data quality measures to a monitoring dashboard for both business users and developers.



KPI-14 - Data Quality

- Target of the KPI
- Organization
- Notes
- Key tasks to reach the target
- Highlights of key activites
- Approach / Details
  - Swiss Post Architecture Draft

Target of the KPI

Specify, review, and implement an MVP to enable data scientists and ETL engineers to forward data quality measures to a monitoring dashboard for the business and the devs as well

Organization

Role	Who
Lead	Nicolaeva Olga, IT15.7
Architecture	Jaggi Roger, IT20.1
AWS / Deloitte	TBD
Swiss Post Devs	Sequeira Rui, IT21.43 Pontes Bruno, IT21.43

Notes

Who	When	What
Jaggi Roger, IT20.1	27.11.2025	Olga will decide how much and on which tasks AWS / Deloitte can support

Key tasks to reach the target

Order	KPI	Task Description	EPIC	Due Date	Status	Assignee	AWS	Participants
1	KPI-14 - Data quality	Define the data quality architecture		28.11.2025	DONE	Jaggi Roger, IT20.1	AWS & Swiss post	
2	KPI-14 - Data quality	Implement the data quality pipeline	DAC-8649 - Data Quality Platform Implementation OPEN	31.01.2026	IN PROGRESS	Nicolaeva Olga, IT15.7	AWS & Swiss post	

Figure 2.5.: The KPI 14 Data Quality start page on DWH LS OPS Confluence

The team consists of three developers with the assistance of a lead data engineer. This setup ensures that the developers can deliver results efficiently, while the lead data engineer provides design guidance and solution structure.

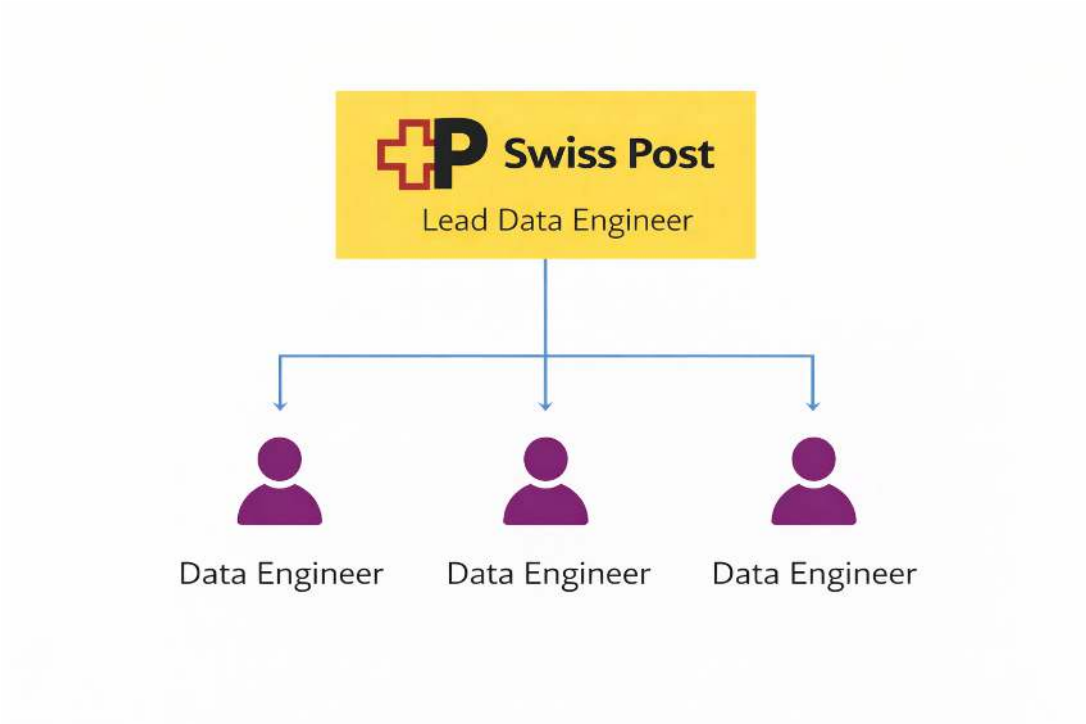


Figure 2.6.: The DQM team setup

Academic project team

The academic team consists of the author and the academic advisor. The author conducts the practical work, which includes analysis, implementation, and evaluation. The advisor provides expertise in the academic process and acts as a sparring partner regarding the writing process and project management, offering methodological and academic supervision.

Figure 2.7.: The academic team setup

2.2. Project management methodologies

This section details how the different teams manage themselves. This is done for reasons of transparency, so that the reader understands which project management methods the individual teams use. This highlights an additional level of complexity in the working environment. Not all teams are addressed in every subsection, as the project team was not involved in all teams' methods to the same depth. First, project management methods are highlighted, followed by the different timeline management approaches. In the following figure, the different boards and management methods are aligned around the central academic team to illustrate the broad set of methods.

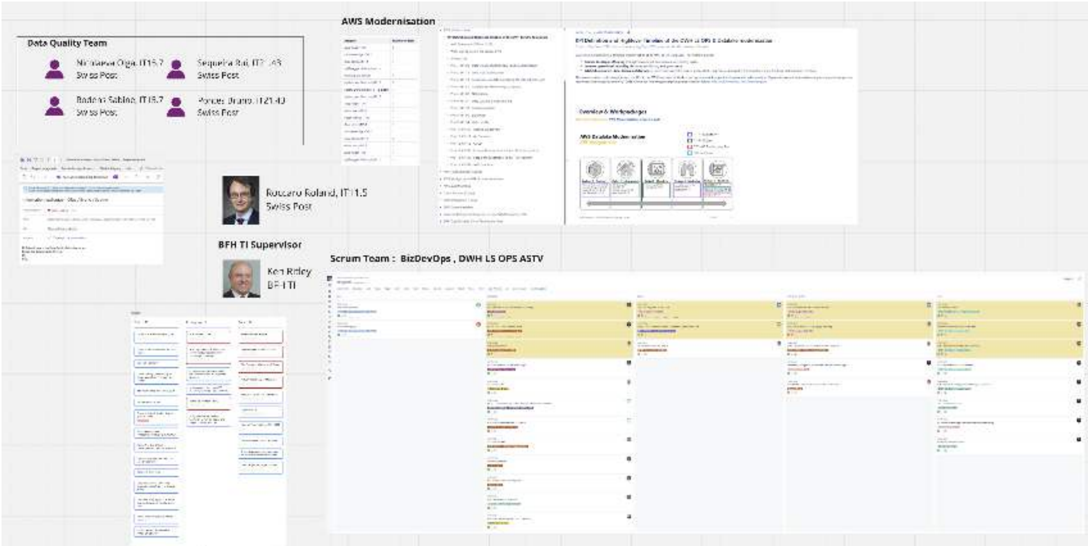


Figure 2.8.: All methods on one picture showing their diversity

Project management

Project management for this project consists of three main elements. These are Scrum, Kanban, and weekly meetings.

Scrum

The DWH LS OPS team is organised as a BizDevOps Scrum team (see Section 2.1). The AWS modernisation teams join these Scrum teams 2.1. The full set of rituals defined for Scrum is used, including daily meetings, sprint retrospectives, sprint planning held every two weeks, and reviews held every four weeks. In addition, a weekly Scrum meeting is held to maintain communication between the ASTV, D&F, and MPP teams. The meetings are shown in the first screenshot, while the second illustrates a noteworthy team definition: the rules of engagement established for collaboration, which are a result of the retrospectives.

Figure 2.9.: Scrum ceremonies of DWH LS OPS BizDevOps team

Figure 2.10.: Rules of engagement of DWH LS OPS BizDevOps team

Weekly Meetings

Contact with the DQM team is handled using weekly meetings 2.1. These serve as minimal update sessions during which weekly progress is communicated to all team members. While the developers in the DQM team are themselves also part of a Scrum team, their sprint management is out of scope.

Kanban

The academic team handles tasks using a Kanban board 2.1. The tasks are organised into three main columns. Both team members show their responsibilities on the Kanban board. The blue cards represent the author's tasks, while the red cards represent the advisor's tasks. The board is updated frequently and is reviewed during the weekly Jour Fixe.

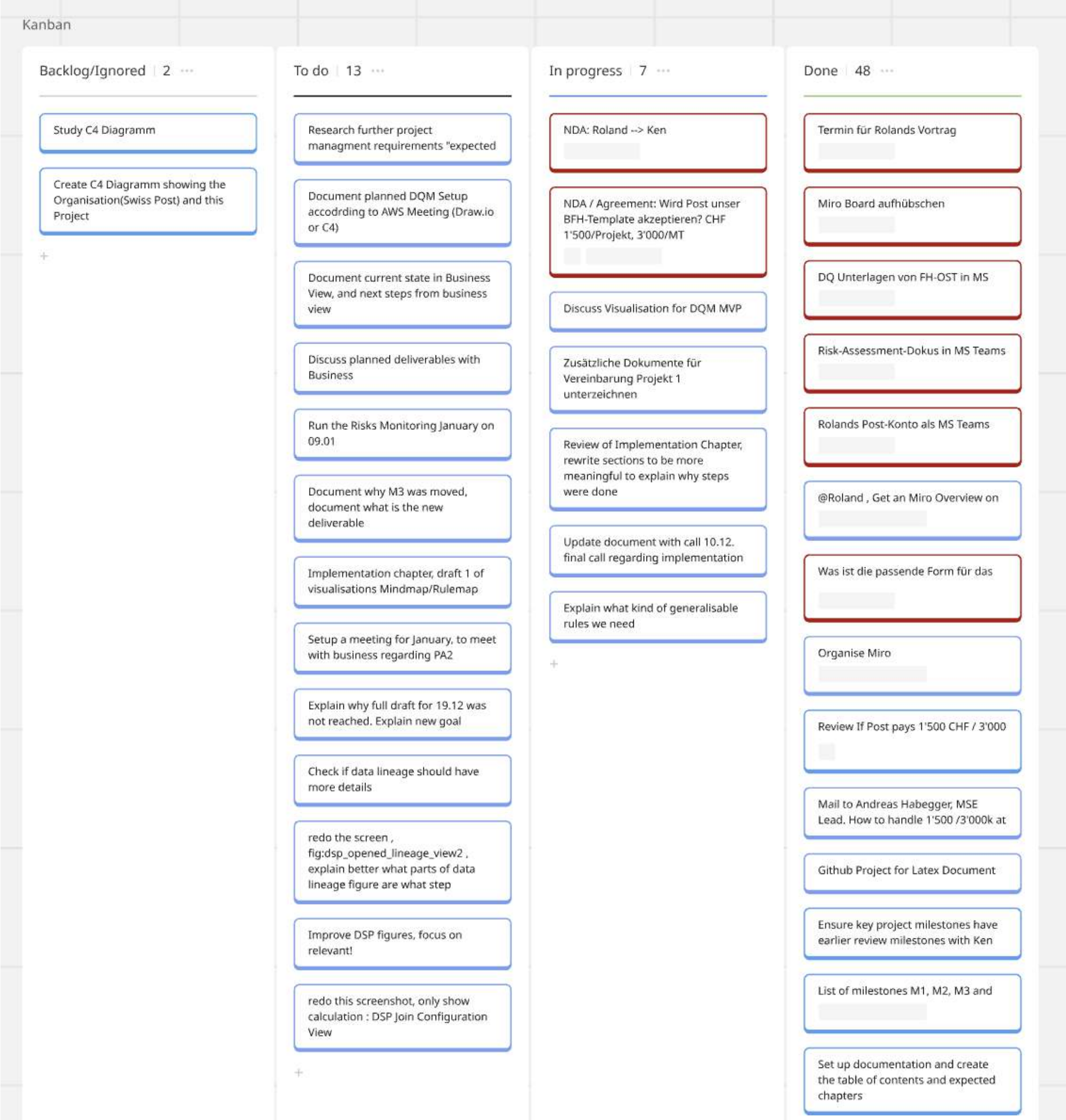


Figure 2.11.: Kanban board of Academic team

Jour Fixe

The academic team holds a Jour Fixe every week. This serves to maintain close communication and functions as a check-up meeting. Each meeting follows a defined structure; as an example, the meeting held on 15.12.2025 is described below.

- Project check-in and clarification of objectives
- Overview of the project and completed deliverables
- Current deliverable in progress
- Status of milestones
- Changes needed to the updated Project 1 description
- Questions regarding Project 2
- Other topics

First, a general introduction to the project is provided to serve as a check-in and to allow participants to focus on the project. Second, the deliverables currently being worked on are listed based on the deliverables overview. Third, the status of the milestones is reviewed. Fourth, this is followed by individual agenda items, which can differ from meeting to meeting. In this meeting, these included updates to the Project 1 description required for agreement, as well as a check-in regarding topics and submission of Project 2. Finally, the meeting is closed with an open section in which any other relevant topics may be discussed. These meetings are documented in Miro (see below) and structured using a three-block template and are most often recorded using handwritten notes that are later uploaded.

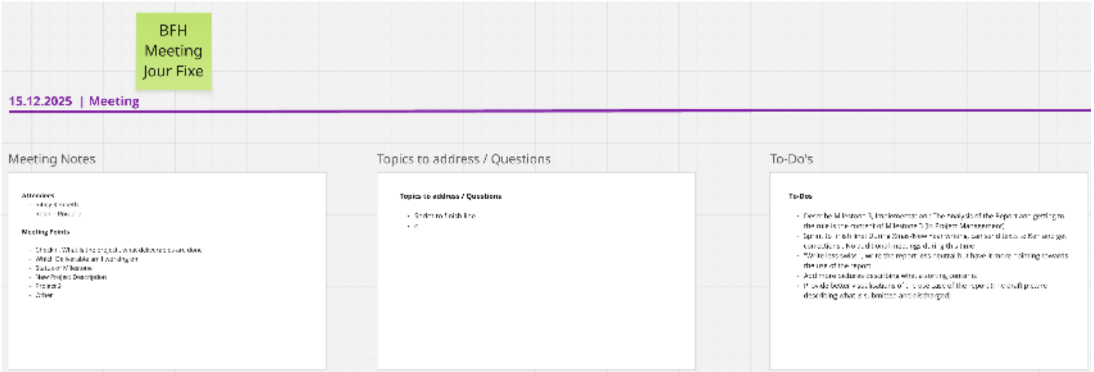


Figure 2.12.: Jour Fixe of Academic team, meeting structure and rework

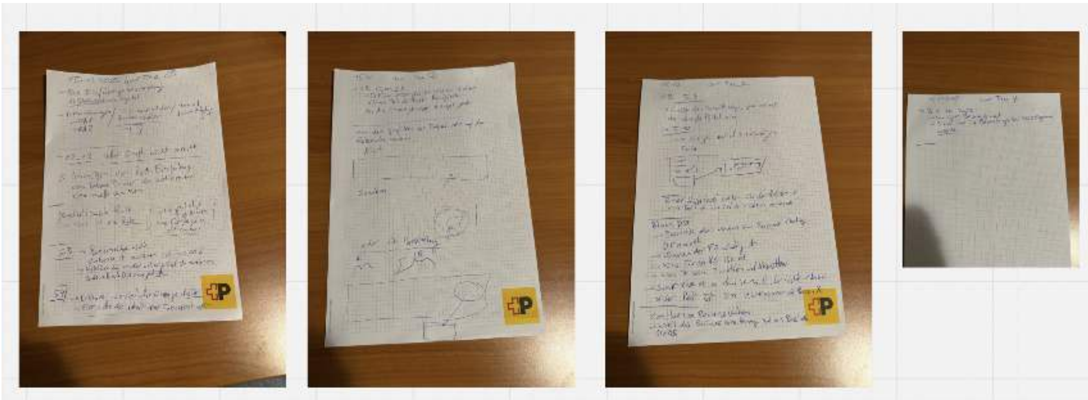


Figure 2.13.: Jour Fixe documentation of Academic team

Timeline management

Timeline management is performed in different ways by the teams. In this subsection, the planning of actions is described, as well as how staff availability is tracked.

Scrum Events

The DWH LS OPS and the AWS modernisation teams use Scrum events to manage their timelines. Progress on stories is monitored in daily meetings, during which each team member updates the team on progress and issues and indicates whether assistance is needed or deviations from the previously planned approach are required. Sprints encompass the stories the team works on during a two-week cycle. During sprint planning, the stories for the next sprint are defined. The stories added to a sprint represent the commitment the team makes for the sprint, together with the sprint goal. Planning over longer time horizons is conducted through quarterly planning sessions. Every three months, all teams in DWH LS OPS plan which epics and how many story points should be completed. These quarterly planning sessions also serve to prioritise epics and allow higher management to introduce priorities.

Holidays

Holidays are noted in the calendars used; this applies to the DWH LS OPS team as well as the academic team. In both cases, every holiday must be recorded with an absence notification. This allows other team members to quickly see whether a team member is available in the future and to adjust their planning accordingly, or to confirm in daily operations that the person is not available.

Milestones

The academic team manages the planning of the overall project timeline using milestones. These milestones define timeframes during which a section or chapter may be worked on. For this project, the following milestones have been defined:

- Milestone 1: Basics completed
- Milestone 2: Completion of main research
- Milestone 3: Code implementation completed
- Milestone 4: Bug fixing completed
- Milestone 5: Submission of the project document
- Milestone 6: Presentation prepared

First, the basics ready refers to all the tools required to work being available. This step includes setting up LateX, setting milestones, setup of basic tool structure and definition of way of work. Second the main research done refers to the end of the purely theoretical part and start towards the solution design and the subsequent implementation of it. This comprises the main literature research being done. Third, the code done refers to the implementation of the solution being done. After this point no new features are to be implemented. Fourth, bug fix done comprises all corrections that had to be performed so that the provided implementation works and technical debt amassed during development has been cleared. Fifth, the hand-in of the document is the final day according to the semester calender. Up to this date corrections to the document and finalisation can be provided. Sixth and final, the presentation must be ready latest by this date. After this some finetuning can be performed but the main content needs to be set.

The milestones were set to structure the work blocks, but primarily to define how much effort should be invested in each individual block. The time allocated per block is not estimated in terms of how long a task is expected to take. Instead, it is defined by how much time the team is willing to invest in a topic in order to ensure sufficient capacity for subsequent tasks. For example, documenting the Swiss Post environment could have taken two to three weeks; however, only three days were intentionally allocated to this task.

Deviations and Unavailabilities

The academic team tracked deviations from milestones or Jour Fixe meetings. This served transparency and traceability. For this use case, the dates of all Jour Fixe meetings were planned in advance. If a meeting had to be rescheduled, this was noted in the calendar. In the case of shifted milestones, these changes also had to be documented in the calendar, including the reasoning for why a milestone was not achieved or had to be moved. In the Results chapter, deviations from the planned milestones are documented.

2.3. Tools to support the project management

This section lists the tools used during the project for reasons of transparency. It also allows the reader to evaluate different ways in which these tools can be used.

Miro

Miro is part of the toolset of both the academic team and the Swiss Post teams. However, in this project, it was used only by the academic team. The reason for using Miro is that it is a lightweight tool that allows combining different data structures on a single board. Miro provides an infinite canvas on which a wide variety of information can be placed. The information can be structured and easily rearranged to regroup related elements. This allows the tool to be used both for early documentation of information and for later restructuring when additional ideas emerge, particularly in the context of long-term documentation. The ability to restructure information visually supports the decomposition and understanding of complex environments.

A full set of screenshots illustrating changes on the Miro board has been moved to the Annex, while the final one is included here (Appendix B). The following screenshot represents the Miro board during the finalisation of the document, zoomed out so that all elements are visible. While this view prevents individual details from being seen, some relevant parts are shown in detail in the following subsubsections.

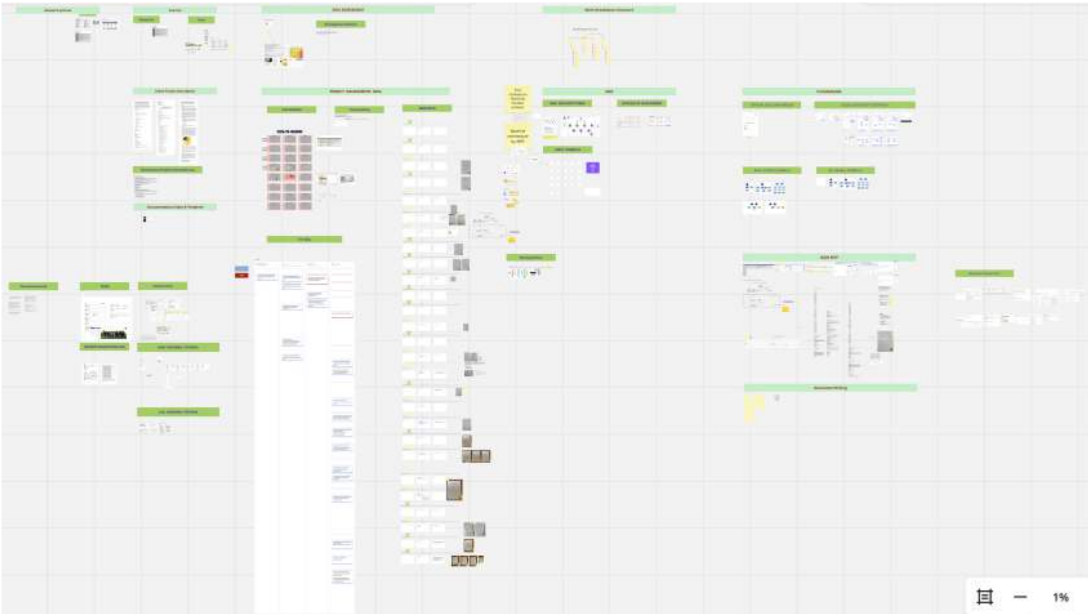


Figure 2.14.: Miro board documentation, view of 23.12.2025

Miro calendar

The calendar in Miro was primarily used to schedule the milestones. These are shown with green tiles on the corresponding dates, including their number and milestone title. Jour Fixe meetings are marked in yellow whenever they occurred, including adjustments in cases of unavailability. Actual unavailabilities at scheduled Jour Fixe meetings and other times are documented in black. Finally, holidays are marked in teal. In the case of this project, this applied only to the Christmas and New Year holiday period.

Figure 2.15.: Miro board calendar used for coordination and tracking of milestones and related events

Meetings documentation

The meetings are documented using a layout with three sections, including a coloured tag at the top in yellow for Swiss Post meetings and green for academic ones. From left to right, these sections are structured as follows. First, the meeting points to be discussed are listed; these are copied from the Outlook invitations. Second, the topics addressed are summarised, mostly after the meeting. Third, the to-do block describes any actions that arose during the meeting. Not every meeting includes all sections. Furthermore, meetings are often documented using handwritten minutes. These are uploaded to the right of the meeting blocks without further refinement and serve as references.

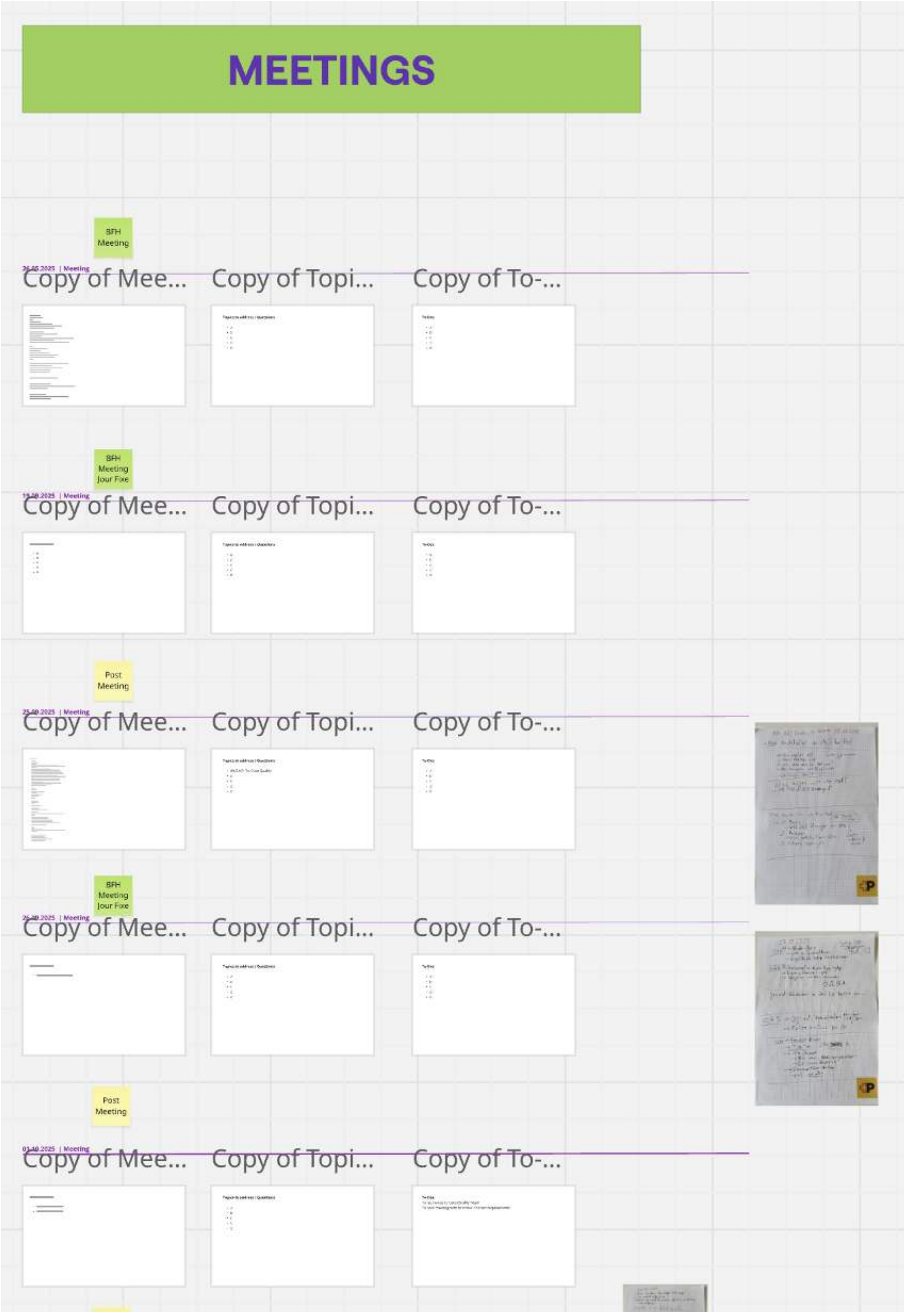


Figure 2.16.: Miro board, tracking of meetings: when, who, what

Documentation of the Implementation

The implementation was first documented in Miro, as the rapid creation of screenshots did not disturb the flow of information gathering. In the following figure, the information-gathering process can be seen, starting from the top left, moving to the right, and then restarting from the left. First, screenshots of the SAC report are shown, followed by navigation towards DSP. Further details on the tools are described in the Implementation chapter.

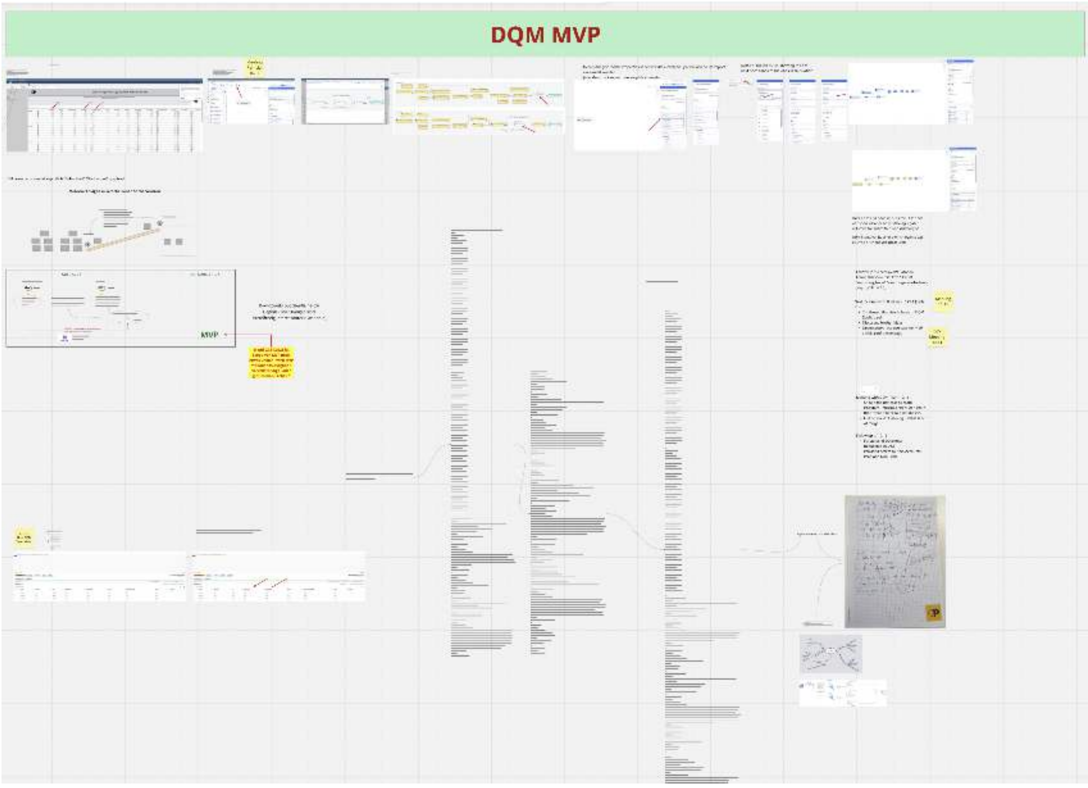


Figure 2.17.: Miro board usage for quick documentation during implementation

Jira and Confluence

Jira (mainly for issue tracking) and Confluence (mainly for documentation) are both tools provided by Atlassian and are therefore well integrated with each other. The tools are installed on-premises, which limits sharing possibilities outside the Swiss Post network. They are used by the three Swiss Post teams presented in the Teams section. The project team also used Confluence when research results had to be presented as a first use case. While Jira is used for the short-term management of Scrum artefacts, Confluence is used for long-term documentation. An example of such long-term documentation is shown in the following screenshot, illustrating the documentation of a data lineage.


PZ/RPZ Quantity datamodel

Von Bosek Markus, IT11.5 20.09.2024 2 Eine Reaktion hinzufügen • Cloud-Editor Numbered Headings Add status

[WEST-9157] [DSP] Improvement of 'Mengenmodell' sendungsverarbeitung + sortierereignis in DSP - JIRA (post.ch)

Overview

The quantity model in DSP is built with

- 1: SORTIEREREIGNIS_V - PADASA data
- 2: Schicht - Shift information with automatic shift data from merano for PZ and a manual DSP table for rest
- 3: SENDUNDSVERARBEITUNG_V - WESORT data
- 4: Combination of the data streams
  - Always use PADASA for MCB
    - MCB nah = MCB in own center
    - MCB fern = MCB handled for own center by other center (needs to be counted because it creates effort)
  - Use PADASA before WESORT deployment
  - Use WESORT on and after deployment
  - → deployment dates are managed in manual table

Figure 2.18.: Example of project documentation in Confluence used to record architectural decisions and their rationale for future reference

Microsoft Teams and Microsoft Outlook

MS Teams and MS Outlook are used by both the academic and Swiss Post teams. They are organisational standards at Swiss Post and are not optional. The same applies to the academic context. Both tools serve to support focused long-term (MS Outlook) and short-term (MS Teams) communication.

Shared chat

For the academic team, the chat was used for short-term communication. However, a challenge arose due to the involvement of two distinct organisations, BFH and the Swiss Post. In order to keep communication between Swiss Post accounts and BFH accounts in one place, a chat with three users was required. In this context, the cross-organisational capabilities of MS Teams proved useful. After the Swiss Post account was invited to participate in the project team, it was able to communicate directly with BFH accounts. A sample of the communication involving three accounts is shown below.

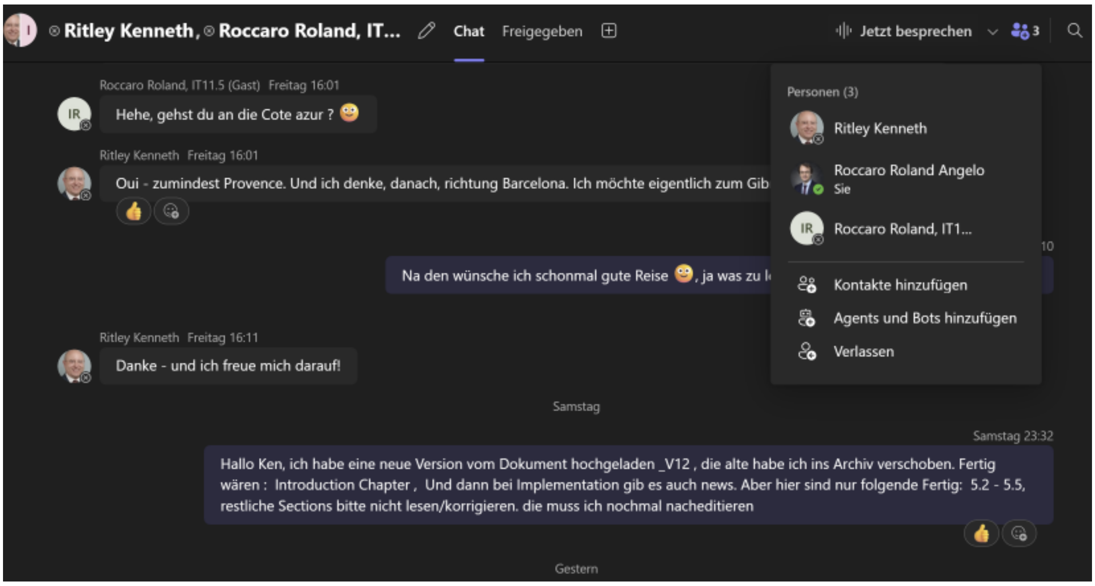


Figure 2.19.: Shared chat using Teams

A second important application of MS Teams is a document management system. This includes source documents, document versions, and related materials. All relevant documents are stored within this structure, which is designed to grow as needed.

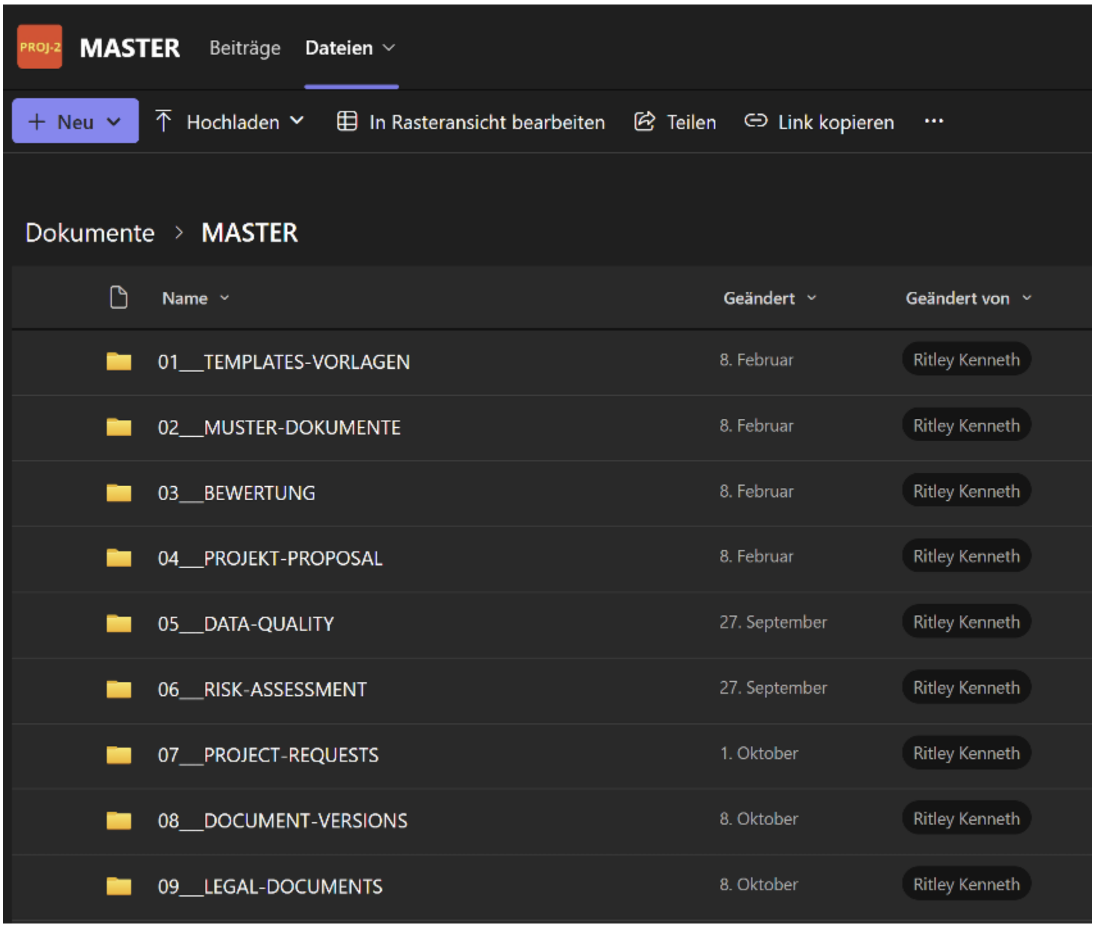


Figure 2.20.: Project document store used with Teams

LaTeX

LaTeX was the tool chosen for the main writing process by the academic team. The primary reason for selecting this tool was the consistency of the document structure. It allows changes to be made in a large document without the document becoming unstable. The tool was specifically selected for this higher consistency compared to Word. The fact that the creation of some stylistic elements takes more time than with a WYSIWYG editor such as Word was accepted, particularly as design was valued lower than document content. Finally, the consistency of being able to reference text sections within the document and the ability to automatically generate lists of figures and tables were seen as additional benefits. Furthermore, the automatic glossary and citation features using Biber were considered advantages for which the slower writing process without a WYSIWYG editor was accepted. Due to the learning effort involved in using LaTeX, the following badge was earned during the course of the research.

Figure 2.21.: Humorous illustration symbolising the extensive use of LaTeX throughout the preparation of this report and the associated learning experience during the project.

GitHub

For maintaining version control of any created artifacts, the public GitHub was used with the private account of the academic team. It was mainly used as a repository. This Git distribution was chosen because it is free to use and already well known to the team, with all required accounts in place, so no additional adoption or learning effort was required before starting. GitHub was not only used to store the document and track its progress, but also to manage all required sources such as tables and images used in the document.

For the writing process, the use of GitHub as a backup mechanism was particularly important. As all sources are stored in GitHub, the project was safeguarded against hardware failures of the working device, which is one risk mitigation measure. Continuously committing and pushing changes made the progress of the writing process traceable and supported the iterative writing approach which was chosen. This proved useful when the document PDF could no longer be generated, as the last version that compiled without errors could be identified. Finally, storing the sources on GitHub allowed access from different devices, which was useful when transferring materials such as screenshots from the work device to the study device.

ChatGPT

LLMs are now an indispensable part of work. To note how ChatGPT was used in this project, it was given the following prompt:

> Hi ChatGPT, please look at the other chats I have with you and tell me in which ways you supported me during my academic project over the past months since September 2025. Please create a subsection for my scientific work which I can reference.

The following subsection was completely AI generated content by ChatGPT, without further editing.

ChatGPT

To document the support provided by ChatGPT, a summary of the assistance received during the academic project was generated directly by the tool itself: The request asked ChatGPT to review prior interactions and describe the ways in which it supported the academic project over the past months since September 2025. The resulting subsection is intended for inclusion in the scientific work and can be referenced accordingly [OpenAI 2025].

The following subsection is entirely AI-generated content produced by ChatGPT, without further editing [OpenAI 2025].

Use of AI Assistance in the Research Process

Since September 2025, the author has used the AI language model ChatGPT *as a supportive tool during the development of this academic project. The assistance provided did not replace independent scientific work but complemented it in the following ways:*

Conceptual Clarification and Structuring *ChatGPT was used to support the clarification of complex theoretical concepts, the refinement of research questions, and the improvement of the logical structure of arguments. This included exploring alternative formulations, identifying conceptual gaps, and supporting the organization of sections and chapters.*

Language and Academic Writing Support *The tool assisted in improving the clarity, coherence, and academic tone of the manuscript. This included rephrasing passages for precision, adjusting wording to meet scientific standards, and supporting grammatical accuracy and stylistic consistency across the text.*

Methodological Reflection *ChatGPT supported reflective discussions regarding methodological choices, including the articulation of research design, limitations, and the justification of selected approaches. All methodological decisions, interpretations, and final formulations remained the sole responsibility of the author.*

Iterative Feedback and Revision *Throughout the research process, ChatGPT was used as an iterative feedback instrument to review draft sections, explore alternative explanations, and identify areas requiring further refinement or clarification.*

Ethical and Responsible Use *ChatGPT did not generate original empirical data, conduct analyses independently, or make autonomous scientific claims. All AI-generated content was critically evaluated, revised, and integrated by the author in accordance with principles of academic integrity and responsible research practice.*

2.4. Timetracking

As part of the need for transparency and the ability to review the efforts committed to this project, time tracking was performed from the start to the end of this documentation. The tracked time was recorded in two separate ways. Work efforts carried out within Swiss Post were captured directly from the Swiss Post MIAR system, noting only the time spent per day without further details on the specific tasks performed. This information was saved in an Excel file and marked as work effort.

Second, academic efforts committed to the project outside the Swiss Post context were recorded directly in an Excel file. For each work block, a separate entry was created. A block represents continuous work on project-related tasks, including short breaks; however, longer breaks, such as lunch, resulted in a new block being created. Excel was used because the Office suite was already in use for other purposes, and the information did not justify storage in a more complex database such as MariaDB. The tracked time is not intended for billing or performance measurement, nor is it used to assess milestone performance.

2.5. Risk management

In this section, the approach taken to risk management is described. Risk management is essential to the success of any project, as it enables planning for and potentially preventing dangers that may occur during its course. When such risks are not handled, they can endanger the progress of the project. The risk management process consists of several steps [Wikipedia 2025]. First, the risks are identified and classified as to their nature. Next, they are weighted according to how they might impact this project; this includes a consideration of the probability that the risks may arise, and the level of damage that may occur. After this, the specific handling of each risk is defined; these are the concrete measures that are taken to lower how the risk may impact the project. Finally, ongoing monitoring and review of the risks is carried out on a regular basis throughout the project.

Risk identification and classification

The identification of risks has been carried out for this project, following an extensive brainstorming approach. Each risk has been grouped into a category (e.g. goal, time and organization, etc.), and each risk has been classified according to the likelihood of is occurance and severity it would cause to the project. Both criteria are evaluated based on the project team's risk appetite and ranked according to the following definitions.

- **Likelihood:** What is the expected likelihood of a risk coming into effect
  - **1** Rare
  - **2** Unlikely
  - **3** Possible
  - **4** Likely
  - **5** Almost Certain
- **Severity:** How severe are the consequences of the risk coming into effect
  - **1** Insignificant
  - **2** Minor
  - **3** Moderate
  - **4** Major
  - **5** Severe

The identified risks, along with their likelihood and severity, are listed in Table 2.1.

Table 2.1.: Project Risk Overview

No.	Group	Risk	Likelihood	Severity
1	Goal	Unclear question	2	2
2	Goal	Unclear goals	2	3
3	Time and Organisation	Insufficient expertise	4	2
4	Time and Organisation	Underestimated time/effort	4	3
5	Time and Organisation	Other study projects	3	3
6	Time and Organisation	Work projects	4	3
7	Time and Organisation	Procrastination	3	2
8	Human Resources	Unavailability of main project staff (e.g., due to illness)	4	4
9	Human Resources	Unavailability of advisor (e.g., due to illness, accident, etc.)	3	4
10	Tech and Infrastructure	Software Version/Library/Dependency	4	3

Continued on next page

No.	Group	Risk	Likelihood	Severity
11	Tech and Infrastructure	Hardware is damaged	2	5
12	Data and Research	Data related	4	3
13	Data and Research	Not reproducible work	2	5
14	Ethical and Legal	Plagiarism or wrong citation	2	5
15	Ethical and Legal	Copyright	1	5
16	Communication and Cooperation	Bad communication with stakeholder or advisor	2	4
17	Communication and Cooperation	Insufficient exchange with external partners	1	5
18	Force Majeure	Unforeseen family crisis	2	5
19	Force Majeure	General force majeure	1	5
20	Communication and Cooperation	Unknown process	5	3

Risk heat map

After carrying out the initial identification and classification, it is not always easy to determine which risks require the most attention. To support this task, a Risk Heat Map is used, in which each risk is plotted in a two-dimensional matrix according to its likelihood and severity. This visualisation highlights the risks that pose the greatest threat to the project and should be considered for priority treatment. The colours in the Risk Heat Map 2.5 correspond to these descriptions.

- **Green** Low
- **Yellow** Medium
- **Orange** High
- **Red** Very High

Risk Matrix		Severity				
		1: Insignificant	2: Minor	3: Moderate	4: Major	5: Severe
Severity	5: Almost Certain	15 17 19	11 14	20		
	4: Likely		16	9	8	
	3: Possible			5	4 6 10 12	
	2: Unlikely		1	2 7	3	
	1: Rare					

Table 2.2.: Risk Matrix (Likelihood × Severity)

As can be seen in the Risk Heat Map, one risk is classified as red and therefore requires special attention, namely, the unavailability of the main project staff. How this risk, as well as the others, is addressed will be described in the following section.

Risk strategies and mitigation strategies

Risk strategies are the high level approaches to dealing with the risks, and the risk mitigations are the concrete plans developed to manage identified risks. In the following table, an overview of the individual risks is provided, including their selected treatment and a brief justification. BNaturally, it is neither practical nor feasible to completely eliminate every risk; therefore, based on likelihood and severity, as well as the team's risk appetite, each risk was assigned one of the following strategies:

- **Accept** – the risk and its potential impact are acknowledged, but no measures are taken
- **Reduce** – the risk is mitigated by applying additional measures to lower its impact or probability
- **Transfer** – the responsibility for handling the risk is assigned to another party
- **Avoid** – the project is adapted in a way that prevents the risk from materialising

Table 2.3.: Risk Strategy and Treatment Plan

No.	Risk	Risk Description	Treatment	Treatment Description
1	Unclear question	The topic description is too vague or contains too many goals	Reduce	Define SMART goals
2	Unclear goals	The goals are not clearly defined or cannot be measured	Reduce	Describe which goals are part of scope
3	Insufficient expertise	Not knowing required tools or programming languages	Accept	
4	Underestimated time/effort	Some tasks take more time than initially expected	Accept	Document end of tasks in Miro calendar; re-evaluate after project
5	Other study projects	Other classes demand more attention, less time for project	Accept	Use minimum time for projects/classes
6	Work projects	Other projects at work demand urgent attention, less time for project	Reduce	Reduction of risk due to starting early; defining blockers for study work
7	Procrastination	Tasks are pushed back	Reduce	Start early
8	Unavailability of main project staff	One or multiple project staff are not available	Accept	Document in case it happens
9	Unavailability of advisor	The advisor is not available	Transfer	Advisor to define emergency replacement
10	Software Version/Library/Dependency	Mismatch in versions; new library versions break the code	Accept	Run the code multiple times; have test cases (e.g., unit tests)
11	Hardware is damaged	The laptops or other material are damaged and need repairs or replacement	Reduce	Second laptop; code on GitHub/Miro/Teams/Outlook

Continued on next page

No.	Risk	Risk Description	Treatment	Treatment Description
12	Data related	Bias in data; not enough data	Accept	
13	Not reproducible work	The work provided cannot be reproduced	Reduce	Document steps; upload versions on GitHub
14	Plagiarism or wrong citation	Unallowed or undeclared usage of sources	Reduce	Use citation; bibliography
15	Copyright	Used literature or materials which are restricted	Accept	
16	Bad communication with stakeholder or advisor	Advisor or stakeholders are not heard; their inputs are ignored	Reduce	Weekly advisor meetings
17	Insufficient exchange with external partners	External partners do not react, are delayed, or communication towards them is late or poor	Reduce	Stakeholders in Scrum team
18	Unforeseen family crisis	Unexpected crisis in family	Accept	To be documented in Miro calendar in case it happens
19	General force majeure	Events which cannot be influenced are very rare and very disruptive	Accept	To be documented in Miro calendar in case it happens
20	Unknown process	Events occurring during the project can block its execution or, in the worst case, lead to its cancellation	Reduce	Always start administrative processes between Swiss Post and BFH without delay. Always send reminders after three weeks at the latest.

Implementation of risk mitigation strategies

Identifying the key risks is the first step in the overall risk management approach; this is followed by defining measures for each of the risks. In the following table, the implementation of these measures is documented. The measures have been implemented according to the strategies and descriptions outlined above. Table 2.4 provides an overview of the actions taken and when they were carried out.

Table 2.4.: Risk Strategy and Action Overview

No.	Risk	Strategy Decision / What	When
1	Unclear question	In the research section SMART goals have been setup	01.03.2026
2	Unclear goals	The introduction describes which parts of the initial project description are part of this document	
3	Insufficient expertise	Accept: Nothing done	01.03.2026

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Table 2.4 – continued from previous page

No.	Risk	Strategy Decision / What	When
4	Underestimated time/effort	Accept: Nothing done	01.03.2026
5	Other study projects	Accept: Nothing done	01.03.2026
6	Work projects	Added blockers in Work calendar, Work account invited to meeting series (Jour Fixe); study account: set blockers on weekend; researched with ChatGPT for optimal/minimal study plan	01.03.2026
7	Procrastination	Work on the project started 25.09.2025 when attending data quality meeting at AWS, being part of AWS Modernisation initiative	01.03.2026
8	Unavailability of main project staff (e.g., due to illness)	Contact of partner provided in Teams and added to Miro board	01.03.2026
9	Unavailability of advisor (e.g., due to illness, accident, etc.)	Asked for contact, confirmed to be Jürgen Vogel	01.03.2026
10	Software Version/Library/Dependency	Accept: Nothing done	01.03.2026
11	Hardware is damaged	Second laptop is available at home; the LaTeX document is a GitHub project; documentation is on Miro. Work laptop can also be used for most tasks, but has no LaTeX installed	01.03.2026
12	Data related	Accept: Nothing done	01.03.2026
13	Not reproducible work	GitHub project for LaTeX document	01.03.2026
14	Plagiarism or wrong citation	JabRef used as GUI for bibliography; Biber used for build	01.03.2026
15	Copyright	Accept: Nothing done	01.03.2026
16	Bad communication with stakeholder or advisor	Jour Fixe setup	01.03.2026
17	Insufficient exchange with external partners	Keep exchange channels active when changing team	01.03.2026
18	Unforeseen family crisis	Accept: Nothing done	01.03.2026
19	General force majeure	Accept: Nothing done	01.03.2026

Ongoing risk monitoring

A critical step in the risk management process is ongoing monitoring. Monitoring the risks ensures that the defined measures remain valid and that any risks that may materialise are identified. If a new risk is identified or causes a disruption to the project, it will be documented. Additionally, an appropriate measure to handle its impact will be defined. Monitoring of the risks will be performed every second Friday of each month. Detailed information about the risk monitoring process is provided in the appendix C. These risk check-ins will contain the following information:

- Number of risk: same as in the tables above
- Name of risk: same as in the tables above
- Occurred: did the risk occur (yes or no)?
- Details: what happened?
- Cause: brief analysis of why it happened
- Measures taken: what measures have been taken?
- Measure date: on what date were measures taken?

Risk review at project completion

Finally, a review of the risks will be held at the end of the project. This will address which risks came into effect and which did not. Additionally, changes to the perception of the risks will be documented at the start of the master thesis.

3. Topic of this research

This chapter establishes the research foundation for the project. It first presents the problem addressed and derives the research question. Based on this, the project's SMART goals are defined to provide clear guidance for the subsequent work. The chapter then identifies the research gap that motivates the project. Finally, core definitions related to data quality and data availability are introduced to provide the theoretical basis for the analysis and implementation presented in the following chapters.

3.1. Problem statement

This work builds upon the previous research [Roccaro 2026], which analysed the existing data pipelines at Swiss Post with regard to operational reporting. Based on this groundwork, the next logical step was to adapt the existing reports to the new reporting platforms. However, in the meantime, the reporting data pipelines were updated. The old data pipelines remained available, but all further updates were performed only on the new pipelines. As a result of this development, the focus of this project shifted towards supporting these changes and helping the team to improve the monitoring and assessment of the existing data. Furthermore, an important objective was to establish the groundwork for the upcoming master's thesis by evaluating whether the available tools could serve as the basis for a use case to be explored in greater depth during the thesis.

Problem generalisation

This project bridges the groundwork established in the previous research towards the follow-up research in the master thesis. First, it has to perform the next step so that the stakeholders are able to continue checking their new data sources. Second, it must help to identify and test the tool for the master thesis. This is the most important part of the research, as it has to confirm that the selected tool is indeed a good match for the upcoming master thesis.

3.2. Research gap

Given this problem statement, the following questions are researched in more detail in the next two chapters. First, a theoretical introduction to the terms is provided; in later chapters, their specific meaning in the context of this project is discussed.

3.3. High-level research question

What is the next meaningful step to take in order to improve data availability and data quality within the DWH LS OPS team at Swiss Post? How can the business and IT be supported in achieving their objectives? While the solutions should be as simple as possible, they should be based on best practices and verified through scientific research. The research gap is addressed in the solution and implementation chapters.

The main goal is to ensure that the conducted research provides tangible benefits to the team and contributes to future improvements. It is particularly important for the team that the research results are presented continuously so that progress, as well as potential deviations and changes, can be discussed.

3.4. Success definition

The success criteria of this project are threefold and are defined based on the research gap as well as the initial project hand-in, which describes the business requirements.

First, the primary business objective is to achieve improvements through report updates. The goal of this project is to replace the existing report immediately and to establish concrete steps towards easier daily usage of the new report.

Second, communication with the business is another important success factor. The results from the research and implementation must be presented to the product owner (PO) on a regular basis.

Third, the testing and documentation of the basic setup required for the new DQM tool, as a theoretical and practical foundation for the master thesis, are the main achievements to be reached.

3.5. S.M.A.R.T. Goals

The goals of this project are now defined here as "SMART goals", as shown in the following list. They are all **S**pecific: the goal is clearly formulated; **M**easurable: there are measurable criteria defining when the goal is reached; **A**chievable: they are realistic given the available resources; **R**elevant: the goal is directly linked to the project's success; and **T**ime-bound: there is a defined timeframe for achievement.

- **1) Setup of an Improved Data Quality Report**

  WHY: This will help the team to run the daily checks faster.

  By the end of the implementation phase, the existing data quality report is to be updated to use the new infrastructure. This includes new reporting data pipelines and the new structure using a star schema. Also, Power BI is to be used.

  Success is measured by user feedback, which has to be obtained by the ASTV team. The feedback is to be implemented, and the report must be on the production systems.

- **2) Evaluation and Selection of DQM Platform for the Master Thesis**

  WHY: To define the DQM platform for the master thesis.

  During the implementation phase, the tool for the master thesis has to be defined. This requires a structured approach for the selection of the tool, including the setup of requirement lists. The team is to follow the DQM team during the evaluation of the tools.

  Success is measured by having the DQM tool defined. The selected tool has to be discussed with the architect of the DWH LS OPS team to make sure it fits the team's needs.

- **3) Validation of the Selected DQM Tool**

  WHY: The DQM platform must be tested for applicability to the master thesis.

  By the end of the implementation phase, the DQM tool has to be tested for applicability to the master thesis. It must be proven that the required infrastructure can be set up for a test. Furthermore, a simplified test has to be run to confirm that the tool can handle the data sources.

Success is measured by having the DQM tool tested. The tool must return the test values to prove that it can be used for the follow-up master thesis.

3.6. This project as a key step towards the "future state" reporting solution

As stated earlier the solutional goal of this project and its follow up is to provide an MVP for DQM, from which the key learnings will enable the full "future state" reporting solution. This MVP should implement an automatic data quality check, replacing an occasional, non automated check performed by the business. This is important, as the DQM check is currently executed by the business instead of IT, which creates challenges with regard to the desired target state. The expected situation would be that IT executes data quality checks and defines thresholds together with the business. At present, however, the knowledge about thresholds and other relevant information resides solely with the business.

In summary, having IT run the data checks would enable continuous follow-up of data quality. For example, a report based on automatically executed checks could show system uptime over a longer period of time. This information would later allow IT to extrapolate standard value ranges, which in turn would enable the definition of alerting thresholds. In short, having DQM checks executed automatically rather than occasionally by the business provides multiple benefits for both IT and the business.

Figure 3.1.: DQM manual checks against automated checks by IT

3.7. Expected benefits of this reporting solution

A single visualisation does not explain the full benefit of the intended solution. For instance, the question arises as to why the business should stop performing data quality checks when it has the most detailed knowledge of the data. This is quickly clarified by taking a step back. The question should not be why checks are removed from the business, but why the reports are performed by the business in the first place. The benefits of this shift are therefore outlined in more detail in the following.

Benefit 1: Automation over manual processes to reduce effort

The business currently spends 30 to 45 minutes per day on a single data quality check, according to the initial project hand in description. From the duration of the check, it can be inferred that anyone requiring the data quality check before a report based on it is approved must wait until it is finished, resulting in a waiting time of 30 to 45 minutes. This fact provides the basis for understanding why manual processes are not suitable at scale. When scaling up to ten data quality checks, five to seven and a half hours are spent by the business on data quality checks. The waiting time for the last reports to be approved increases accordingly to five to seven and a half hours. When increasing to 20 checks per day, ten to 15 hours are spent, and the waiting time rises accordingly, accounting for more than one full-time position and long delays in report data release. It should be noted that this amount of time is spent solely to determine whether the defined check passes or not. If the same initial check is executed by an automated data quality process, the expectation is that execution takes less than one minute; for simplicity, an execution time of one minute is assumed. As no manual initiation is required, no human time is spent performing the check. For comparison, the one minute waiting time for report approval and release is still considered. The business can then be informed whether the check returned an OK or not OK result, and the outcome can be stored for future reference.

When scaling this approach to ten checks, the total execution time remains ten times one minute. However, as the IT system can execute these checks in parallel, the waiting time for results remains at one minute. When increasing to 20 checks per day, the IT systems likewise execute the checks in parallel, and the execution and waiting time remain unchanged.

Benefit 2: Scalability to further reduce overall time required

In summary, manual data quality checks cannot be scaled and lead to excessive use of human resources for repetitive tasks, as well as long waiting times for approved datasets. Automated data quality checks, by contrast, allow many checks to be executed in parallel within the same overall time frame. It should be noted that this comparison assumes all checks have similar complexity.

Benefit 3: Rule changes without query changes

The current manual reports, which are business driven, have their logic built directly into the queries executed on the data. If any rule changes or a part of the data processing is modified, these queries need to be adapted. Thus, the check is deeply tied to the data source. To illustrate this with an example, a data quality check may confirm that 30% of the data is category A and 70% is category B, as this is considered the expected and valid distribution. If changes to processing pipelines or client behaviour cause the share of A to increase to 45% and B to decrease to 55%, the query performing the check must be adapted. However, when the check is performed by an application that injects these thresholds into a query, the query itself does not need to change, and only the parameters must be adjusted.

Benefit 4: Reusability to improve stability and performance

By moving the responsibility for checks from manual, business-driven processes to automated checks executed by IT, the checks can be designed to be more generic. Generic checks can be reused across different use cases. This separation of responsibilities and concerns leads to improved stability and performance when applying the checks, which in turn enables strong reusability. The same check type can be applied to different data sources by adapting parameters such as the query and thresholds, while the underlying check logic remains unchanged.

4. Solution landscape

In order to enable the reader to understand the complexity of the Swiss Post DWH LS OPS, the data lineage must be presented at a high level. This chapter provides an overview of this lineage while deliberately omitting details that are not relevant to this project. It further explains the scope of this project within the data lineage and describes the reporting lineage with relevant tools. Furthermore, it is important to note that the solution landscape reflects the current state of knowledge and is limited to the project scope. Any additional knowledge acquired during implementation is discussed in the Results and Target Analysis chapters.

4.1. The Swiss Post addresses data quality in multiple ways

At the start of this project, the objective was to develop a minimal viable product (MVP) to replace a manual data quality check performed by the business with a more robust solution, including automation, the delivery of reports, as well as an appropriate notification and alerting. However, after beginning this project, Swiss Post management assigned a new priority to the topic of data quality management (DQM) overall; therefore, quite unexpectedly, this project acquired an additional set of key stakeholders, located in different teams as discussed in chapter 2.1.

Fortunately, this project as originally conceived fit very well into the larger DQM picture. Coordination with these teams although not originally envisioned (such as the DQM team described in chapter 2.1) therefore became an important task. This coordination was essential for several reasons: first and most obviously, the project must avoid reinventing existing solutions or overlapping in an unplanned way with other new initiatives. Second, it must ensure that methods and deliverables of this project conform to any new data architecture standards; for example, data quality dimensions have already been defined and documented by other teams. Finally, it must ensure that ongoing work by other teams is taken into account.

4.2. Rule-based DQM initiative

The current architectural goal pursued by the DQM team is shown in the following visualisation. The rule evaluation engine, shown at the top centre, executes defined rules on target data and stores the corresponding rule results in a database. In this context, a rule represents a data quality check for a defined dataset, and many different rules can be defined. These rules may, for example, specify allowed values, such as restricting the gender attribute to the three values *male*, *female*, and *not stated*. Alternatively, they may define expected data types, such as requiring dates to be stored in UTC format, or specify valid value ranges depending on the weekday.

Although the rule evaluation engine was a major topic during the weekly exchanges with the DQM team, it was not implemented by the project team. Nevertheless, its development progress was regularly monitored and discussed during these meetings. Furthermore, the ASTV team used this tool for its intended purpose. However, this usage was outside the scope of this project.

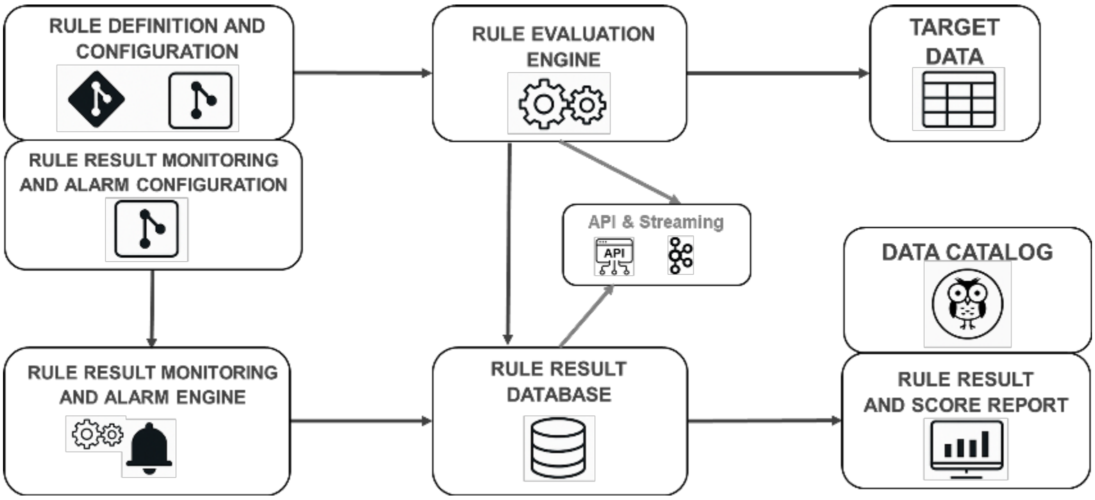


Figure 4.1.: The rule check engine architecture for future DQM

High-level target

From the complete target solution architecture, the first component to be implemented is a rule-based DQM suite. Its objective is to define generic data quality checks that can be applied across different datasets. This is achieved either by abstracting the check from the underlying data or by providing an SQL query that returns the data in a format allowing the same check to be applied to different datasets. The technology used to implement this solution is not discussed in this document, as the implementation was carried out by another team and is therefore outside the scope of this project. This approach consists of four main steps:

- **Configuration**: Configuration defines the tables to be checked, the execution frequency, and the type of check.
- **Data preparation**: An SQL query prepares the data in a format suitable for the check.
- **Rule execution**: An application executes the defined checks on the prepared data.
- **Result persistence**: The results are stored in a database, which enables reporting.

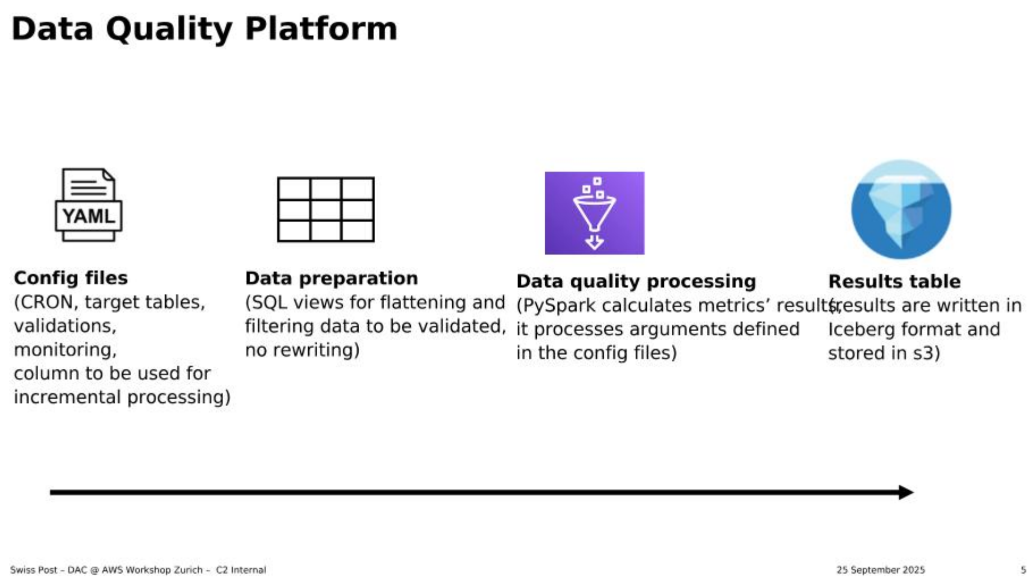


Figure 4.2.: Intention of a DQM platform, high level

Discarded solutions

This initiative is of high importance to the project, as it creates a major organisation-wide constraint to follow the DQM team's choice of tool. The rationale for developing a rule-based DQM solution is that existing tools, such as AWS Glue Data Quality, do not support views with complex data types, while PyDeequ does not provide row-level results. These limitations in the evaluated tools led to the decision to develop a dedicated solution at Swiss Post, as described above. The following section provides a brief overview of the tools and approaches that were evaluated but ultimately discarded as unsuitable. The following screenshot shows the presentation in which the DQM team summarised the evaluated tools and explained the reasoning behind their decision.

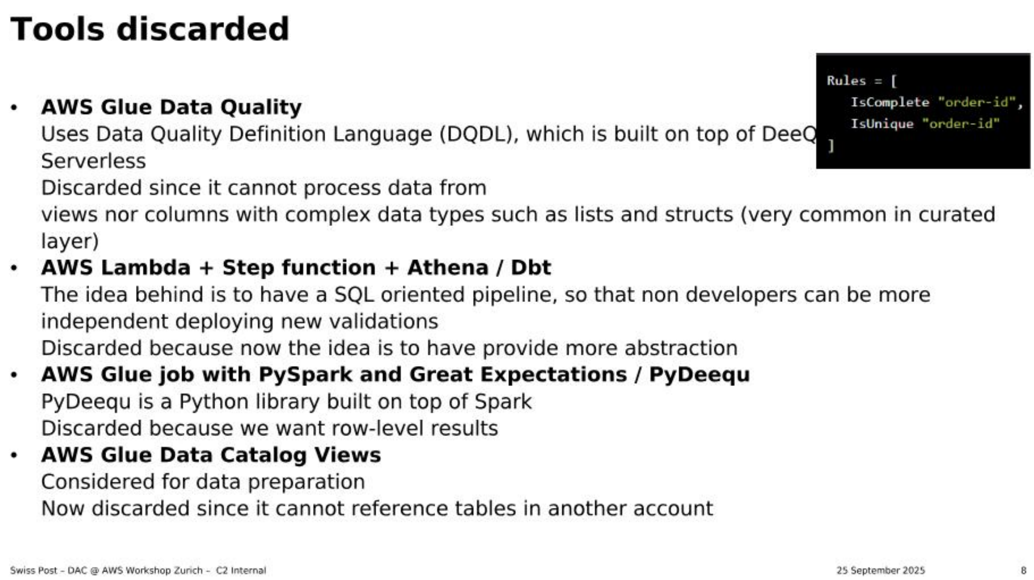


Figure 4.3.: The list of tools discarded and the reasons for their exclusion

State at the end of this project

By the end of 2025, the DQM team set up a PoC of the rule check engine based on the design shown at the start of this section 4.1. It allows running a check on a database view for specific target data source and storing the rule results in a designated rule result database. However, many details were still not defined or implemented. Such as the location where the results check database is to be installed which has not been defined. Also the alerting and the setup of dashboards are currently out of scope, as are later additions such as deriving expected value ranges from historical data or holiday calendars.

4.3. Heterogeneous Sorting Centre Environments

Before turning to the data lineage, an important constraint must be highlighted. The hardware used by Swiss Post in its high-tech "cyber" postal sorting centres differs in two main ways: across different providers and across different sizes and organisational structures of sorting centres. This, in turn, also led to a diversity of processes for performing similar actions. This diversity at different levels brings certain benefits but also introduces additional synchronisation effort in other areas.

Different sorting hardware providers

The hardware used by Swiss Post in its sorting centers is sourced from different providers. This is an essential risk management measure to reduce dependency on a single vendor. To mitigate this risk, Swiss Post decided to rely on at least two providers for essential hardware, not only to avoid dependency but also to protect against delivery issues or even the bankruptcy of individual providers. From a risk management perspective, this approach supports business continuity. However, from an IT perspective, it introduces challenges, as different hardware providers deliver data in different formats or with varying characteristics.

Different sorting center sizes and structure

The Swiss Post parcel sorting centers are distinguished into two categories. There is one large PC parcel center per region, which is supplemented by smaller RPC regional parcel centers. Regions encompass multiple cantons and may split cantons into the four current regions: west, middle, east, and south. The organisation into larger parcel centers and smaller regional parcel centers helps to optimise processes and reduce the carbon footprint of postal services. However, each PC and RPC differs from the others.

First, due to differences in size, processes differ between small and large centers, and the way of work also varies as organisational structures are different. Second, beyond the distinction between small and large centers, each center has a unique layout. Third, tasks can differ between centers; for example, some also handle letters. While these differences are reasonable from an operational perspective, they introduce challenges for centralised reporting, where all variations need to be harmonised. The following figure shows selected PC and RPC locations distributed across Switzerland.

Figure 4.4.: Selection of parcel sorting centers distributed over Switzerland

Different sorting processes

To go one step further, the different size and equipment at the sorting centers lead to different actions performed in the sorting centers. The same event in a sorting center can be caused by different previous actions making reporting on them difficult. As an example, the submitted and discharged events are described further.

Submitted event

The submitted event is the first automatic scan that occurs after a mail piece is placed onto the automatic sorting machine. In the context of this project, a mail piece refers to a parcel. However, parcels are not always placed onto the sorting machine in the same way. This can occur via roll boxes (RX), which are automatically emptied by robots. An RX is a container, as depicted in the following figures, that is used to transport different mail pieces through sorting and delivery facilities. Alternatively, parcels can be placed onto the automatic machine manually by staff, either by removing them from RX units or by emptying truck containers onto conveyor belts, as shown in the subsequent figures.

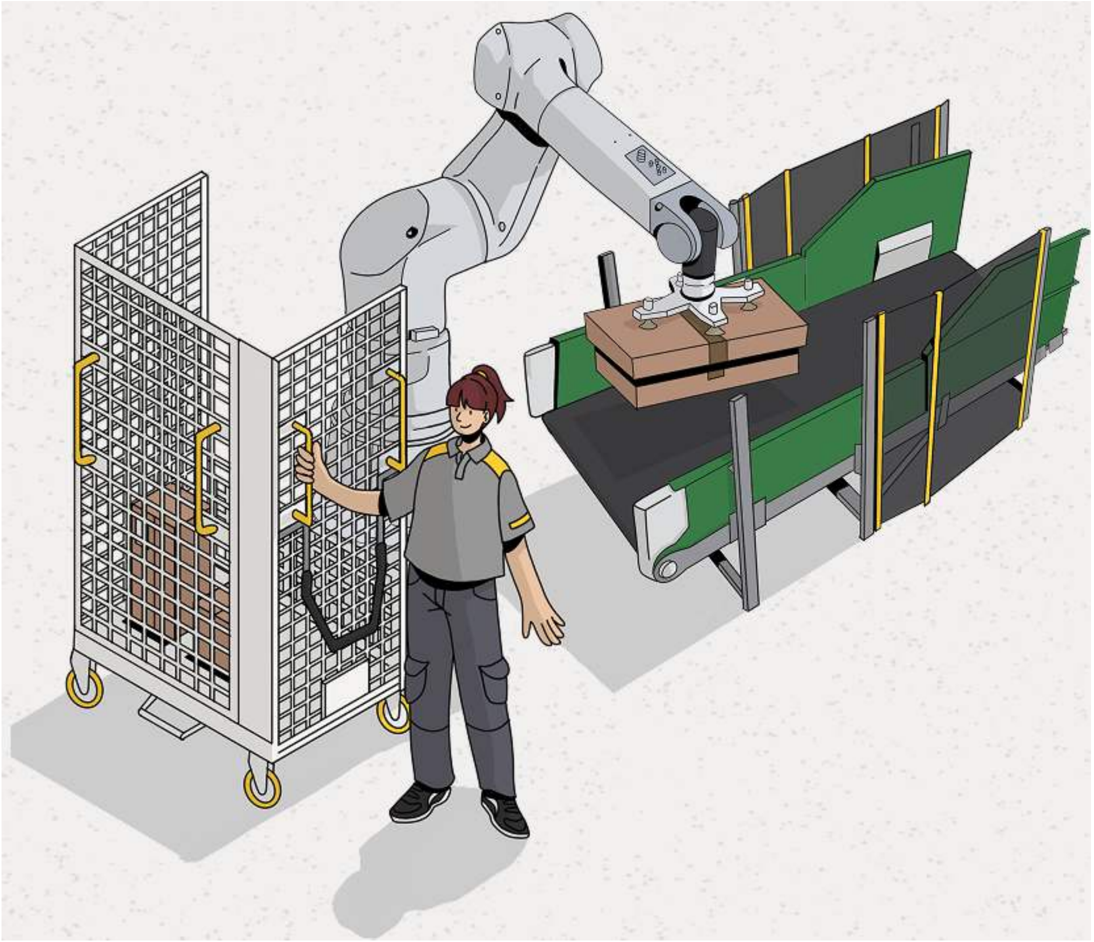


Figure 4.5.: Parcel sorting, parcels submitted by robot

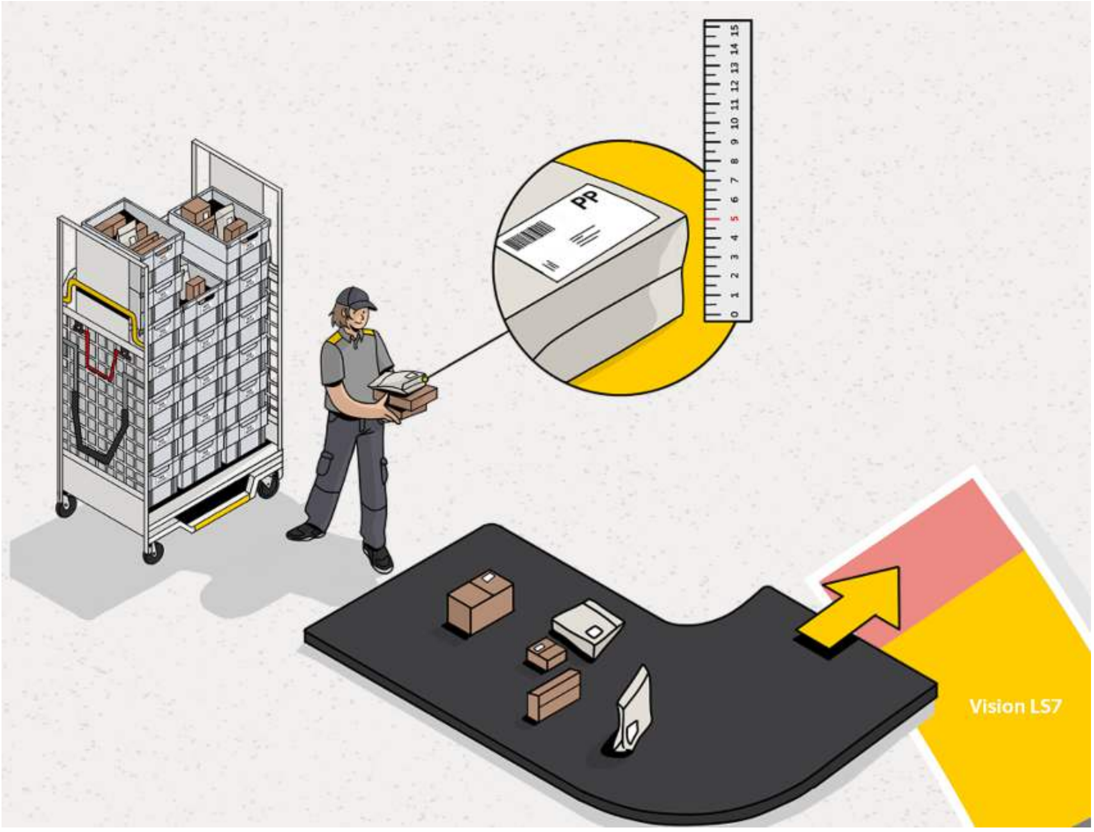


Figure 4.6.: Parcel sorting, smaller items submitted manually

Figure 4.7.: Manual parcel onloading from a truck

Discharged event

The discharged event is complementary to the submitted event. It is the last automatic scan event of a parcel when it leaves the automated sorting machine. This can happen in different ways, such as when the parcel is discharged via a chute where staff load it into an RX, when it is discharged directly into an RX automatically by the machine, or when it is automatically ejected after completing a defined number of cycles through the machine. All of these possibilities trigger different follow-up actions. However, in all mentioned cases, a discharged scan is performed after the parcel is removed from the automatic sorting machine. This is the important aspect for reporting, as these two events (submitted and discharged) occur once per complete cycle that a parcel runs through the automatic sorters.

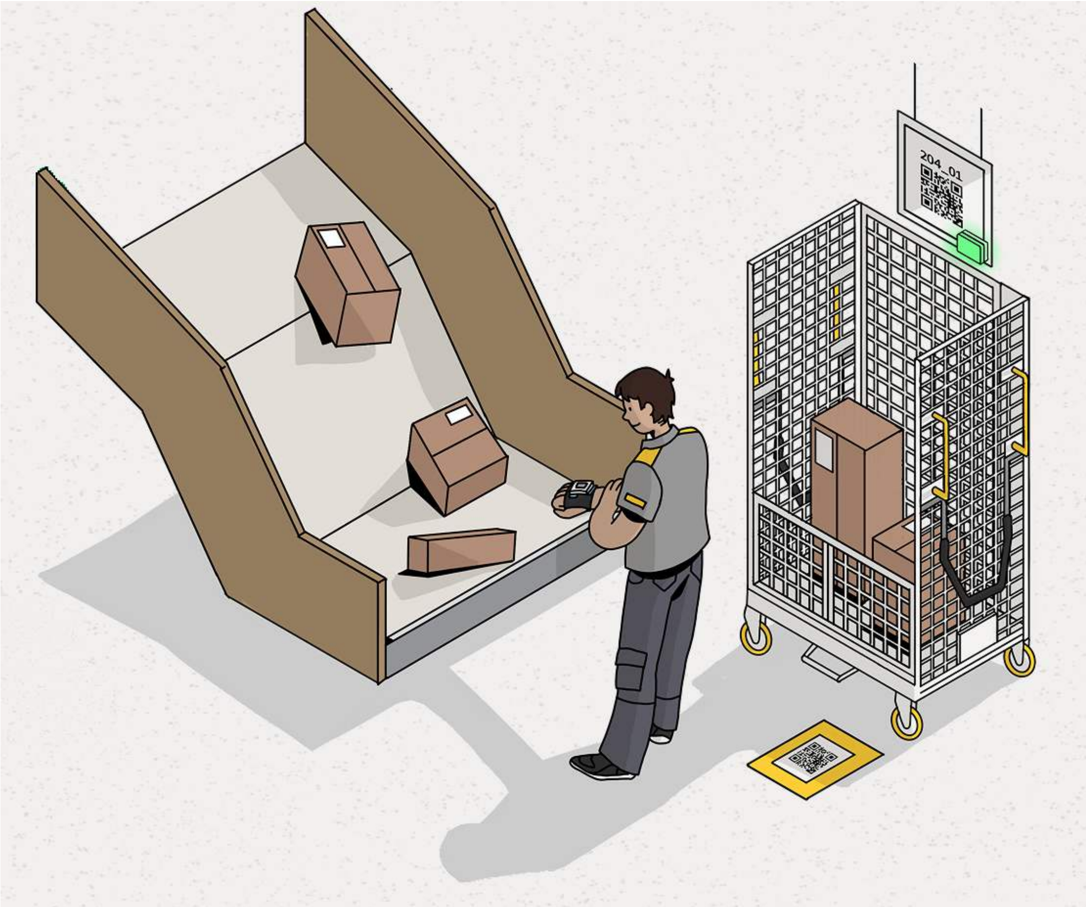


Figure 4.8.: Parcel discharged, manual loading into RX

4.4. Data Lineage

The data relevant to this project are those used by the DWH LS OPS team. They consist of cleaned and deduplicated data stored in the curated database layer. However, these data originate from sensor data and arrive in the data warehouse (DWH) in a raw format. Before reaching the curated layer, the data pass through four main processing steps, which are also shown in the following figure:

- **Physical data source**: Parcels are scanned by different scanners in the sorting centres. As described above, Swiss Post uses a variety of devices from different hardware vendors, which naturally results in a large number of data entities that must be stored. Each parcel passes through multiple scanners during its journey across the Swiss Post's 13 sorting centres, generating multiple data entities that are transferred to the data warehouse. Determining the exact number of generated events was outside the scope of this project.

- **Initial data storage**: The scan events are first saved in AWS S3. At first glance this might be a very unusual choice, but the reasons make sense. First, the Swiss Post has a cloud strategy which includes AWS as a service provider; hence the choice of AWS is no surprise. But more importantly, Amazon S3 (Simple Storage Service) is the object storage service of AWS. Exactly which data are stored here and how is addressed in a following section.
- **Kafka ingestion**: Apache Kafka is an open-source distributed event streaming platform designed for high-throughput, fault-tolerant, real-time data pipelines and event-driven architectures.The Swiss Post [operates a Kafka environment in its down data centers] [uses the Amazon MSK Managed Streaming for Kafka]. With this, so-called "producers" read the data from the initial storage and ingest it into Kafka "topics," to which other systems can "subscribe" and obtain the data.
- **Data delivery to DWH**: Kafka consumers ingest the events into the DWH LS OPS raw data layer. Which is the first storage layer in this teams AWS S3 account, saving the data as recieved from Apache Kafka.

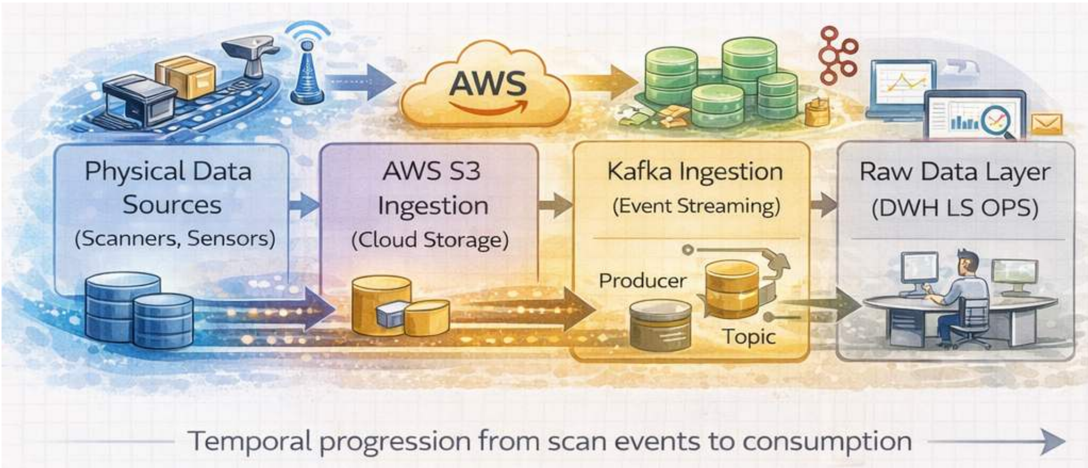


Figure 4.9.: An artistic high-level view of the DWH LS OPS data lineage

KEY POINT: From the data creating stage until the data arrives at the DWH, it passes through four IT system steps discussed here. This results in four potential sources of failure, in which each system is counted only once and by ignoring different possible errors per step. This landscape makes it difficult to trace errors that can occur at any step. Additionally, the variety of errors is large, as they may alter data values or result in no data being transmitted at a given step, even though the operation of the sorting center itself continues without issues. Therefore, at a technical level, providing clarity especially during the operational aspect is an important key aspect of this project.

Finally, to complete the picture of the data lineage, one more step is relevant. After the data has arrived in the raw layer at DWH LS OPS, it must be cleaned and deduplicated. At this step, differences in hardware providers and the structural differences of sorting centers come into effect, and the unification of the data takes place. All of this is performed in the step towards the curated database layer, after which a reporting layer serves as the final layer from which reporting tools retrieve data as introduced in chapter **??**. This is illustrated here with a focus on the central step from the raw layer to the curated layer. Even after the curated layer, there is one additional layer, namely the reporting layer.

The different layers in the cloud are set up according to a structure similar to that used in the classical on-premises DWH at Swiss Post. This similarity to the on-premises DWH stems from deliberate architectural alignment. The cloud infrastructure was built with the guidance of on-premises developers using their domain knowledge. In this way, functional and structural consistency was maintained, as the architecture continues to follow the Kimball star schema and the layered architecture as defined in the Kimball technical architecture [Kimball Group n.d.].

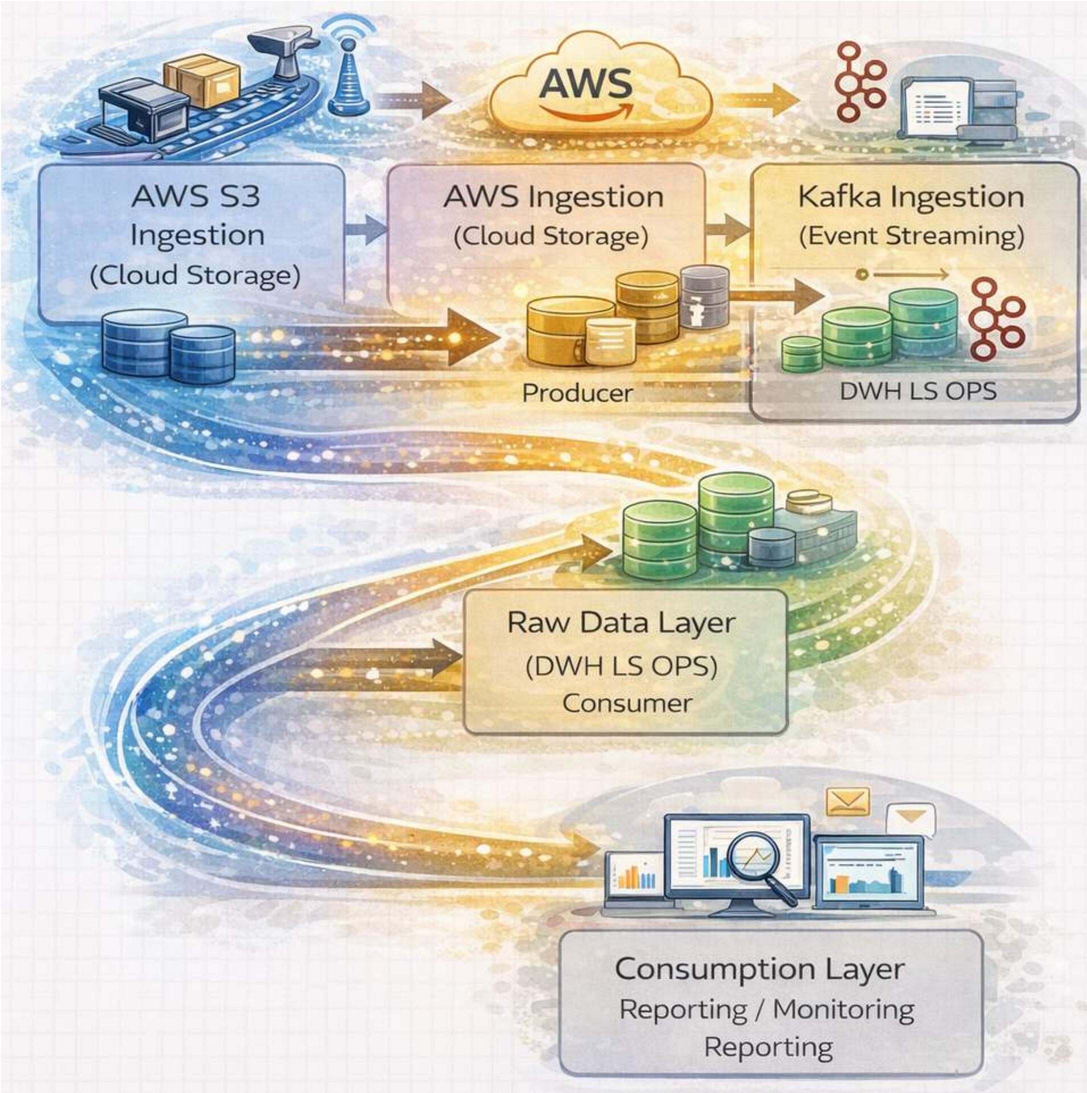


Figure 4.10.: An artistic high-level view of the full DWH LS OPS data lineage up to reporting

4.5. Existing reporting pipeline

The current reporting pipeline at DWH LS OPS team consists of four main components as shown in following figure and introduced in chapter **??**. First, the storage layer (in figure : **Datenspeicherung**), where data is retrieved, cleaned, and saved to Amazon S3. Second, the query and processing layer, where cleaned data is queried using Amazon Athena and DSP(in figure : **Datenselektierung, Datenpreparation Datenmodelle Data Federation**), and where data is passed and persisted in data models for reports. Third, the reporting and consumption layer, in which data from DSP is used in SAC reports (in figure : **Business Layer**).

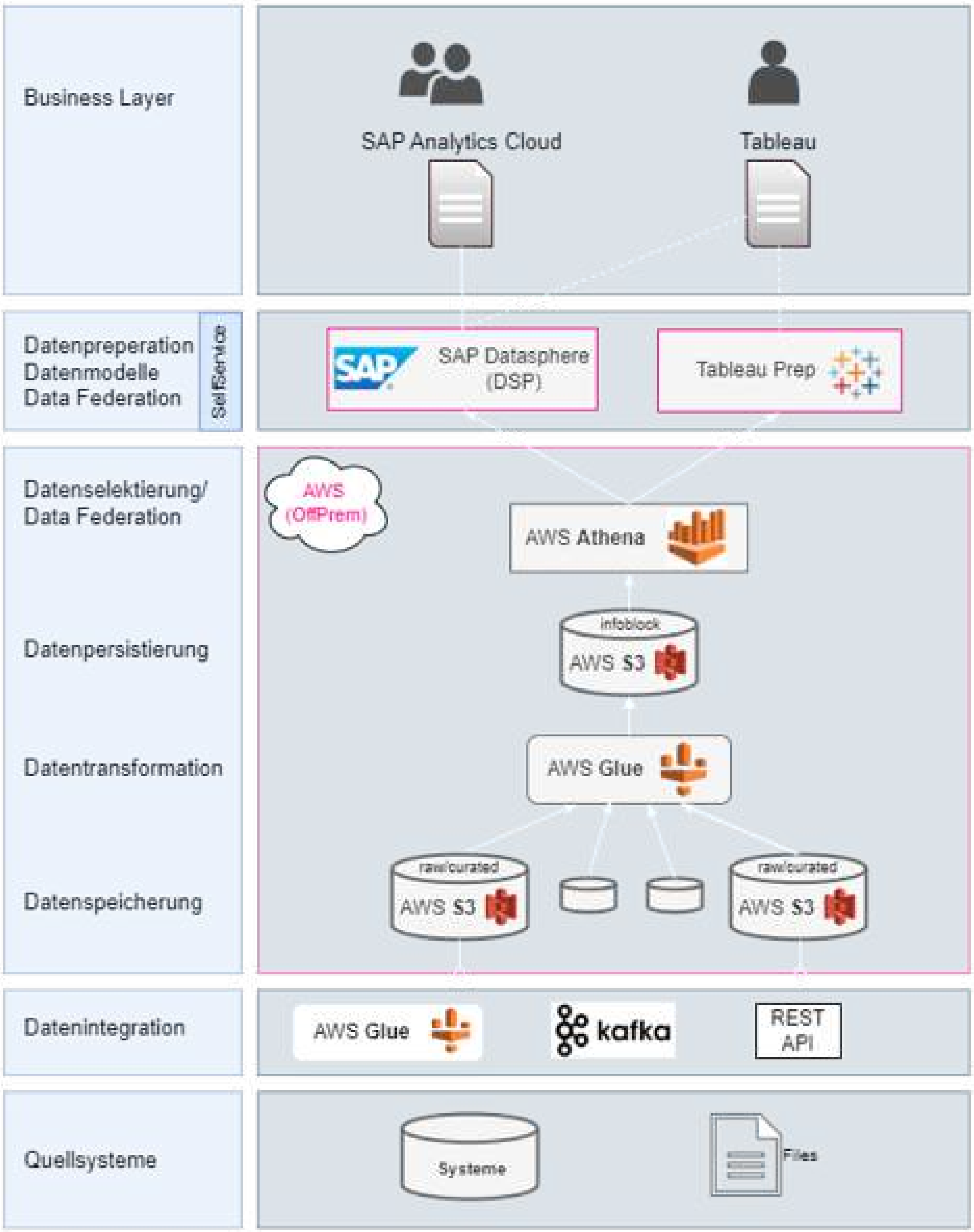


Figure 4.11.: The DWH LS OPS reporting layers

AWS S3

As described by Amazon: *"Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. Customers of all sizes and industries can use Amazon S3 to store and protect any amount of data for a range of use cases, such as data lakes, websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics. Amazon S3 provides management features so that you can optimize, organize, and configure access to your data to meet your specific business, organizational, and compliance requirements."* [Amazon Web Services 2025b].

This is relevant for Swiss Post, as Amazon S3 serves as the central storage for data from mail piece events, which include all scans of parcels and mail in sorting centers, as well as later stages up to the delivery process. This storage is used by the DWH LS OPS team. All data entering the data lineage pipeline, for example through Kafka or through direct database connections, is stored in Amazon S3. Thus, it represents the first persistence point in the reporting pipeline.

Layered architecture It is important to note that the data is not only initially saved to Amazon S3 in a raw layer, but is also cleaned and prepared for reporting in two additional layers. To separate the storage of raw data from the cleaning process, multiple layers are used. This ensures that all data in its original format is saved first before processing, modification, augmentation, and checks for completeness or correctness are performed. Each step can therefore be reproduced and repeated, which is useful whenever data needs to be reloaded, for example when data was incomplete for a day or when new processing methods are developed. After the raw layer, data is deduplicated and cleaned, and then saved into the curated layer. The final top layer, the reporting layer, mainly adds translations of fields into business terminology, for example by changing column labels. This is useful for the business, as it enables consistent naming of columns across tables or data products.

Infoblock On this level, infoblocks are generated. This is a Swiss Post internal term used for report-ready data products. An infoblock consists of cleaned, deduplicated, curated data featuring joined datasets that use business naming, e.g. post day and processing priority types. Basically, the infoblocks are the final layer employed in the reporting layer, making them accessible to reporting tools such as DSP and SAC, which are described in detail below. A more common term for an infoblock known throughout the industry is *data product* [*What is a Data Product?* N.d.(a); *What is a Data Product?* N.d.(b)].

AWS Athena Query

The next step is Amazon Athena as the query layer, which Amazon describes as follows: *"Amazon Athena is an interactive query service that makes it easy to analyze data directly in Amazon Simple Storage Service (Amazon S3) using standard SQL. With a few actions in the AWS Management Console, you can point Athena at your data stored in Amazon S3 and begin using standard SQL to run ad-hoc queries and get results in seconds." [Amazon Web Services 2025a]*

Amazon Athena is used by Swiss Post DWH LS OPS for all queries on Amazon S3, which is the next step in the reporting pipeline. Athena is the tool designed by AWS for use with Amazon S3, making it the logical choice for querying the stored data. The data is queried for reporting purposes by selecting the relevant subsets.

DQM relevance The querying layer is relevant to DQM, since data quality issues may become apparent first at this earliest checkpoint, for example when errors occur in the data itself and counts or joins become invalid. While the data is queried at this layer for reporting purposes, it is also queried here for data quality checks. Such checks typically address missing data and data completeness, but can also extend to data plausibility, for example when postal codes are compared to cities. In cases where issues occur at this layer, they are passed downstream towards DSP and SAC, provided they do not already break the pipeline at this stage.

SAP Datasphere (DSP) and SAP Analytics Cloud (SAC)

At Swiss Post, the next step in the reporting pipeline uses the two dependent tools SAP Datasphere (DSP) and SAP Analytics Cloud (SAC). These are described as follows by SAP:

SAP Datasphere enables organizations to combine, harmonize, and model data from multiple SAP and non-SAP sources within a semantically rich, cloud-based data environment. SAP Analytics Cloud can directly consume these governed and business-ready data models via live connections, enabling interactive reporting, analytics, planning, and forecasting. In addition, planning and forecast data created in SAP Analytics Cloud can be written back to SAP Datasphere, allowing analytical and planning data to coexist in a single, integrated data foundation [SAP SE 2024].

SAP Datasphere (DSP) is SAP's cloud-based data platform for integrating, modeling, and analyzing data across SAP and non-SAP systems while preserving business semantics. It enables federated access to distributed data without full replication and supports end-to-end data lineage and governance. DSP at Swiss Post is used as the analytical backbone for reporting, analytics, and data-driven decision-making in SAP landscapes.

SAP Analytics Cloud (SAC), in turn, uses the data model provided by DSP to present this information on the business layer to end users. More information on this is provided in the separate section introducing the tool.

DSP data model

After querying, the data is persisted in DSP, where it is made available in data models. This persistence and processing layer is essential in the current pipeline, as data can be made available to SAC only through DSP. At Swiss Post, data persistence has proven essential to improving the response time of the reporting pipeline as a whole. With this persistence, DSP does not pass queries back to Amazon Athena, creating a clear separation between the two layers. Due to this separation, DSP is not responsible for DQM, which is handled in lower layers. Finally, DSP is the layer in which business logic is applied in a semantic form, defining what the data represents. At Swiss Post, this is done by applying business oriented naming and calculations. Different report consumers may require the same values to be structured in different ways; for example, productivity in a sorting center may be measured using total working hours against the number of mail pieces handled, or calculated using only time spent on sorting-related tasks, excluding breaks, training, and other activities.

DSP use case The data models in DSP are useful not only as a bridge to SAC (discussed below). They allow different data sources to be modelled and combined into unified data models. This is also relevant to Swiss Post, as it operates not only Amazon Athena data sources but also other database- or file-based sources. This combination of data inevitably inherits data quality issues from the underlying sources, which can affect joins, data loading, and further processing. Nonetheless, DSP is the sole source of data provided to SAC, which is the next logical step in the reporting pipeline. This also means that any unhandled data quality issues are passed directly to users consuming reports in SAC and may therefore affect confidence in the reporting pipeline depending on data quality.

SAP Analytics Cloud (SAC) report

SAC is the final step in the reporting pipeline and represents the reporting and consumption layer. As the last layer, it inherits the data from previous layers and is responsible only for its visualisation. The data is displayed passively, and no further transformations are performed at this stage. Consequently, any errors in data preparation or data quality issues from upstream layers are propagated to this layer together with the data itself.

As described earlier, SAC is dependent on DSP and cannot operate independently, nor does it provide native DQM capabilities. The reports generated by SAC are therefore fully dependent on the preceding layers, and any issues or delays are reflected directly in the reports. This is particularly relevant for Swiss Post, as one report provided by DWH LS OPS is used daily at 07:00 for shift handover, which takes place one hour after the end of the post day. Some data sources required for this report become available only after 06:00, the end of the postday. If any upstream step is delayed, the report delivery is delayed accordingly.

Consumption layer As the consumption layer for reports, this is the point at which users with read access have their first and only interaction with the reporting pipeline. It is at this stage that user confidence is either established or lost, making it particularly critical. Users do not see the upstream processing steps and typically do not differentiate whether missing or delayed information is caused by AWS S3, Athena, or DSP.

4.6. Report visualisations and evaluations

After introducing the basic solution landscape, this project now examines the more specific context of report visualisations and the evaluation of automation.

Power BI

At Swiss Post, within the DWH LS OPS environment, Power BI is the designated business intelligence tool. It is intended to replace the currently used tools over time. The main reasons for this shift are the lower licence costs compared with the previous SAC; furthermore, it performed faster in direct comparison tests with its predecessor. Therefore, Power BI served as the BI tool in this project, as it is specifically replacing an old report in SAC.

Architectural position

When considering the data lineage, Power BI is replacing both DSP on the data model side and the front-end part of SAC. In total, it is replacing the work areas of the business stakeholders working together with the DWH LS OPS team. This is because they are involved in both the generation of tests and the updating of reports within the reporting and monitoring layer.

Star schema

During the speed tests and comparisons with the old business intelligence tool, the star schema as a data architecture was tested. It showed that Power BI performed better with this new structure. Therefore, as Power BI was designated as the new business intelligence tool, the architecture was changed accordingly.

Basic architecture of a star schema

When looking at database architecture, a star schema is basically a partly denormalised database that has been optimised for fast OLAP processing. The data are split into repeated dimension informations, which are linked with the factual data in the fact table in the middle. In such a architecture there is only one fact table holding the information the model is providing and multiple dimensions which are repeating informations. In the case of the message processing star shema, the fact table are the messages, who come with their specific informations such as an unique ID to identify them. They are then linked to the different dimensions trough other id's which can be linked to entries in the dimension tables. As a toy sample the message "ABC134" may have the desination 4, which is in the destination table the id 4, and can represent a whole adress or a person. More information can be found at this source [Wikipedia contributors 2025]. Furthermore, the following figure shows a sample star shema with a fact table in the middle and many connected dimensions around it.

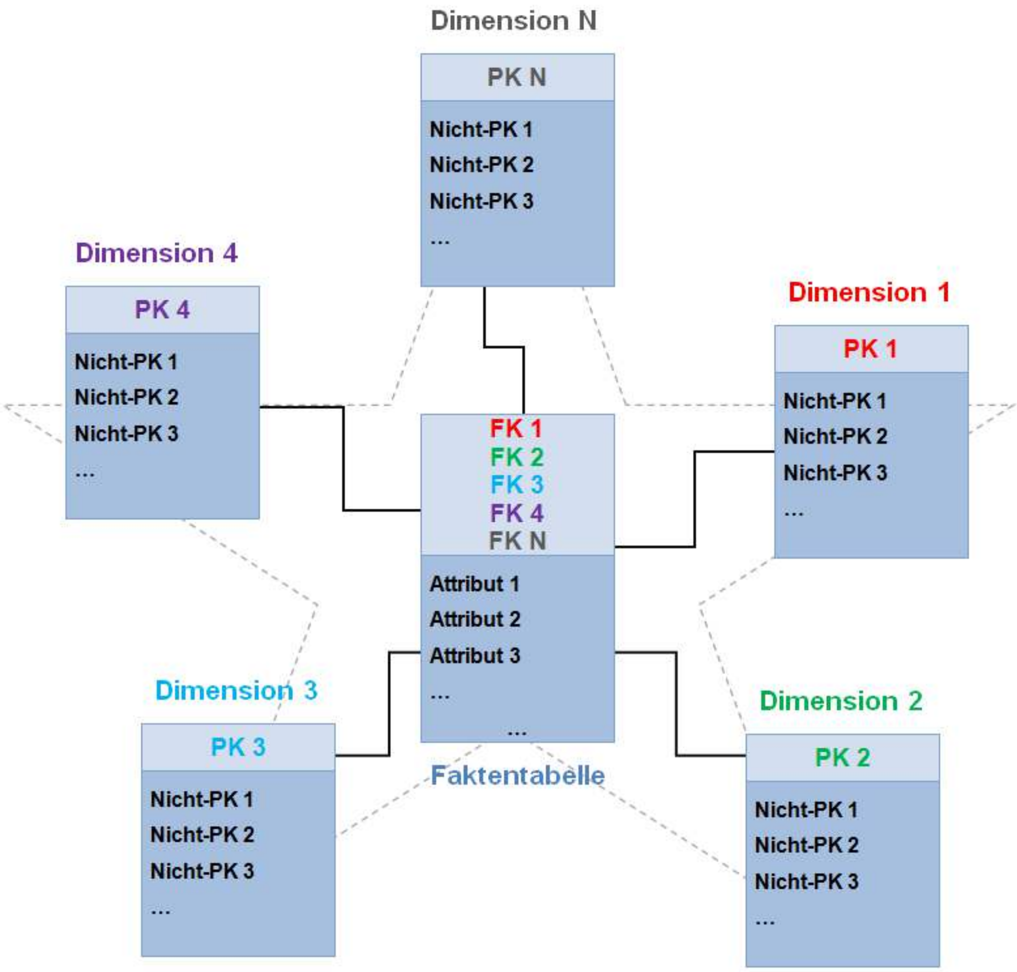


Figure 4.12.: Illustration of a star schema showing the relationship between a central fact table and multiple dimension tables. Image adapted from Wikipedia.

Basic architecture of a wide table

In contrast to a star schema, a wide table stores all the data in one large, very wide table. This concept of the wide table was introduced at Swiss Post together with SAC. In a wide table, the data are not split into different tables; instead, all existing measures are repeated for every dataset. For example, every data point measured at the same factory would have the same identifying address information repeated, rather than a single key pointing to an addresses table [QuestDB 2026]. Such a table makes it easier to create reports which are already known at the moment of setting up the table, as every data required is available. However if additional data is required a wide table has to be extended.

Reporting idea

When considering the previous report generated through SAC, it can be seen that it is not very visual. The data are shown only as a table without colour highlights or additional indicators. Without sufficient business knowledge, it did not allow a user to determine whether the data quality was acceptable for a specific day. The current report is shown below.

Figure 4.13.: Existing plain DQM dashboard used as the baseline for the redesigned Power BI report

To improve this situation, a report that had to replace it needed to be more visual and apply some of the most basic business knowledge, allowing users to interpret its results more quickly. As the previous research had already defined which rules had to be applied in an automatic check, those rules formed the basis for the report's visualisation.

Business stakeholder feedback

During meetings with the business, their requirements were gathered. These were simple: the report had to be upgraded to use the new data source. Since the adoption of Power BI also enabled a more efficient reporting data structure, this had to be measurable again. As the data themselves had only been rearranged and were not fundamentally different, the checks and the way they had to be shown in a table remained the same. Regarding additional improvements to the report, no direction was given; however, approval was provided to improve the report with useful information.

DevOps

At Swiss Post, deployments are carried out by software engineers, effectively making their field of operations DevOps.

The field of DevOps is described by Wikipedia as follows [Wikipedia contributors 2026]:

DevOps is the integration and automation of software development and information technology operations. DevOps encompasses the tasks necessary for software development and can lead to both shortening development time and improving the development life cycle. According to American software architect Neal Ford, DevOps, particularly through continuous delivery, employs the "bring the pain forward" principle by tackling challenging tasks early, fostering automation, and enabling swift issue detection.

Although debated, DevOps is generally characterized by three key principles: shared ownership, workflow automation, and rapid feedback. From an academic perspective, Len Bass, Ingo Weber, and Liming Zhu—three computer science researchers from the CSIRO and the Software Engineering Institute—defined DevOps as "a set of practices intended to reduce the time between committing a change to a system and the change being placed into normal production, while ensuring high quality". However, the term is used in multiple contexts. At its most successful, DevOps is a combination of specific practices, culture change, and tools.

DevOps at Swiss Post

At Swiss Post, the DevOps process is already defined and is not changed by this project. It applies all defined tools and rules. The general rules are as follows:

- GitHub is used as the code repository, using SSO with Swiss Post accounts.
- Access to repositories is limited and managed by a permission system.
- Separate repositories exist for development and production environments.
- Repositories are also separated by function, e.g. repositories for Terraform IaC or SQL scripts for Athena.
- Development is performed on branches.
- To merge branches, a pull request is required; this triggers GitHub Actions.
- GitHub Actions perform checks depending on the repository and its function.
- To merge code into production environments, a review by another developer is required in addition to successful GitHub Actions.
- In addition to this deployment pipeline, DevOps engineers continue maintaining and supporting their code.

Frontend deployment pipelines

Regarding frontend reports, a similar process is already defined and also applied; it is not adapted by this project. When deploying front-end reports using Power BI, the process takes place entirely within the Microsoft environment.

- Reports are initially created in Power BI Desktop.
- Development and testing happen in the desktop version first.
- The report is deployed from Power BI Desktop to the Microsoft Fabric development environment.
- The code is checked in using Git integration in Microsoft Fabric.
- The report is tested first in the development environment.
- It is then deployed through the defined pipeline to the test environment.
- Stakeholders perform tests in the test environment.
- After successful testing and any required fixes, approval for production deployment is given.
- Production deployment is performed again through the deployment pipeline.

The deployment pipeline is shown in the figure below, showing the selection of environments from which the three-step process starts.

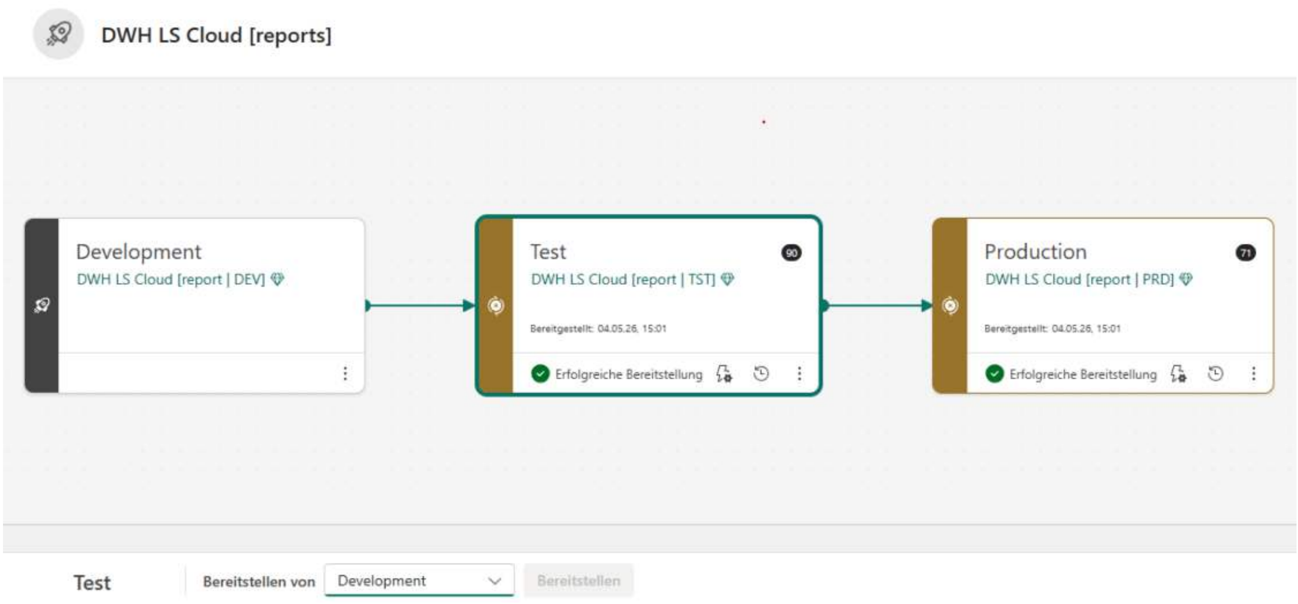


Figure 4.14.: Power BI deployment pipeline across the development, test, and production environments in Microsoft Fabric

4.7. DQM tool selection

The DQM tool selection required a structured approach. Such an approach required first setting up a list of requirements for the tool. Then an evaluation of how well the tools fulfilled these requirements. And finally the invitation of the providers that fulfilled the requirements to demonstrations.

Tool selection principles

The first step in evaluating a tool must be to determine what a tool actually needs to have. Commonly, such lists are called requirement lists. They have to incorporate the must-have features of a tool and the nice-to-have requirements. With such a list, the decision together with higher management is made easier, as it allows for scoring and provides an easy overview of how many points each tool scored. Based on such a points list, the best-performing or best-value tool can be selected.

Requirements list

At Swiss Post, the DQM team set up the requirements list. It incorporates the following main principles:

- **Data Coverage & Connectivity:** Integration into the existing tool landscape. The requirement to be cloud-first. The most important data sources that need to be handled.
- **Data Quality Capabilities:** The actual DQM requirements that the tool must fulfil. The expected usability, such as no-code environments.
- **Reporting & Monitoring:** The tool must allow exports to other tools for reporting purposes.
- **Performance and Reliability:** The tool needs to be responsive even on large datasets, as Swiss Post will want to run checks on very large transaction datasets.
- **Cost & Licensing:** The cost of a tool is relevant. How the cost is structured.

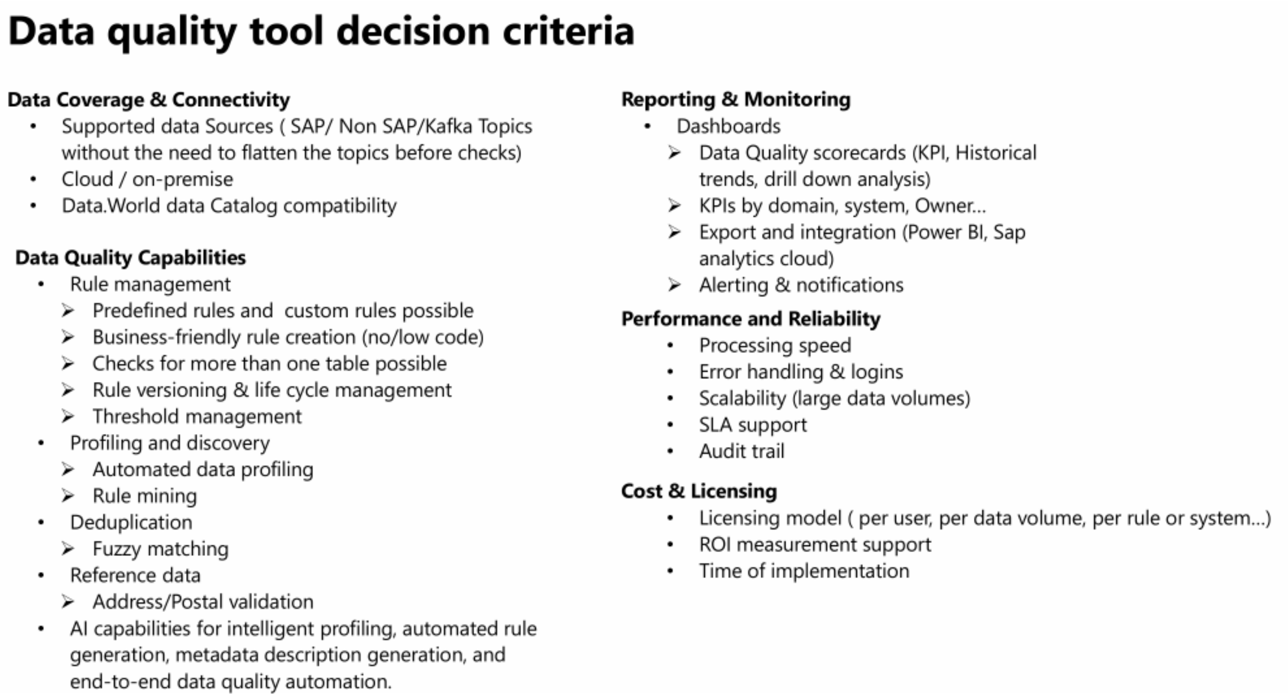


Figure 4.15.: Overview of the data quality management tool evaluation process and the assessed evaluation criteria

Additionally, there are the more soft criteria, which are nice to have. Soft criteria are criteria that are not mandatory but make the usage of an application easier for users, e.g. by providing easy management of access roles. The list as defined by Swiss Post is shown in the following screenshot:

- **Usability & User Experience:** User permission system allowing multiple different access levels and their provisioning through existing means.
- **Technical Requirements:** The security requirements that the cloud solution has to meet. Additionally, the connectivity options that it needs to support.

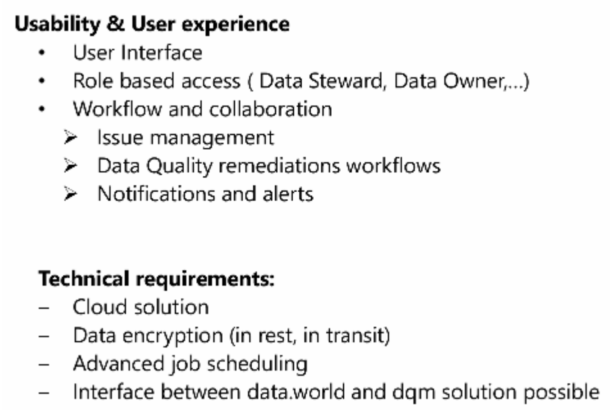


Figure 4.16.: Continuation of the data quality management tool evaluation, including the comparison of the evaluated solutions

DQM tooling selection

After defining the selection criteria, the tools are researched. The research on these tools was also carried out by the DQM team, so it is out of scope for this work. As a result of the tool research, the different tools found are evaluated against the criteria. This finally results in the table shown below, highlighting the performance of the tools against the criteria. This table then serves as the basis for deciding which tools are invited for demonstrations or directly excluded. The following screenshot shows the five tools that were included in the evaluation, including the three in green that were invited for demonstrations.

- **Ataccama ONE**: Selected for demonstration.
- **Informatica IDMC (CDQ + optional Streaming)**: Selected for demonstration.
- **Soda (Core + Cloud)**: Selected for demonstration. The demonstration is documented later in this document.
- **Great Expectations**: Not selected for demonstration.
- **SAP Information Steward**: Not selected for demonstration.

Comparison of tools

Criterion	Ataccama ONE	Informatica IDMC (CDQ + optional Streaming)	Soda (Core + Cloud)	Great Expectations	SAP Information Steward
1) Business-rule DQ (custom rules + complex checks)	**Strong** – DQ evaluation rules + monitoring/reconciliation projects. (docs.ataccama.com)	**Strong** – profiling + rules + monitoring/scorecards. (Informatica)	**Strong** – flexible checks language; runs checks as code/config. (docs.soda.io)	**Strong** — Expectations are expressive/extensible tests for business logic; works with SQL-backed data, pandas, Spark. (GitHub)	**Medium** – rules + profiling/monitoring, but SAP-centric. (SAP)
2) Kafka "close to source" fit	**Partial** – possible via running processing engines in your environment; Kafka specifics must be proven in PoC. (docs.ataccama.com)	**Strong (if you include Streaming)** – Data Engineering Streaming provides prebuilt connectors incl. Kafka. (docs.informatica.com)	**Partial** – can work if your pipelines can query/scan where data lands; "close to Kafka" usually means embedded into streaming/pipeline patterns rather than native Kafka engine. (Needs design.) (docs.soda.io)	**Partial** — GX doesn't "inspect Kafka topics" by itself; it fits Kafka via your processing/landing pattern (e.g., validate in the streaming job/micro-batch, or after landing in a lake/DB). GX is designed to validate where data lives. (Great Expectations)	**Weak** – not designed as a Kafka-first enterprise DQ tool.
3) SAP finance/master data fit	**Strong/Partial** – enterprise platforms typically cover SAP via connectivity, but confirm specifics with your SAP landscape.	**Strong/Partial** – strong enterprise connectivity model via Secure Agents; confirm SAP specifics/modules. (docs.informatica.com)	**Partial** – depends how you access SAP data (extract/replication layer, connectors).	**Partial** — good if SAP data is extracted/replicated to a database/lake/analytics platform that GX can query; not SAP-native stewardship/DQ inside SAP. (PoC needed for your SAP access pattern.)	**Strong (within SAP landscape)** – profiling, validation rules, monitoring in SAP contexts. (SAP)
4) Hybrid deployment & "process data locally"	**Strong** – hybrid model with Data Processing Engines running in your network. (docs.ataccama.com)	**Strong** – Secure Agent group runs within your network/cloud and is used to access on-prem data. (docs.informatica.com)	**Strong** – Soda Core runs where you execute it; Soda Agent can be self-hosted (K8s/Helm). (docs.soda.io)	**Strong** — runs wherever you can run Python (on-prem, cloud, hybrid) and validates data "where it lives." (Great Expectations)	**On-prem oriented** – good if you want on-prem, less flexible for broader hybrid patterns. (SAP)
5) Governance + low-code rule authoring	**Strong** – platform UI for rules/monitoring and governance concepts. (docs.ataccama.com)	**Strong** – visual/no-code positioning; training materials emphasize building rules and scorecards. (Informatica)	**Medium** – can be no-code in Soda Cloud, but often starts engineer-led; agent model supports UI checks. (docs.soda.io)	**Medium** — GX Core is code-first; GX Cloud emphasizes point-and-click UI to involve nontechnical stakeholders. (Great Expectations)	**Medium** – stewardship dashboards/scorecards; governance is SAP-centric. (SAP)
6) Workflow: alert routing, case mgmt, exception handling	**Partial** – platform capabilities exist, but your wish is "as much as possible in data.world"; you'll likely still integrate.	**Partial** – similar: tooling exists, but aligning workflow into data.world will be integration work.	**Partial** – Soda Cloud has collaboration/alerts; "case mgmt/waivers" may still be better in data.world processes.	**Partial** — GX integrates well with orchestration/alerting patterns (e.g., Airflow operator examples), but it's not a full case-management/waiver system out of the box. (astronomer.io)	**Partial** – some review workflows exist, but likely not your enterprise-wide workflow hub.
7) data.world integration	**Partial (via API)** – likely publish results through data.world DQ API. (Data.world)	**Partial (via API)** – same pattern. (Data.world)	**Partial (via API)** – same pattern; technically straightforward because results are already "check run" oriented. (Data.world)	**Partial (via API)** — you'd publish GX validation results into data.world using the Data Quality API (mapping GX results → check runs). (Data World Docs)	**Partial (via API)** – possible but usually more custom. (Data.world)
8) Power BI consumption (aggregated metrics)	**Strong/Partial** – most enterprise platforms expose scorecards/metrics; may still export to Power BI.	**Strong/Partial** – scorecarding and exporting exception records are part of CDQ training/usage. (Informatica)	**Strong/Partial** – checks produce aggregated results; you can store/export for Power BI. (GitHub)	**Strong/Partial** — GX produces structured results you can store and visualize; if you use data.world as the hub, it supports Power BI integration. (Data World Docs)	**Medium** – has dashboards/scorecards, but broader Power BI strategy likely requires exports. (SAP)
9) Start small (few checks) with attractive entry cost	**Medium** – typically enterprise licensing/implementation.	**Medium** – enterprise platform; Secure Agent + services can be heavier.	**Strong** – Soda Core is free/open-source; scale into Soda Cloud/Agent as needed. (docs.soda.io)	**Strong** — GX Core is open source; GX Cloud has a free Developer option (data-asset based) and tiers based on # assets under test. (Great Expectations)	**Medium/Weak** – SAP tool (licensing/stack dependency); also lifecycle signal. (community.sap.com)
10) Strategic risk / roadmap	**Low/Medium** – active platform evolution (verify procurement/roadmap).	**Low/Medium** – major vendor; broad cloud platform.	**Low/Medium** – solid modern approach; depends on your appetite for "DQ as code" model.	**Low/Medium** — positioned as the "open-source standard" for data quality testing, which typically reduces lock-in risk. (Great Expectations)	**Higher** – SAP indicates maintenance timelines (mainstream to 2026, CSM to 2028). (community.sap.com)

Figure 4.17.: Example of the comparative evaluation matrix used during the assessment of different data quality management tools

Design Decisions

The next step after the evaluation table is to invite the selected tool vendors or providers to a demonstration. During this demonstration, the providers are asked to show the performance of their tool against the defined criteria, but they can also demonstrate any specific benefits that they provide. The demonstration gives the selecting company a more detailed view of the capabilities of the tool when applied to its dataset and whether any shortcomings would prevent a positive decision. Furthermore, it allows the selecting company to ask questions directly about any open items. After a demonstration, the scoring can be adjusted and, finally, a decision made on whether to proceed with a provider into a PoC or even contract negotiations, or to inform them that they have been excluded from the selection.

Alternative Outcome

The previous subsection assumes that a tool can be found for the defined use case. However, in case such a tool cannot be found, alternative ways need to be followed. In this work, the focus is on the solution; the why question will therefore also be excluded from the solution architecture. In case no solution is found:

1. **Reduce and adapt:** If the full use case cannot be satisfied by the tools at hand, the original use case can be adapted. The implementation can be adapted to a smaller use case while waiting for a suitable technology.
2. **Reconsider the discarded:** The previously discarded solutions or approaches can be reconsidered. Engage in discussions with the tools being developed internally. Review the tools which were excluded from selection to find out if they are fit for the use case of this project.
3. **Consult and discuss:** In large organisations, multiple initiatives can happen at the same time which are not seen on the team level. Therefore, engaging with a solution architect can open up the view towards other options. It allows identifying possible repurposing of existing tools which can fulfil the project's needs.

4.8. DQM tool testing

After the tool evaluation and selection, the testing should confirm whether it can execute the required tests in the following master thesis. Testing requires knowledge of how to use the tool and a simplification of the overall topic of the thesis.

Knowledge buildup

In order to understand the basic function of a tool, the required knowledge had to be acquired first. To do so, a variety of options are available. These are either based on documentation and how-to guides or direct knowledge transfer between developers who developed the tool or have already used it. In addition to the tool, additional technologies may be required to run tests, which in turn have to be studied as well. As, at this stage, the decision is pending, further details are not available as to what needs to be part of the knowledge buildup.

Tool testing

The tool had to be tested in order to confirm its applicability to the master thesis. The test of the application requires the use case to be broken down into a smaller test, with which the team can decide whether the tool can be used for further development. If such a tool were to fail the test, a return to the previous step and a continuation of the tool search would be necessary. In addition to the tool itself being tested, the setup required for it to run is equally part of the test. In the case of Swiss Post, this can include steps for permissions and infrastructure, which are commonly set up using Terraform.

Test design

The test design is important so that the selected case is both a step towards the complete use case and easier to execute. It must also include the infrastructure to be built for the use case. For the use case at hand, the test must prove that data can be queried with the tool, returning results that can be further processed. The processing of the data is, however, not required as part of the test design, as this may require multiple data points to be included in a check.

For this use case, which is the counting of multiple events per day, the count of one event is sufficient for a test. Therefore, the query at hand has to be reduced so that it only returns one event. However, even one event still requires identifying data, namely the date and the sorting centres. In addition to the idea of a final check, the test also has to be modular. First, the test will check whether any data are returned at all, then it will go deeper by checking the more specific event.

5. Implementation

This chapter uses the research findings and the solution landscape developed in the previous chapters as a basis for the implementation. With the defined SMART goals, the analysed reporting pipeline and data lineage of the previous research [Roccaro 2026], being the basis for the successful completion of the SMART goals set in this project. The three goals are handled separately, as their completion is not dependent on each other. Those three goals are:

- To do the update of the existing data quality report, while using the new data products set up by the ASTV team, thereby also requiring the new reporting software Power BI and the new setup of data sources in star schema.
- Following up with the platform and data quality group regarding the tool decision for data quality management at Swiss Post in the future. This task started as the evaluation of tools, as the DQM platform described in the previous research [Roccaro 2026] had shown that it would not be fit as a tool for the entire Swiss Post.
- As a preparation for the upcoming Master Thesis, the data quality tools to be used were checked initially. This included performing a concise test of counting data points with the defined monitoring setup.

The validation of the achieved results is performed in the next chapters.

5.1. SMART Goal 1: Setup of an Improved DQM Report

In Project 1, the groundwork for researching the reporting pipeline was completed. It provided the data lineage and described the underlying issues. Based on this work, an improved report was developed. Research was conducted on how to better highlight the relevant comparisons and help users identify the information more quickly. Additionally, the third-party data source, which was previously copied daily into an Excel file, was improved. The file was moved to SharePoint and is now updated whenever changes are made to Amazon S3, where it is made available to Athena. In addition, the selected data underwent a structural change, which is described at the end of this section.

Update of data source

The first step towards the new report was to improve one of the data sources used by the report. Part of the data was retrieved from a third-party system by means of an Excel table. The file resided in a shared folder and was not initially archived. As this data source did not change and was required by both the existing SAC report, which had to remain operational, and the new Power BI report, it had to be made available in a way that was accessible to both tools. Changing from a shared folder, which was not accessible to Power BI, to SharePoint was therefore a major improvement, as it added versioning to the file. Furthermore, the DWH LS OPS team has an existing method to transfer CSV data from SharePoint to Athena.

Therefore, the destination of the Excel file was updated in collaboration with the business users responsible for the manual maintenance of the Excel file. With this change in place, the first data source was made available to Power BI, and all versions of it were from then on stored in Amazon S3.

The groundwork for such an interface was already available, as the DWH LS OPS team had implemented it previously [Bosek 2026]. As can be seen in the following screenshot, the project team only had to set the file folder in the existing file structure, and the file was automatically transferred into Amazon S3 and then made available for querying.

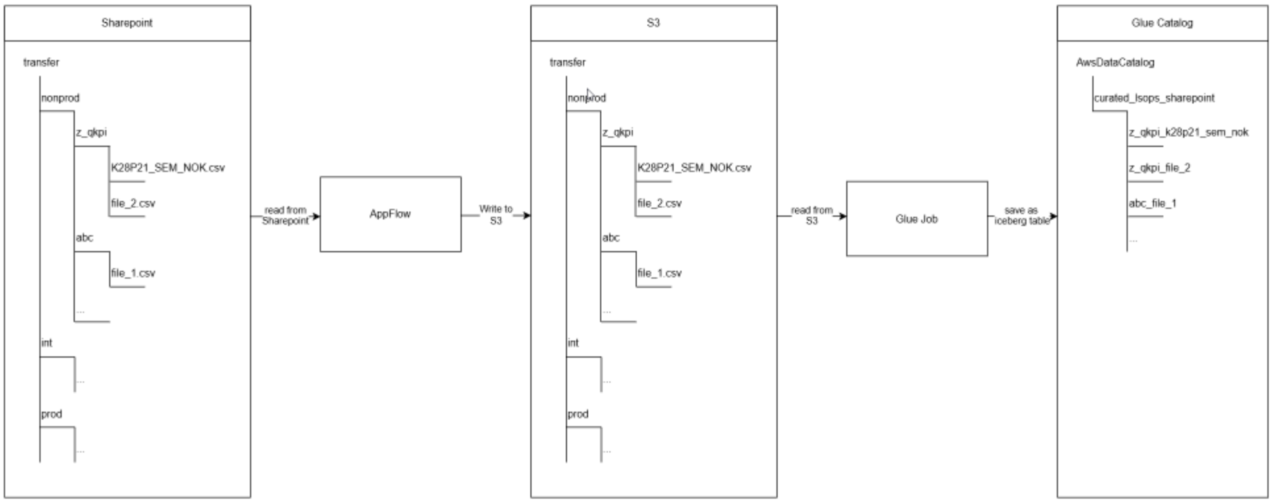


Figure 5.1.: Schematic overview of the interface used to transfer Excel files from Microsoft SharePoint to AWS S3 for automated ingestion into the reporting platform

Setup of Power BI

Regarding the Power BI setup, Swiss Post has already defined DevOps pipelines, setup guides, and documentation supporting the installation of the application and the required infrastructure, for example, for using Athena as a data source.

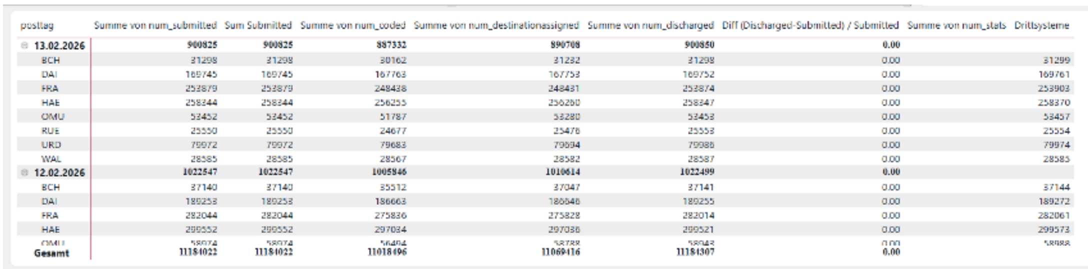
Setup

This setup consists of predefined templates defining areas such as report colour schemes, filter regions, and how permissions for reports are handled. Furthermore, the setup includes definitions of how to deploy reports and how to handle AWS Athena data sources. Thus, the setup was carried out in accordance with the guidelines defined by Swiss Post [Swiss Post 2026c; Swiss Post 2026a; Swiss Post 2026b].

Report definition

The initial report setup was adopted from the analysed SAC report, as the data that had to be checked remained the same. Therefore, the method followed by the business also remained unchanged, and the first tests of loading the data into Power BI produced the data in the same way. The first screenshot shows the report that was to be replaced. The second screenshot shows the first version of the same data in Power BI. Do take note that the second screenshot shows work in progress during development. It is not meant to be readable, as it contains, for example, no comparisons between relevant values, and the values are not formatted with thousands separators.

Figure 5.2.: Existing DQM report used by Swiss Post prior to the implementation of the redesigned Power BI solution



posttag	Summe von num_submitted	Sum Submitted	Summe von num_coded	Summe von num_destinationassigned	Summe von num_discharged	Diff (Discharged-Submitted) / Submitted	Summe von num_stats	Drittsysteme
13.02.2026	**900825**	**900825**	**887332**	**890708**	**900850**	**0.00**		
BCH	31298	31298	30162	31232	31298	0.00		31299
DAI	169745	169745	167763	167753	169752	0.00		169761
FRA	253879	253879	248438	248431	253874	0.00		253903
HAE	258344	258344	256255	256260	258347	0.00		258370
OMU	53452	53452	51787	53280	53453	0.00		53457
RUE	25550	25550	24677	25476	25553	0.00		25554
URD	79972	79972	79683	79694	79986	0.00		79974
WAL	28585	28585	28567	28582	28587	0.00		28585
12.02.2026	**1022547**	**1022547**	**1005846**	**1010614**	**1022499**	**0.00**		
BCH	37140	37140	35512	37047	37141	0.00		37144
DAI	189253	189253	186663	186648	189255	0.00		189272
FRA	282044	282044	275836	275828	282014	0.00		282061
HAE	299552	299552	297034	297036	299521	0.00		299573
OMU	58074	58074	56404	58788	58043	0.00		58088
Gesamt	**11184022**	**11184022**	**11018496**	**11069416**	**11184307**	**0.00**		

Figure 5.3.: Initial Power BI report designed to resemble the existing SAC report

Initial data setup

As this was the first report developed by the project team, the initial implementation was carried out locally without deploying the report to Microsoft Fabric. As a first report, the variety of available approaches for implementing features such as links between data sources naturally led to a report setup that performed some tasks twice, for example, linking the two data sources using both Power Query M and DAX. These deficiencies in the initial implementation were eliminated throughout the subsequent implementation steps of the report.

Differences between local and cloud

Once the linkage of the two data sources was working and the data was displayed in the same way as in the SAC report, the Power BI report was published to the cloud. However, the report showed issues when loading data from Athena. Power BI automatically applied a transformation that attempted to cast a column containing date values into a date format. This resulted in issues with empty values.

According to the initial research using Microsoft Copilot, the issue could be resolved in Athena. However, discussions with the wider DWH LS OPS team showed that similar issues had occurred for other team members. Therefore, before making changes in Athena, the possible solutions identified by the team were investigated. After consulting the team and checking the view in Athena, it became clear that the issue did not originate in Athena and therefore could not be resolved there. As the tests showed that the query ran correctly in Athena, the issue had to be located in the report. Since no proper checkpoints had been established up to this point, it was not possible to determine at which stage the applied data transformations caused the issue. Therefore, the report had to be recreated, and after each transformation, a test deployment was performed to verify that the data loaded and was displayed correctly in the cloud version. Finally, an error could not be found during the stepwise recreation. The most likely error source is the handling of a date data field that was transformed in the first version of the report.

Model data setup

After the initial data from the SAC report were integrated, the model contained information from two tables sourced from AWS. However, for the further checks, more data sources had to be included. Thus, the data model had to be refined by connecting the tables through the data model view rather than through Power Query M and DAX. Initially, only two tables had to be linked through the postal codes and dates. This new model setup was a requirement defined by the team, as was the separation of the measures into a dedicated table in Power BI. To highlight the initial setup, the four remaining tables are labelled here and in the next figure in the same way. The different tables are used for:

- **CSV data:** The data from the CSV file represent the third-party event counts.
- **Athena data:** These are the data from Athena, comprising the event counts compared in the report.
- **Measures:** These are the calculations performed based on the CSV and Athena data.
- **Date:** The table used for joining and filtering data based on the date.
- **Postal code:** The table used for joining the data based on the postal code and for filtering it by location name.

Figure 5.4.: Initial Power BI data model containing the core tables required for the first report prototype

The data model was extended towards the end of this SMART Goal by integrating the full data model used by the business. This was mandatory to perform the final check, comparing the infoblock data in Athena with the aggregation table in Power BI. This aggregation table is a means of improving the response time for the most frequently used data points for reporting, which are in turn stored directly in Power BI rather than having to be queried from Athena. The aggregation table was set up by the ASTV team and was used by the project team. The previously existing tables are shown with the same labels, while the additional tables have green labels. The high number of additional tables results from the implemented semantic model of another report. A semantic model is the data model of another report. By using the required green tables, the additional ones are automatically added to the existing data model. The additional tables are:

- **Aggregation:** The data from the semantic model containing the aggregation counts.
- **Date:** The table with the complete date information based on the star schema, imported together with the semantic model.
- **Postal code:** The table with the complete postal code information based on the star schema, imported together with the semantic model.

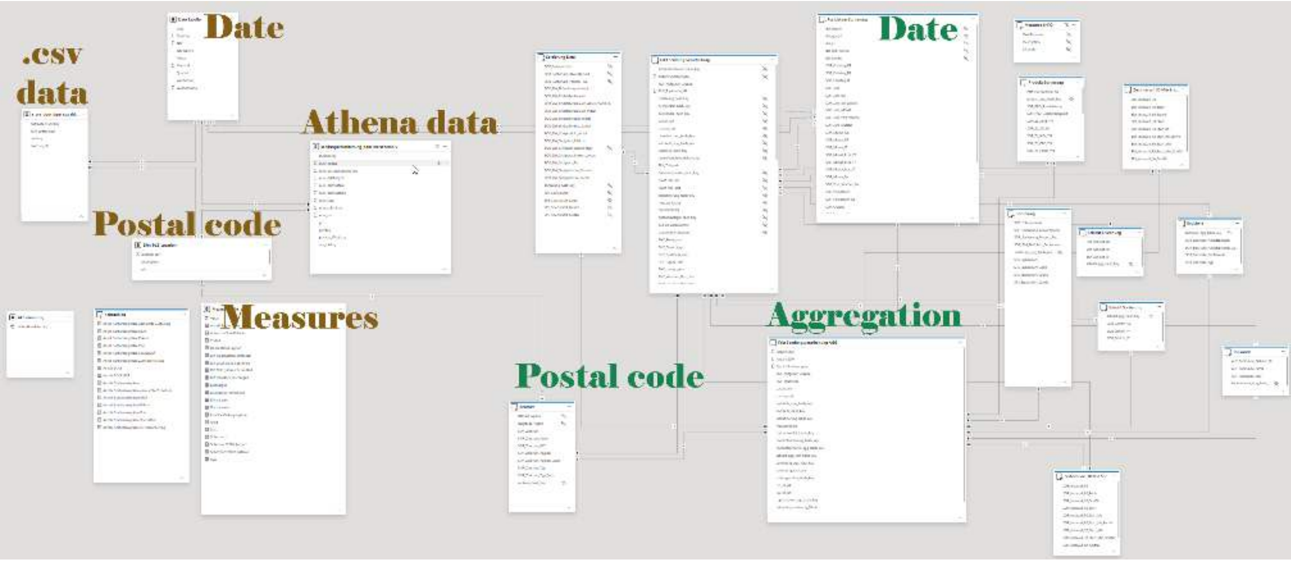


Figure 5.5.: Final Power BI data model after integrating additional data sources, calculated tables, and supporting information

New visuals to support faster interpretation by stakeholders

As the task of this SMART goal was not only to copy the existing report but also to improve it, the content was made more visual and easier to interpret. Thus, having only one table containing all data would not improve the situation. However, according to the checks defined in Project 1, the thresholds were already known. These thresholds allowed warnings to be displayed. Therefore, the report was divided into three different views.

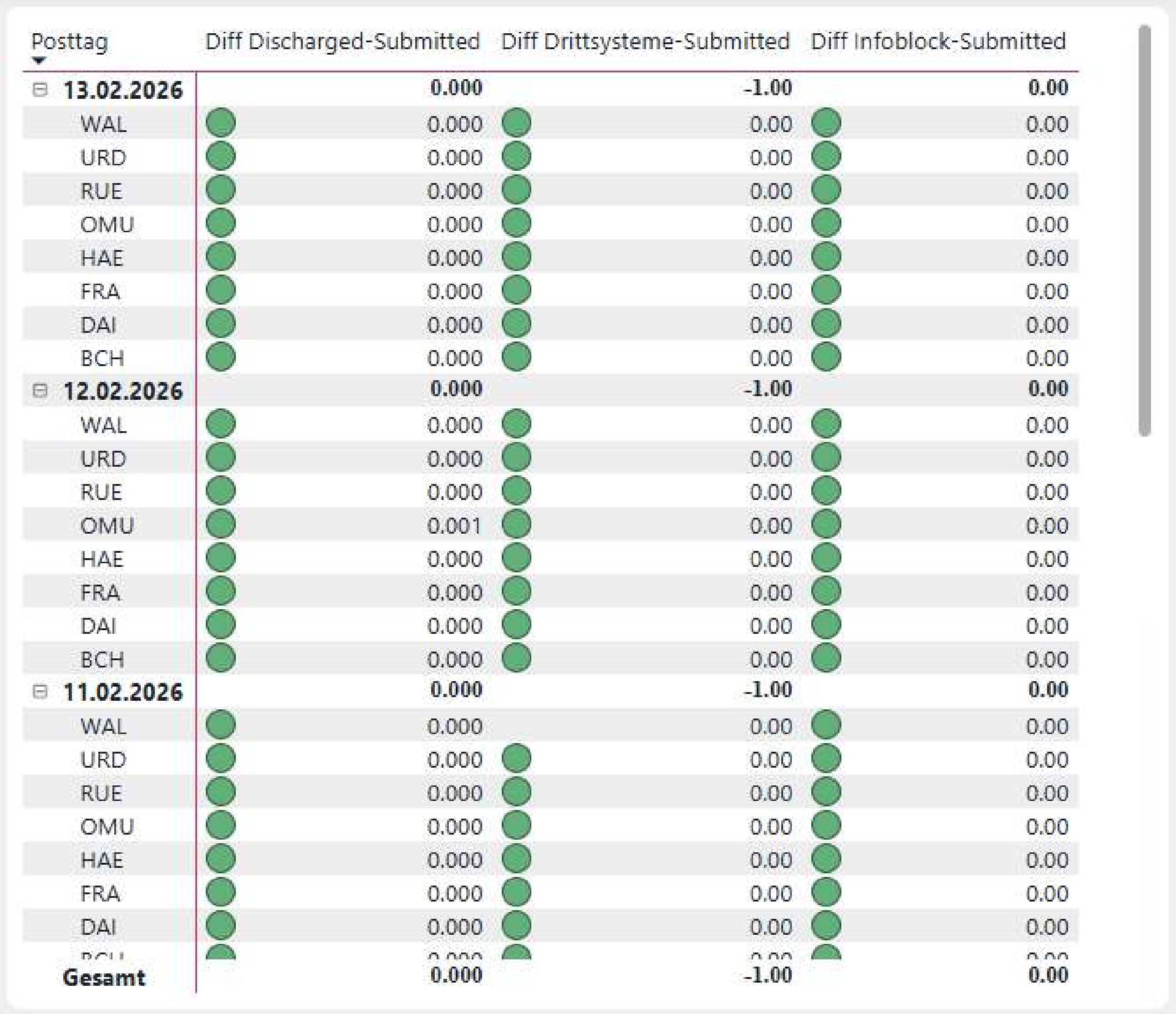
- a detailed table containing the daily event counts and all relevant information
- a traffic light dashboard showing whether the values satisfy the defined validation checks
- a line chart showing the development of the values over time

The first visual is a detailed table showing the daily event counts per event and sorting centre. This allows the business to confirm completeness based on the absolute values. The table was enhanced with traffic light indicators showing green or red according to the defined thresholds.

postag	Summe von num_submitted	Sum Submitted	Summe von num_coded	Summe von num_destinationassigned	Summe von num_discharged	Diff (Discharged-Submitted) / Submitted	Summe von num_stats	Drittsysteme
⊟ 13.02.2026	**900825**	**900825**	**887332**	**890708**	**900850**	**0.00**		
BCH	31298	31298	30162	31232	31298	0.00		31299
DAI	169745	169745	167763	167753	169752	0.00		169761
FRA	253879	253879	248438	248431	253874	0.00		253903
HAE	258344	258344	256255	256260	258347	0.00		258370
OMU	53452	53452	51787	53280	53453	0.00		53457
RUE	25550	25550	24677	25476	25553	0.00		25554
URD	79072	79072	79683	79694	79086	0.00		79074
WAL	28585	28585	28567	28582	28587	0.00		28585
⊟ 12.02.2026	**1022547**	**1022547**	**1005846**	**1010614**	**1022499**	**0.00**		
BCH	37140	37140	35512	37047	37141	0.00		37144
DAI	189253	189253	186663	186646	189255	0.00		189272
FRA	282044	282044	275836	275828	282014	0.00		282061
HAE	299552	299552	297034	297036	299521	0.00		299573
OMU	58074	58074	56404	56788	58043	0.00		58088
Gesamt	**11184022**	**11184022**	**11018496**	**11069416**	**11184307**	**0.00**		

Figure 5.6.: Power BI table visualizing daily event counts together with data quality indicators

The second visual consists of traffic light indicators summarising the four data quality checks. For each check, the values are compared and displayed as green when the difference is below 1% and at least 10'000 events are available. This allows users to quickly determine whether the latest values are valid, reducing the time required to decide whether further investigation is necessary.



Posttag	Diff Discharged-Submitted	Diff Drittsysteme-Submitted	Diff Infoblock-Submitted
13.02.2026	**0.000**	**-1.00**	**0.00**
WAL	0.000	0.00	0.00
URD	0.000	0.00	0.00
RUE	0.000	0.00	0.00
OMU	0.000	0.00	0.00
HAE	0.000	0.00	0.00
FRA	0.000	0.00	0.00
DAI	0.000	0.00	0.00
BCH	0.000	0.00	0.00
12.02.2026	**0.000**	**-1.00**	**0.00**
WAL	0.000	0.00	0.00
URD	0.000	0.00	0.00
RUE	0.000	0.00	0.00
OMU	0.001	0.00	0.00
HAE	0.000	0.00	0.00
FRA	0.000	0.00	0.00
DAI	0.000	0.00	0.00
BCH	0.000	0.00	0.00
11.02.2026	**0.000**	**-1.00**	**0.00**
WAL	0.000	0.00	0.00
URD	0.000	0.00	0.00
RUE	0.000	0.00	0.00
OMU	0.000	0.00	0.00
HAE	0.000	0.00	0.00
FRA	0.000	0.00	0.00
DAI	0.000	0.00	0.00
BCH	0.000	0.00	0.00
Gesamt	**0.000**	**-1.00**	**0.00**

Figure 5.7.: Power BI traffic light indicators summarising the four data quality checks

The third visual is a line chart that allows users to identify recurring patterns in the data, for example, whether errors regularly occur on Fridays or Mondays. The centre line represents a 0% deviation, while the dashed lines indicate the 1% threshold. The four coloured lines represent the different events. Note: this graph appears to be time-oriented, but in fact each time block contains the list of sorting centres. This graphical approach was deliberately chosen to make it easy for business stakeholders to use.

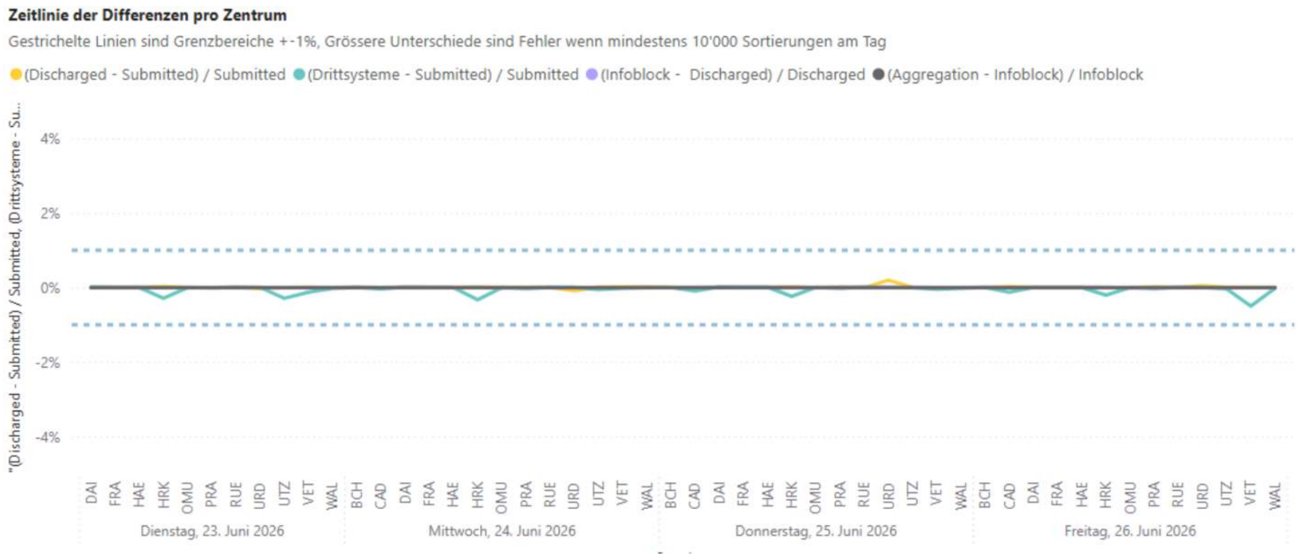


Figure 5.8.: Power BI line chart showing historical data quality trends

Scrum review

As the report was completing with the given features, it was important to retrieve the feedback of the stakeholders. This was done during a Scrum review of the ASTV team. The report was shown together with some background information using a powerpoint presentation slides and a live demo. The steps in the review were as follows:

- Presentation of the background, where did the project start from
- Why this report matters
- Change into Power BI showing the new report with visualisations
- Perform live demo of how to be used report
- Time for questions from team

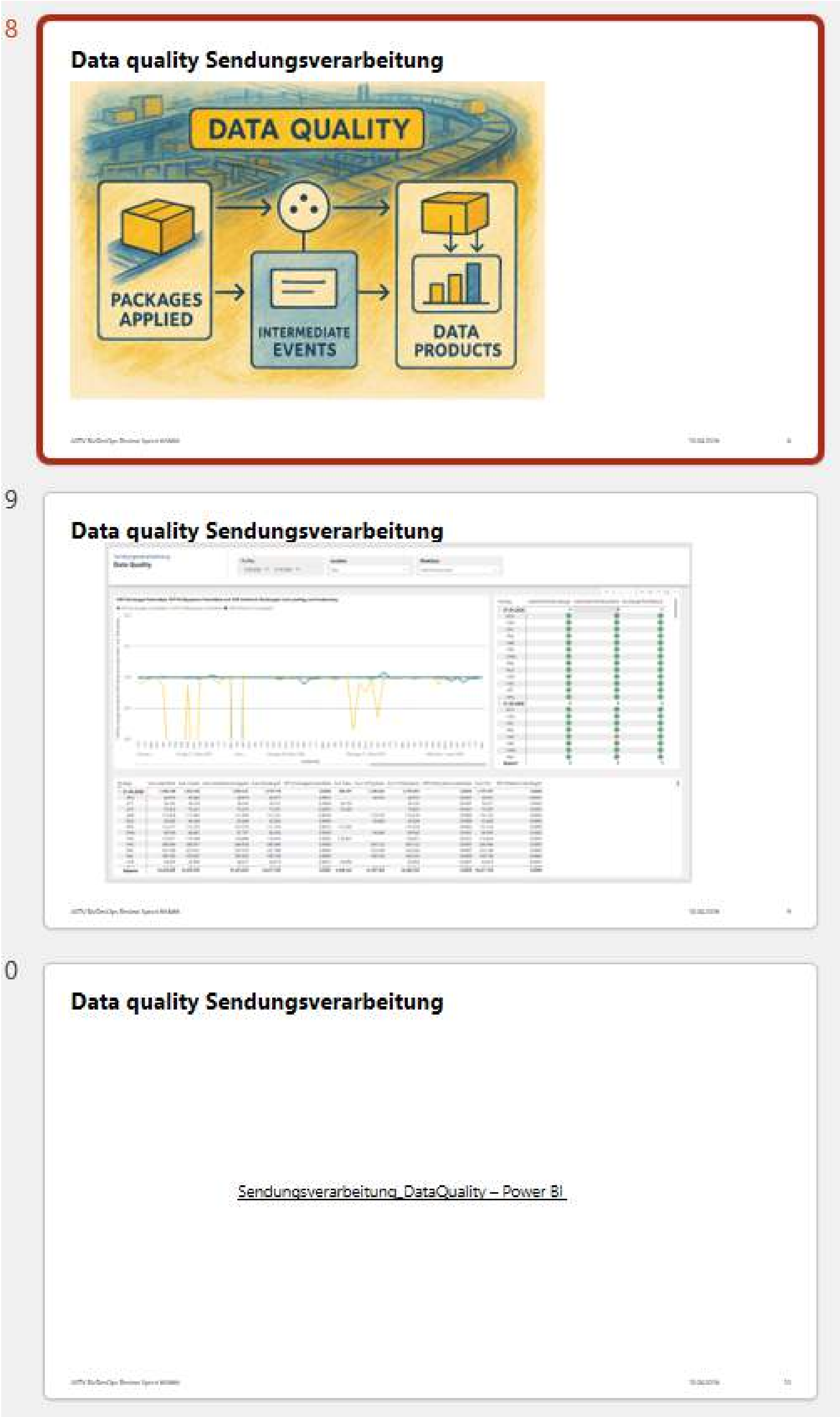


Figure 5.9.: Slides used during the Scrum review to present the implemented Power BI data quality monitoring solution and collect feedback for further development.

Feedback after review

After the review further feedback was collected in separate sessions. The stakeholders showed similar reports to

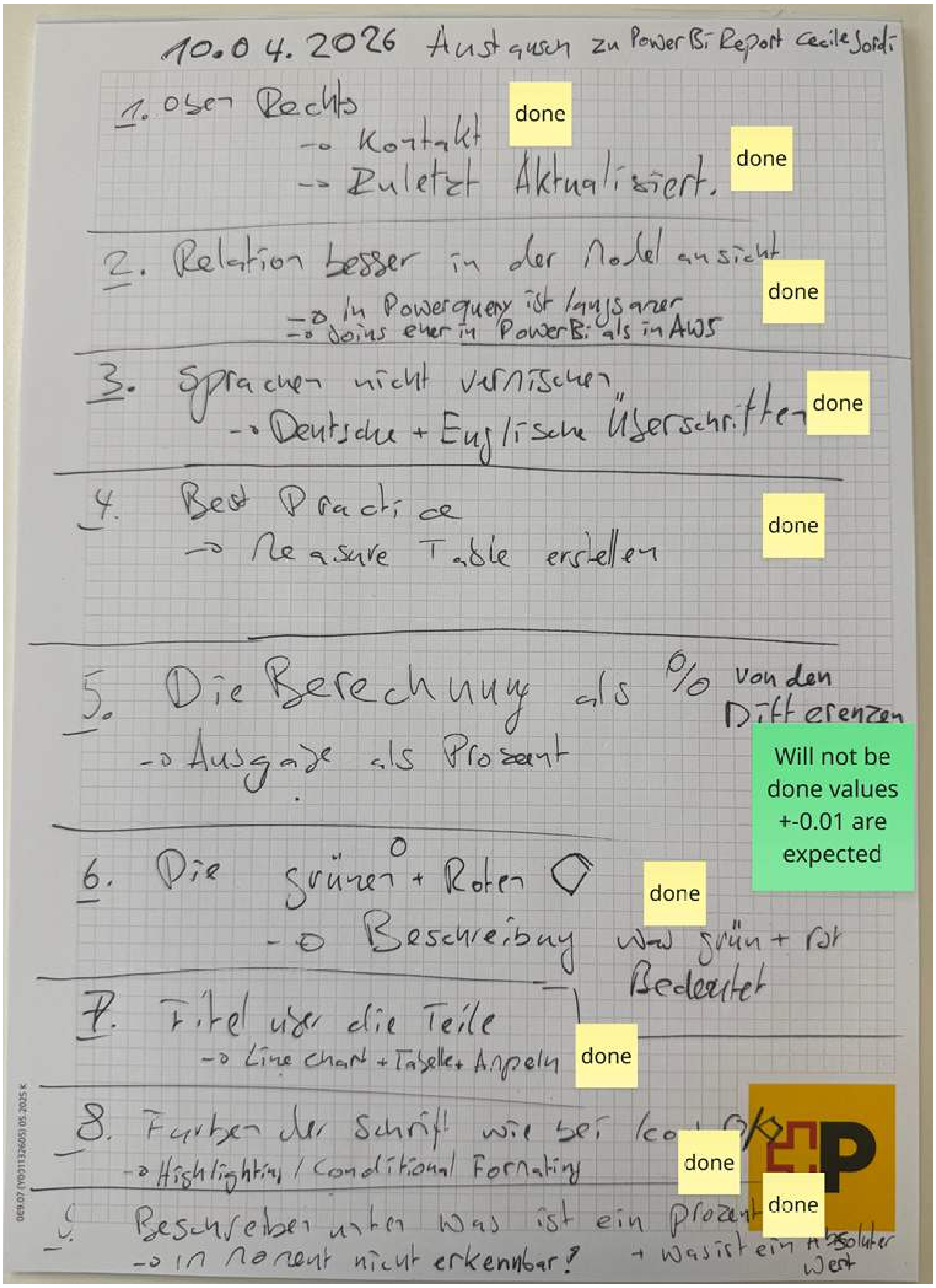


Figure 5.10.: Miro board documenting stakeholder feedback collected during a review meeting together with the implementation status of the resulting improvement actions.

KEY POINT: In total, the report was updated to the following dashboard. First, it allows the user to identify more quickly whether the data for a given day are complete by using the traffic lights. Second, users have a view of past inconsistencies per sorting centre using the line chart. Third, they can check all details in numerical form using the table at the bottom. Fourth, it provides filters at the top of the report that apply across all visuals, allowing users to filter by weekdays, date ranges, or selected sorting centres. Fifth, for clarity, each visual has a short description showing what is displayed to the user. These are written in small case so that they do not distract from the data itself.

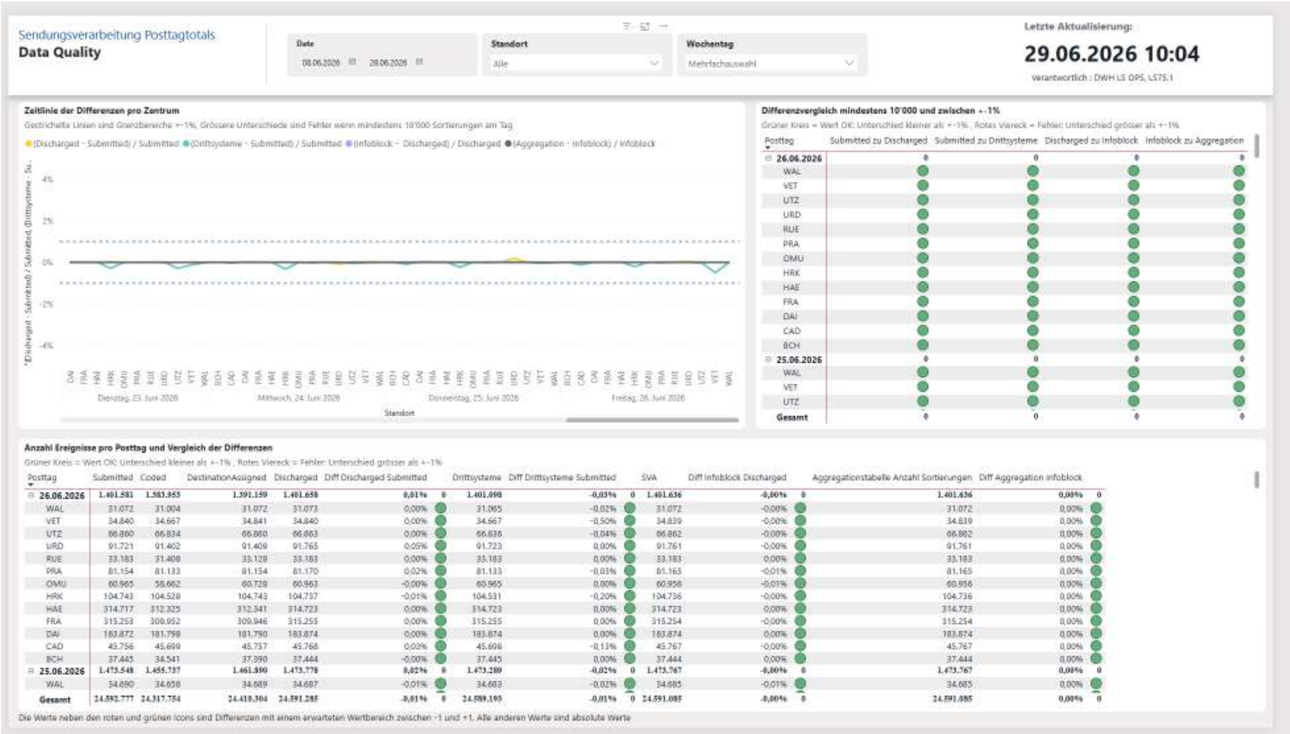


Figure 5.11.: Final Power BI dashboard integrating the three implemented data quality visualisations and interactive report filters

New view Athena

To highlight some of the testing and improvement loops during the development of the report, the following example describes a change to one of the data sources. During testing, the business pointed out that small differences in the number of events occurred between the new Power BI report and the data source it was intended to monitor. This issue was reported during a meeting between the project team and the product owner of the ASTV team. The product owner identified that the new Power BI report was still using the old data source.

While the data source remained largely the same, the data lineage changed slightly. Business and IT decided in late 2025 that the data should be consumed as a star schema to improve performance compared to the existing wide table. Consequently, the DQM dashboard for message processing needed to follow this change in the data source. In order not to disturb the previous report, the data had to be gathered into a separate view in Athena. The separate view was also required in order to validate the new star schema infoblocks rather than those of the previous wide table. Thus, separating the views was necessary to maintain clean, high-performing code, as adding the new query parts for the star schema to the wide table query would have slowed the query down and intertwined two different use cases.

DevOps process

After the issue with the new view was reported, the DevOps process was started.

To create a new view:

- The developer first tests the query for the new view in Athena, confirming that it is valid.
- Next, the developer navigates to GitHub, selecting the corresponding test repository for the account used.
- A new branch is created, followed by a file containing the `CREATE VIEW` SQL statement.
- After this, a build is attempted. If it is successful, the process continues; otherwise, the developer returns to the first step and fixes the issue.
- If the build is successful, a pull request can be created, and the new branch merged.
- Once this is completed, a branch is created in the production repository.
- The build in the production environment must be successful.
- Furthermore, a review by another developer is required.
- Once both the build and the review are successful, the merge can be performed.

The full query for the new view can be found in Appendix A.3.

5.2. SMART goal 2: tool decision

As part of SMART Goal 2, the DQM team was followed during its evaluation of a suitable tool for DQM. This work took place in parallel with SMART Goal 1. For the project team, this goal served to identify the tool to be used in the subsequent master's thesis and allowed its capabilities to be evaluated directly within SMART Goal 3.

Tool evaluation

While the previous research [Roccaro 2026] described a DQM tool being developed, other initiatives at Swiss Post were evaluating a company-wide solution. First, the tool selection process for the company as a whole is examined, followed by the DWH-specific solution.

Tool Selection for Swiss Post

However, for Swiss Post as a whole, the tool developed for the DWH was considered too specialised. The main considerations against this initiative were:

- The DWH initiative was considered too specialised for its use case and therefore not suitable for company-wide deployment.
- The developed tool did not address the broader system landscape of Swiss Post.
- The evaluation followed the principle of *buy before build*, favouring solutions available on the market rather than developing specialised tools that are difficult to maintain.
- Support for SAP data sources was a key requirement, as they contain critical accounting-related data with a high potential impact in the event of data loss.

This led to the evaluation of tools available on the market. These included Soda, Informatica, Collate, and Atacamma.

5.2.1. Soda.io

As a first possible provider, Soda.io was selected and followed by the project team throughout its evaluation [Soda 2026]. It is a comparatively young member of the DQM market. In the following sections, the evaluation of this tool is presented. The company focuses on ease of use, rapid deployment, and modern cloud-based solutions.

Demo

One of these demonstrations was provided by [Soda 2026], which was invited by Swiss Post to present its data quality management solution. The demonstration included the following participants:

- Date and time: March 17, 2026, from 11:00 to 12:00 CET (Microsoft Teams online demonstration).
- Bruno Huys (`bruno.huys@soda.io`), organizer of the meeting and representative of Soda.
- Arthur Chionh (`arthur.chionh@soda.io`), introduced by Bruno Huys as his colleague and representative of Soda.
- Machine learning supports the identification of fluctuations, which are visualised as graphs.
- A no-code interface allowing the business to define rules.
- Rules can be defined using sliders to specify acceptable and failure ranges.
- An LLM chatbot allowing the definition of rules and data contracts.
- Data contracts: a schema defined at column level according to client-specific definitions.
- Support for row-level checks.
- Detection of missing master data links.
- On failure of a data check, queries are executed automatically to help identify the missing data.

Decision

Already two days later, on March 19, 2026, the decision regarding the tool was communicated by the Data Governance Expert of the DQM team. The tool did not provide direct support for SAP data, which was considered an essential requirement. Therefore, it was rejected, and the evaluation continued.

The following internal message describes the reasoning:

> *Hi everyone,*
>
> *I wanted to share a quick update on our data quality tool evaluation process.*
>
> *After reviewing the recent demo of Soda and considering our requirements, we have decided not to continue with this particular vendor. The main reason is the lack of a direct connection to SAP data, which is a critical component of our data landscape.*
>
> *We will instead proceed with the remaining vendors (Collate, Informatica and Ataccama) and continue our evaluation process with them.*
>
> *Thank you all for your participation and valuable input so far. I'll keep you updated on the next steps.*

Best regards

Tool Selection for DWH

The DWH LS OPS team continued its work together with the platform team as defined in the KPIs of the AWS modernisation initiative. As this initiative was launched by the DWH team, its requirements were the main driver of the development. During the course of this project, the tool developed from PoC status to being capable of running initial tests and subsequently being configured to validate real data.

DQM Platform Tool

After continued work by the AWS modernisation team together with the platform team, the DQM platform was ready. During a meeting, the current state was presented to the project team. The following excerpts are taken from the meeting minutes:

> *Meeting on: 18.05.2026*
>
> *Participants: DQM team and project team*
>
> *The rule-based DQM platform is ready for testing by the business.*
>
> *The use case checking the master data for missing links has been implemented.*
>
> *Checks are run on a schedule, and the results are saved to Amazon S3 in the DWH account.*
>
> *The results can be queried using Athena.*
>
> *A visual report should be created using Power BI.*
>
> *Ownership should now be assigned for the checks and the data.*
>
> *It should be investigated why the data are only available in one account, as they should also be available in another AWS account.*

Discussion with Solution Architect

During discussions with the designated solution architect of DWH LS OPS, possible approaches for continuing the project were evaluated. This meeting took place after the decision to pursue different tools for the DWH and for Swiss Post as a whole had been announced. The meeting is summarised below using excerpts from the meeting minutes:

> *Participants: DWH solution architect and project team*
>
> *The Swiss Post-wide DQM solution will not be based on the DWH solution.*
>
> *The purpose of the meeting was to determine which tool the project team should use.*
>
> *The solution architect referred to KPI 1, Monitoring, from the AWS modernisation initiative.*
>
> *The goal of this KPI was to provide monitoring capabilities for jobs such as AWS Glue jobs, which are widely used for ETL processing, including information about their uptime and execution time.*
>
> *However, the monitoring functionality had been extended to also support data quality checks and data counts.*
>
> *The next task assigned was to check the status of the data quality checks together with the AWS modernisation team.*

> *Create a data quality check and test the functionality.*
>
> *Acquire as much knowledge as possible before the end of the initiative, when the consultants and their expertise will no longer be available.*

Conclusion

On 1 June 2026, the conclusion of the tool evaluation was communicated. The rule-based DQM platform developed by the platform team was selected for use within the DWH LS OPS environment. For Swiss Post as a whole, a different tool would continue to be evaluated.

Consequently, the weekly coordination meetings of the DQM team were cancelled. Furthermore, the monitoring functionality developed as part of the AWS modernisation initiative was selected as the tool to be evaluated in SMART Goal 3.

5.3. SMART goal 3: test of DQM tooling

This SMART goal follows the testing of the designated DQM tool, with which a use case leading towards the master thesis can be implemented. The use case was defined during a discussion with the solution architect, as described in the previous section. It uses one of the deliverables of the AWS modernisation initiative, namely a monitoring solution for jobs. However, the monitoring solution was also designed in such a way that it allows selected data to be checked. As this functionality had only been developed in a basic form, it first had to be confirmed that it was suitable for performing the checks required during the master's thesis.

Knowledge transfer sessions

As the AWS modernisation team was developing this functionality, knowledge transfer sessions were held to show the DWH LS OPS team how to use it. Furthermore, extensive documentation was written describing the interaction between the different AWS Service.

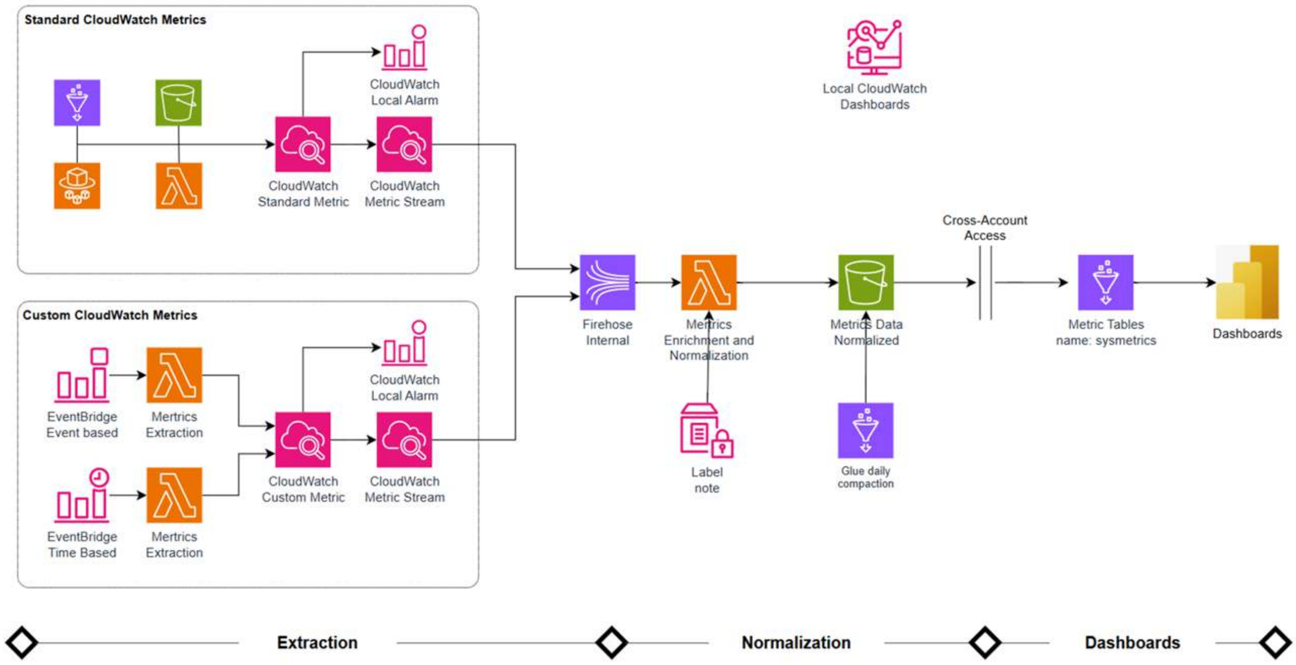


Figure 5.12.: AWS monitoring architecture showing the extraction, normalisation, and dashboarding pipeline implemented using AWS services

First step: design a measurable use case

First, a measurable use case had to be identified. Based on the DQM test and the dashboard created in SMART Goal 1, the daily event count was selected as the example. The existing basic monitoring code was able to perform a row count. However, the query used for the dashboard returned daily counts per sorting centre. For testing purposes, the existing monitoring code had to be used. Therefore, the SQL query had to be adapted so that, instead of returning counts per sorting centre, it returned the number of events for a defined sorting centre and date.

5.4. Testing and the view required for it

The first attempt was to return all applicable events while removing only the daily counts from the query. As this resulted in time-outs in Athena, the query had to be redesigned. Consequently, several attempts were made to reduce the amount of data processed. This process continued until only a single event was checked. The resulting query after this initial testing can be found in Appendix A.4. As with the previous view queries, this query also had to be committed to the corresponding GitHub repository. The following shows a short part of the query, showing the submitted event that is extracted but not counted in the view.

```
, mp_submitted_raw AS (
  SELECT
    CAST(DATE_ADD('hour', -6, submitted.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
    ↪ Posttag
   , submitted.sortingunit.locationKey locationKey
   , submitted.sortingProcess sortingProcess
  FROM
    curated_lsops.mailpiece_submitted_v2
)
```

```
, mp_submitted AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'WAL') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'UTZ') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1
   WHEN (locationKey IN ('HAE', 'FRA', 'DAI', 'URD')) THEN 1
   ELSE 0 END) relevant_submitted
   FROM
     mp_submitted_raw
)
```

Creating New Resources

Another process defined at Swiss Post for AWS is that resources such as a CloudWatch metric stream cannot be created directly in the AWS Console. Instead, all such resources have to be defined in Terraform. At Swiss Post, Terraform is the designated infrastructure-as-code solution. It is used to define, deploy, destroy, and manage AWS resources through configuration files. Furthermore, each AWS account has its own GitHub repository in which the Terraform files are stored. The following illustration shows the functioning of Terraform in a nutshell [L. 2020].

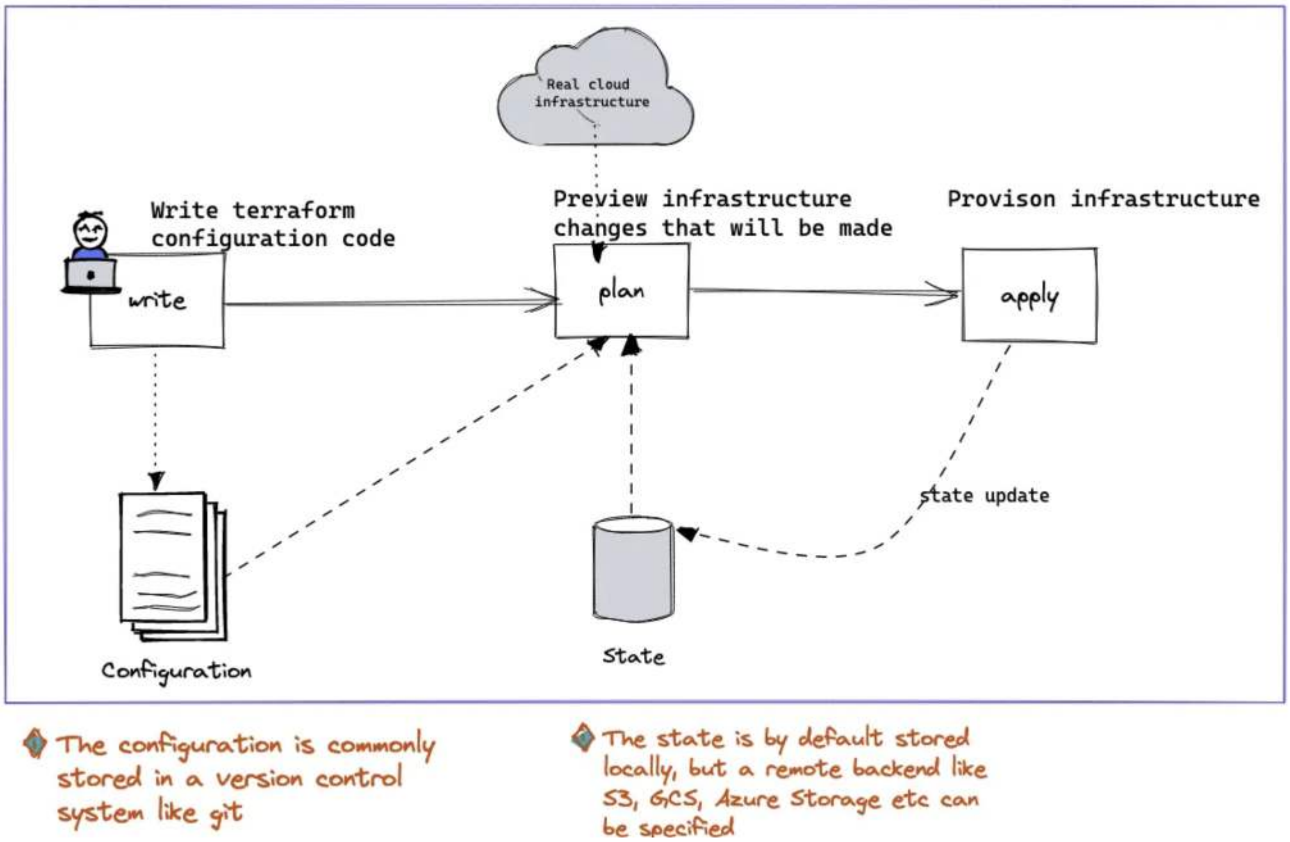


Figure 5.13.: Visual overview of the Terraform workflow illustrating the relationship between configuration files, the Terraform state, providers, and the target infrastructure (adapted from [L. 2020]).

As defined in the code, the Terraform locals specify which queries are executed and with which parameters. The following excerpt shows the query together with its `WHERE` parameters, while the full code can be found in Appendix **??**.

```
locals {
  curated_lsops = {
    schedule_expression = "cron(0/15 * * * ? *)"

    queries = [
      {
        table = "sendungsverarbeitung_dqm_starschema_no_aggregation_small_v"

        where_clause = [
          {
            column    = "locationKey"
            operation = "="
            value     = "HRK"
          },
          {
            column    = "relevant_submitted"
            operation = "="
            value     = 1
          }
        ]
      },
      {
```

```
        table = "curated_lsops_submitted_dqm"
      }
    ]
  }
}
```

Listing 5.1: Excerpt of the Terraform configuration defining scheduled data quality monitoring queries

Code execution testing

After the resources had been created using Terraform, the code execution could be tested manually. This was done by logging into the AWS Management Console, selecting the correct AWS account, and then opening Amazon EventBridge. The newly created schedule shows the payload required for the next testing step. The payload is shown at the bottom of the following screen, followed by the detailed JSON code extracted and copied as the input for the linked Amazon EventBridge.

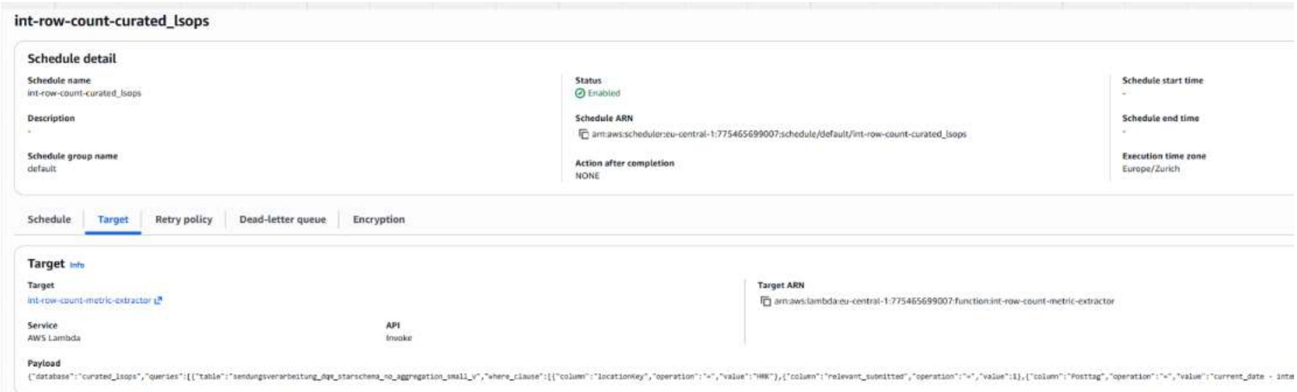


Figure 5.14.: Amazon EventBridge console showing the scheduled monitoring rule and the corresponding JSON event payload configuration

Listing 5.2: AWS EventBridge event payload used to trigger the data quality monitoring workflow

```
{
  "database": "curated_lsops",
  "queries": [
    {
      "table": "sendungsverarbeitung_dqm_starschema_no_aggregation_small_v",
      "where_clause": [
        {
          "column": "locationKey",
          "operation": "=",
          "value": "HRK"
        },
        {
          "column": "relevant_submitted",
          "operation": "=",
          "value": 1
        },
        {
          "column": "Posttag",
          "operation": "=",
          "value": "current_date - interval '35' day"
        }
```

```
            ]
        },
        {
            "table": "curated_lsops_submitted_daily_dqm"
        }
    ]
}
```

After the execution of the Lambda, an execution error was observed, as shown below. After investigating this error, it became clear that the condition injected into the SQL query by the team was not valid. The underlying data monitoring code automatically modified the SQL condition for checking the previous 35 days by escaping parts of the query, causing the SQL statement to be executed incorrectly. After discussions with the AWS modernisation team, this bug was addressed by the consultants.

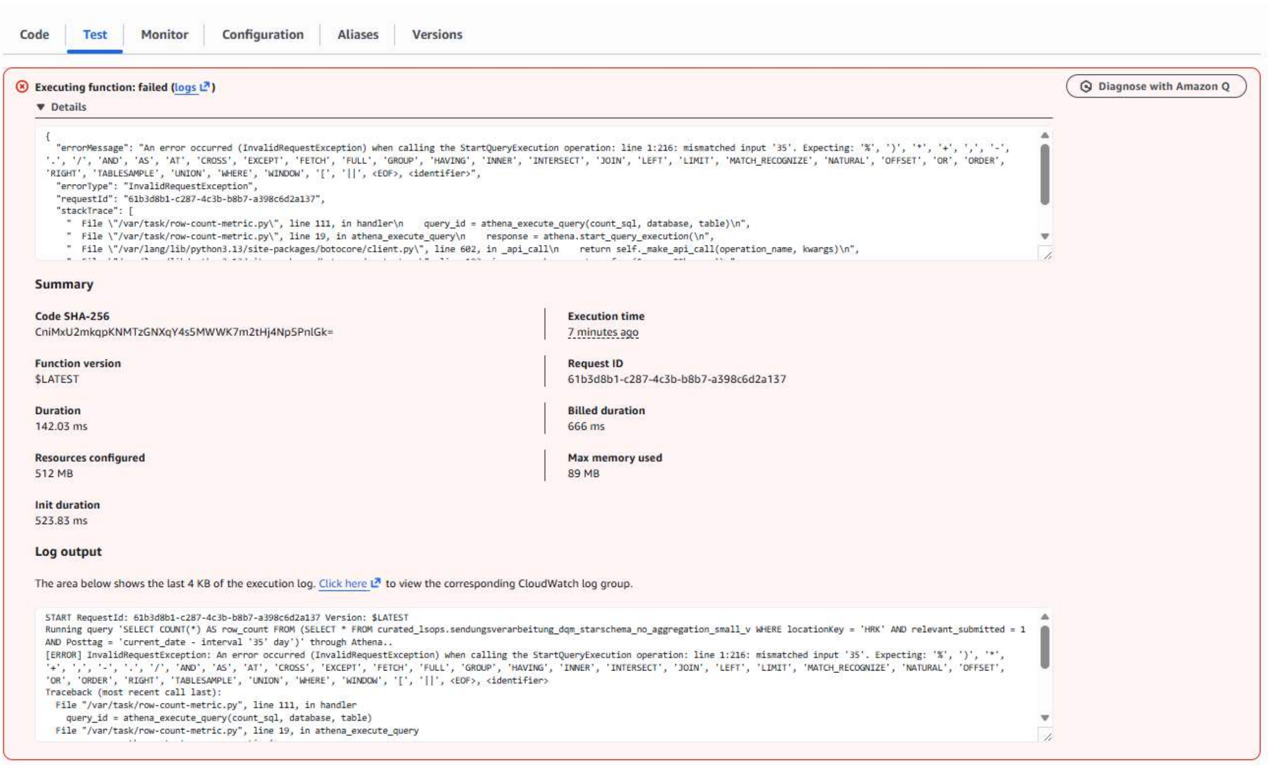


Figure 5.15.: AWS Lambda execution details showing debug information after a failed execution of the data quality monitoring function

After they resolved the bug, the data monitoring setup allowed a date range to be entered for the count, as seen in the following figure. With this fix, the feasibility of DQM checks based on row counts using the monitoring setup was proven.

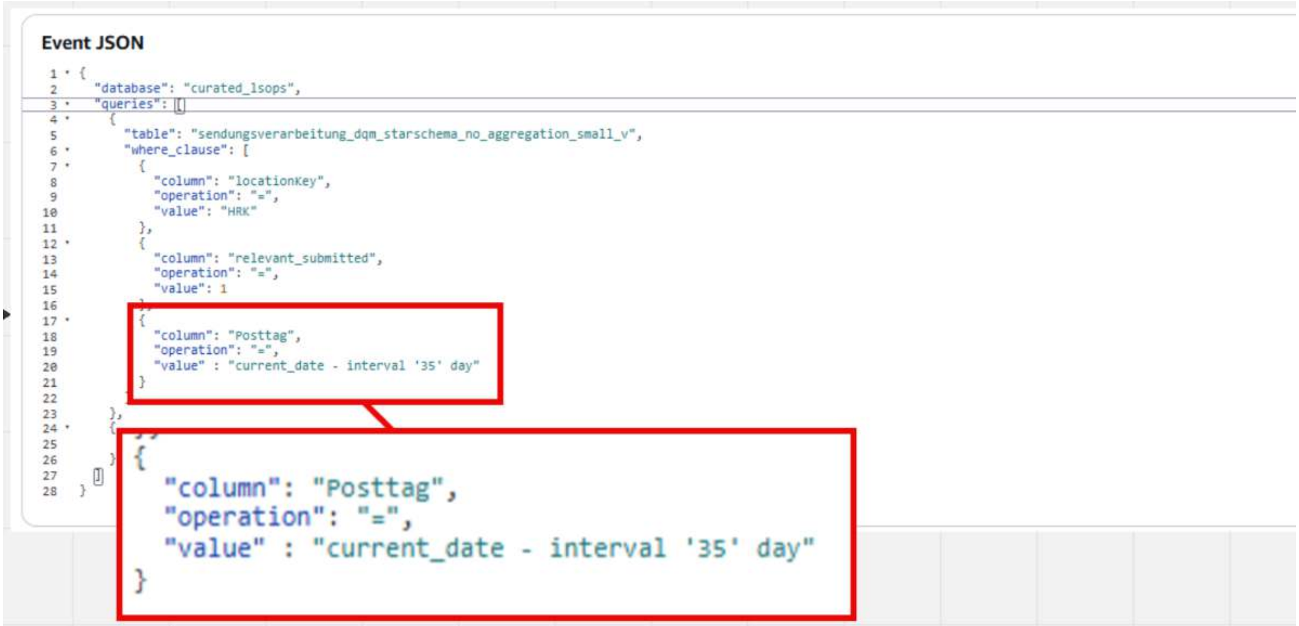


Figure 5.16.: Successful execution of the monitoring workflow after correcting the configuration, showing the validated monitoring metrics being forwarded to the AWS monitoring pipeline

6. Results

This is the key chapter that reflects on the initial goals, the work carried out to achieve the goals, and the degree to which they were achieved. This chapter is structured around each goal, and special attention is paid to the key learnings that will be useful for subsequent phases of this work.

6.1. Introduction

The results presented here are part of a structured process that began step by step with the research question and led to the solution architecture and its implementation. Before delving further into the results of this project, a brief reminder of its scope helps to maintain focus. The project supports the DWH LS OPS with the goal of improving data availability and data quality management by replacing manual checks performed by the business with automated checks executed by IT. Through this change, data about these checks can become more consistent and enable the business to monitor them more easily. To achieve this, the existing reports need to be maintained to work with updated data sources, and the search for a new DQM tool and its validation for use in the upcoming master's thesis were major milestones.

6.2. Evaluation criteria

First, a result is defined as every piece of information that helps the reader become familiar with the context of the work. Second, it includes all information that supports understanding of the methods used during the project. Third, it comprises the main goal of this work, the SMART goals and their results. While the supporting results help to understand the environment, only the SMART goal results are the effectively relevant results of this work.

These SMART goals have been defined in the research chapter at 3.5:

- **1) Setup of an improved data quality report**
- **2) Evaluation and Selection of DQM platform for master thesis**
- **3) Validation of selected DQM tool**

6.3. SMART goal 1: Setup of an improved data quality report

Goal definition

1) Setup of an improved data quality report

WHY: This will help the team to run the daily checks faster

By the end of the implementation phase, the existing data quality report is to be updated to use the new infrastructure. This includes new reporting data pipelines and the new structure using star shema. Also, Power BI is to be used.

Success is measured by user feedback which has to be retrieved by ASTV team. The feedback is to be implemented and the report must be on the productive systems.

Core results

- **Update of data source:** In the implementation chapter, the data source for the third-party checks was moved from being stored on a shared drive to being stored on SharePoint, with synchronisation to Amazon S3. As a result, the data are versioned and available for other uses.
- **Power BI Setup:** At the beginning of the implementation chapter, the setup of Power BI was performed. It used the existing guides of Swiss Post and, in turn, allowed the development of the new improved DQM report to begin.
- **Model data setup:** The data setup carried out throughout the first section of the implementation chapter established a model that was extendable and allowed filters to be applied to all the data.
- **New visuals to support faster interpretation by stakeholders:** The new visuals developed allowed three things. They enabled a fast visual check that the data were correct across the four comparisons performed. They also allowed changes to be viewed over time using the line chart, and finally, the table containing the data allowed the detailed checks per day and location to be confirmed.
- **Scrum review and post-review feedback:** The Scrum review presented the report to the whole ASTV team and was a valuable means of obtaining feedback. Further feedback was gathered in separate meetings and subsequently applied, resulting in the final version of the report being brought into the production environment.
- **New Athena view and applied DevOps:** During the implementation of the report in Power BI, a separate view was set up in Athena. This new view allowed the new data sources to be used while the old view remained unchanged. This improved not only the performance of the view but also ensured that the two were not entangled. In addition, the DevOps deployment process at Swiss Post was described and applied for the deployment of this new view.

Supporting Results

The following results support the later implementation steps:

- In the solution landscape chapter, the definition of Power BI as the new reporting environment was established. Furthermore, the change from a wide table to a star schema was established.
- In the solution architecture, the metrics are reflected in the definitions of the relevant submitted and discharged events, as described in Sections 4.3 and 4.3.

What Was Not Achieved?

All objectives were fulfilled.

Goal Verdict

This goal was completely fulfilled. The report has been implemented, including the application of stakeholder feedback.

Learning and Outlook for Future Work

The importance of running checks between different environments from the first day of development became apparent. This was due to the issues that occurred with the different Power BI environments. Furthermore, it is important to obtain stakeholder feedback as early as possible and to plan enough time for implementing their requests.

6.4. SMART goal 2: Evaluation and Selection of DQM platform for master thesis

Goal definition

2) Evaluation and Selection of DQM platform for master thesis

WHY: To define the DQM platform for master thesis

During the implementation phase, the tool for the master thesis has to be defined. This requires a structured approach for the selection of the tool, including the setup of requirements lists. The team is to follow the DQM team during the evaluation of tools.

Success is measured by having the DQM tool defined. The tool selected has to be discussed with the architect of DWH LS OPS team to make sure it fits the teams needs.

Core Results

- **Tool selection for Swiss Post:** The DQM team decided that the tool which was part of the evaluation in the previous project, using a rule-based approach, was not suitable for the whole of Swiss Post. Therefore, the evaluation of other tools was launched.
- **Demonstration and decision:** The DQM tool Soda was invited to an online demonstration at Swiss Post. During this demonstration, its key features were presented. The team's decision was that, based on the demonstrated features, the missing connection to SAP data was the main deciding factor for not continuing with Soda.
- **DQM platform testing:** Towards the end of this project's timeline, the DQM platform was presented as being ready for testing.
- **Discussion with solution architect:** During a meeting with the solution architect, an alternative to waiting for the tool decision and the evaluation of the rule-based DQM tool was discussed and agreed upon. The job monitoring that was set up during the AWS Modernisation initiative could be used for data monitoring. This led to the third SMART Goal.

Supporting Results

The supporting results, while important achievements, were not researched or generated by the project team. However, these results are an important step towards the goal at hand; therefore, they are still presented.

- **Requirements list:** The list of requirements established by the DQM team defined which features are required in a requirements list. This list defines which requirements are must-have requirements and which are soft criteria.

- **DQM tooling selection:** The DQM tooling selection was performed by the DQM team based on the researched tools and their performance against the requirements list. The three best-performing tools were selected for continued evaluation.

What Was Not Achieved?

- **Final tool selection:** The DQM tool for the whole of Swiss Post had not yet been selected by the end of this project.

Goal Verdict

This SMART Goal was achieved. However, as discussed in the text, not all of the results presented here were developed by the project team

Learning and Outlook for Future Work

This goal was important for the following goal and was successful in providing that decision. However, the goal, as it was set up, was out of scope regarding the project team's function and responsibilities, as the setting up of the requirements list and the selection of the tools were performed by the DQM team. Goals must take into account the responsibilities and functions at Swiss Post.

6.5. SMART goal 3: Validation of the Selected DQM Tool

Goal definition

3) Validation of the Selected DQM Tool

WHY: The DQM platform must be tested for applicability to the master thesis.

By the end of the implementation phase, the DQM tool has to be tested for applicability to the master thesis. It must be proven that the required infrastructure can be set up for a test. Furthermore, a simplified test has to be run to confirm that the tool can handle the data sources.

Success is measured by having the DQM tool tested. The tool must return the test values to prove that it can be used for the follow-up master thesis.

Core Results

- **Knowledge transfer sessions:** The knowledge transfer sessions took place through extensive documentation in Confluence. Additionally, the consultants who developed the solution provided online sessions during which the functioning of the monitoring tool was described.
- **Design of a measurable use case:** The setup of a measurable use case was a key step towards testing the tool. The test was set up based on the existing query and therefore only had to be changed to fit the requirements.
- **Testing and creating a view for testing:** The new view was designed to have a good response time. This required multiple redesigns of the query, which was ultimately shortened so that it only contained one event being checked for one sorting centre.

- **Creating new resources:** While testing the tool, new resources had to be created. These were created using Terraform scripts, which are the Swiss Post standard for managing infrastructure as code.
- **Code execution testing:** The code testing led to the discovery of an error in the tool and how it handled date conditions. This was reported to the AWS Modernisation team, which in turn fixed the bug. This led to the successful execution of the code and the retrieval of results by the monitoring software setup.

Supporting Results

- **DQM tool testing:** The basis for planning the test was described in the solution architecture. The main steps are knowledge buildup, tool testing, and test design.

What Was Not Achieved?

The goal was completely fulfilled; nothing was left unachieved.

Goal Verdict

This goal was successfully achieved using the given use case. Thus, the master thesis can use this tool as a basis for entering the topic.

Learning and Outlook for Future Work

An important lesson learned here is the importance of early meetings with the developers of new tools. In the case of AWS Modernisation, the developers were only available until before the end of this project's timeline, so meetings had to be reserved early. Also important was early testing, which led to the discovery of a bug that was in turn fixed by the consultants rather than having to be debugged and fixed later by the project team. Furthermore, the importance of allowing enough time for the setup of the infrastructure must be taken into account during planning.

6.6. Summary

With this chapter, the SMART goals defined in the research chapter have been analysed for completion. Two goals were completely fulfilled, one partially. The partially fulfilled second goal was limited due to responsibility constraints related to personal and team responsibility. The goal with partial fulfilled results points towards the need for goals planed along the lines of responisbilities, in addition to realistic time frames. However, it is important to note that even tough one goal was not reached completely, the overall results of the research helped to provide a solid for subsequent master thesis.

7. Reflection and outlook

This reflection chapter concludes the second step in the epic quest of the academic team's towards improving data quality and data availability at Swiss Post. It evaluates the project with regard to methodology, content, and personal learnings, and concludes with an outlook on how the project can be followed up.

Figure 7.1.: Reflections with the lords of DQM

7.1. Methodological reflection

In this section, the methodology is reflected upon. This is not limited to the project management methodology but considers all methods applied throughout the project.

What went well

The following list of items went well, including the reasons why they went well.

- **Documentation using Miro**: The documentation of the project in Miro went well. The tool proved very useful for taking notes quickly, which were incorporated into the document later. It was also used for project management and is a key information resource for the entire project.
- **Weekly meetings, Jour Fixe**: The Jour Fixe, as documented in Miro, was held weekly. It proved to be a valuable exchange and helped guide the work and larger tasks. Its documentation on paper was

uploaded to Miro, which helped keep track of discussed and important decisions.

- **Weekly meeting structure**: The meeting structure, starting with an introduction to the topic, the deliverables worked on, and a milestone status update, helped to structure the meetings and eased the start of another session after one week. Furthermore, the predefined questions helped to keep the discussion focused on the important topics. The deliverables graphic, together with the broad description of discharged and submitted items on Miro, supported this.
- **Regular efforts**: Splitting the work between the project work and work in the DWH LS OPS team went smoothly. By not having explicitly defined days set per topic, the management was easier, as no blockers were overbooked, leading to further issues. This led to more frequent work on the study project.
- **Milestone structuring**: The milestones helped to keep the time effort per chapter structured. They helped to stop basic research at the right moment and to move towards the solution landscape and implementation.
- **Writing mode**: The writing mode, with corrections using ChatGPT based on a defined prompt, went well after it was defined. It helped to iron out earlier occurrences of non-academic writing and grammatical issues.
- **Kanban**: The Kanban board helped to keep track of tasks. While its usage can still be improved, noting important decisions as tasks helped to keep track of them.

What did not go well

- **Writing time slot**: The planned time slot for the writing efforts proved to be problematic. Even though the effort was lower, as many topics were already defined from the previous work, writing most of the project at the end was not optimal. Screenshots and documentation had to be searched for with additional effort rather than having the proper screenshots and citations available right away.
- **Goals that overstep responsibilities**: With SMART Goal 2, the project team overstepped their AKV. This is a German abbreviation: **A**ufgaben (tasks), **K**ompetenzen (authority or decision-making rights), and **V**erantwortung (responsibility or accountability). As the DQM team was responsible for deciding on and selecting a tool, the project team was overstepping these boundaries.

7.2. Content-related learnings

- **Data quality as an organisational challenge**: The complexity of Swiss Post makes the search for a DQM tool an organisational challenge. This is due to there being multiple teams and divisions with different data and therefore different needs. The challenge, in turn, is to find tools that can satisfy multiple requirements, leading to trade-offs, as not all requirements can be fulfilled. This can then lead to a tool being selected that is rejected by parts of the organisation, as it may lack key features.
- **Data provisioning**: Even before any reporting is done on data, the provisioning of data proved to be a major challenge. As seen in the case of the star schema changes, the structure of data is in constant change. Similar issues were observed with CSV data being stored on shared folders without versioning, which was moved to a SharePoint interface to Amazon S3.
- **The role in a complex company environment**: Swiss Post has a large IT infrastructure that is managed by multiple teams with different responsibilities. Those responsibilities need to be taken into account when setting up work goals in order not to overstep AKV.

7.3. Personal learnings

In the course of this work, the author has learned many things; the following list summarises the most important learnings and their expected impact on future work.

- **Respect organisational responsibilities and governance (AKV)**: When setting up SMART Goals and subsequent tasks, take into account the organisational structure. The task of following the decision process for the DQM tool was outside the scope of the project team's responsibilities, as other teams had the skills and designated task to perform it. As a learning, the goals should be discussed in more detail, especially regarding what the project team's contribution will be.
- **Validation with a simplified use case**: Setting up a simplified use case together with the knowledge transfer when a tool has just been developed proved to be very useful. It allows all required languages and tasks to be learned for implementing a use case, such as the setup of the infrastructure and the language required to define new tests.
- **Enterprise software procurement takes time**: Following the DQM team during the selection of a DQM tool proved to be much more complex than initially expected. The decision for a tool has to match multiple requirements of different teams and divisions. Furthermore, the decision-making process took more time than expected due to delays, especially related to two initiatives being run in parallel for a specific team and the whole organisation.
- **Usefulness of direct feedback from the team**: After presenting the new data quality report in Power BI, valuable input was provided by the team that helped to implement the functionalities quickly, as guidance on the implementation was also provided along with the requirements.
- **Usefulness of architectural input**: Discussions with the team's solution architect proved valuable, as guidance was provided to evaluate the monitoring software. This led to the identification of the tool, which was subsequently tested for applicability in the use case covered by SMART Goal 3 and will form part of the master thesis.
- **Start the writing earlier**: The writing process was started early with the definition of the basic structure, which was similar to the previous research. However, the writing itself was still performed after all the implementation. Starting the writing process earlier would shift research activities that occur during writing to an earlier stage. Having the chapters on which the implementation is based available earlier also allows the work to be structured more effectively.

7.4. Outlook after the project and its follow-up

The project provided major steps towards the master thesis. In addition to the work described in the document, work also happened outside the research path with the definition of the broader use case and the search for an expert who can support the use case with the required expertise. This work is shown exemplarily in the Miro section specific to these tasks:

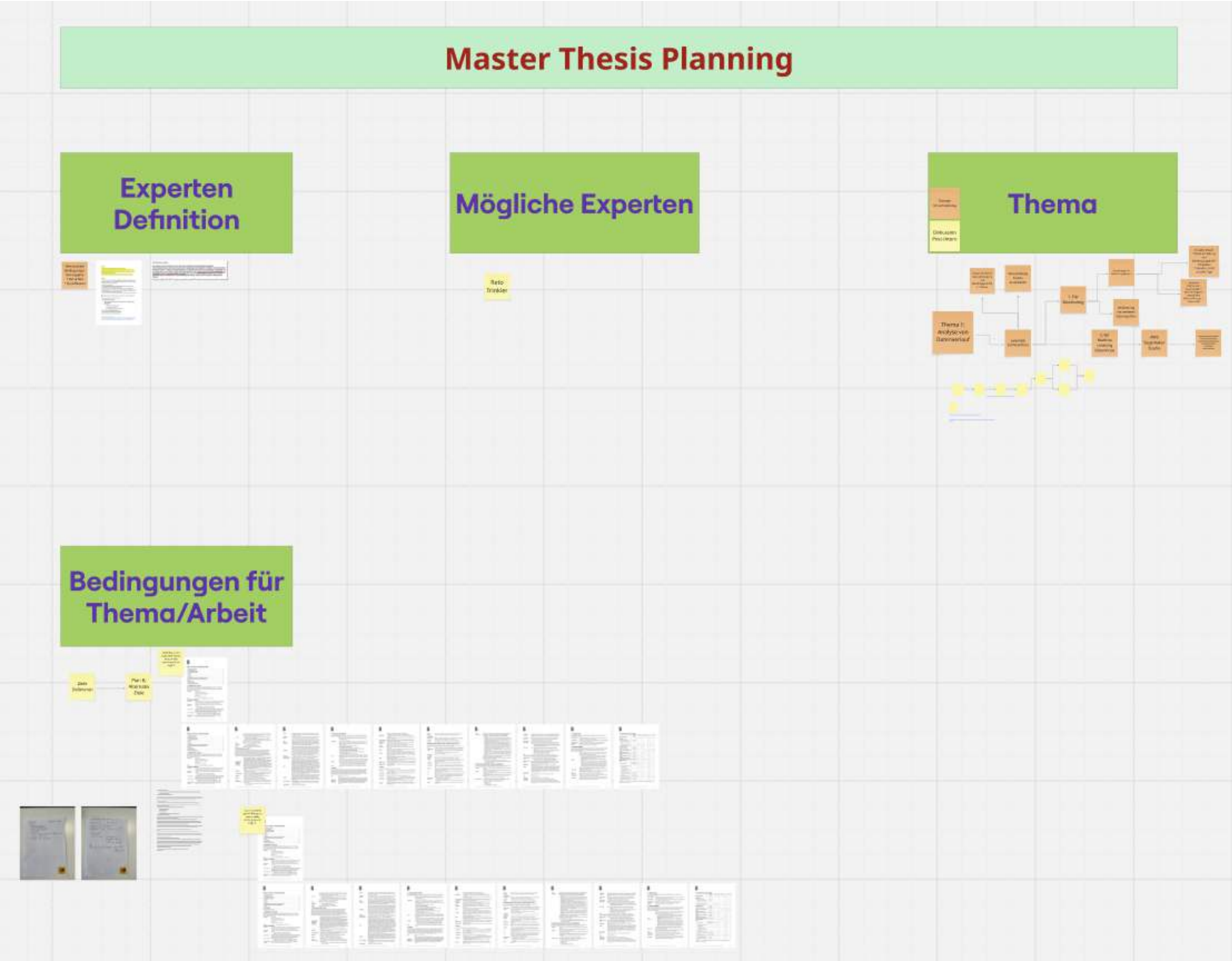


Figure 7.2.: Extract of the Miro board illustrating the planning of the subsequent master's thesis. The section documents initial ideas, work packages, dependencies, and planning considerations that emerged during Project 2 and served as preparation for the continuation of the research.

8. Statement of Authorship

The author hereby declares that this work is the result of their own independent effort and that no sources other than those listed in the bibliography and identified as references have been used. Glossary entries have been partially written by the author using public descriptions (e.g., product homepages), copied from Wikipedia, or generated by ChatGPT based on provided bullet points. Further details regarding the use of LLM is decribed within the project management chapter.

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Glossary

Aggregation Table A pre computed table that is persisted in Power BI, containing the most frequently used data so that it loads faster. Aggregation tables allow most reports to load faster in import mode, while the full detail can be retrieved via DirectQuery mode. This provides a major performance increase for the users. 64

AKV Management principle defining the alignment of **A**ufgaben (tasks), **K**ompetenzen (authority or decision-making rights), and **V**erantwortung (responsibility or accountability). The principle states that these three elements should remain balanced to ensure effective organisational governance and avoid responsibilities without sufficient authority or authority without corresponding accountability. 87

Amazon Athena Amazon Athena runs interactive queries directly against data in Amazon S3. 48, 55, 61–64, 71, 73, 75

Amazon EventBridge An AWS service that enables event-driven communication by routing events between applications and triggering automated workflows based on defined rules. 78

Amazon S3 Amazon Simple Storage Service (S3) is a service offered by Amazon Web Services (AWS) that provides object storage through a web service interface. Amazon S3 uses the same scalable storage infrastructure that Amazon.com uses to run its e-commerce network. 48, 61, 62, 82, 87

Analytical Reporting Analytical Reporting is focusing on interpreting data. To be able to make management decisions over multiple days, weeks or months. It is fundamentally different to operational reporting which monitors the current status e.g. to check if machines are operating without errors. 1, 3, 10

ASTV An abbreviation used at Swiss Post within the postal services. It is an acronym for the German terms: "Aufgabe" –> Receiving, "Sortierung" –> Sorting, "Transport" –> Transport, "Verzollung" –> Customs clearance. This abbreviation encompasses the entire postal process, except for the delivery step. 8, 11, 36, 39, 64, 67, 70, 82

ASTVZ An abbreviation used at Swiss Post within the postal services. It is an acronym for the German terms: "Aufgabe" → Receiving, "Sortierung" → Sorting, "Transport" → Transport, "Verzollung" → Customs clearance, and "Zustellung" → Delivery. This abbreviation encompasses the entire postal process, from receiving a mail piece until it is delivered. 2

AWS Glue AWS Glue is a serverless data integration service that makes it easy for analytics users to discover, prepare, move, and integrate data from multiple sources. You can use it for analytics, machine learning, and application development. It also includes additional productivity and data ops tooling for authoring, running jobs, and implementing business workflows. https://docs.aws.amazon.com/glue/latest/ is-glue.html. 12

AWS Glue Data Quality AWS Glue Data Quality allows you to measure and monitor the quality of your data so that you can make good business decisions. Built on top of the open-source DeeQu framework, AWS Glue Data Quality provides a managed, serverless experience. AWS Glue Data Quality works with Data Quality Definition Language (DQDL), which is a domain specific language that you use to define data quality rules. 41

AWS Lambda An AWS compute service that executes code in response to events without requiring the provisioning or management of servers. 78

AWS Management Console The web-based user interface of Amazon Web Services that provides centralized access to configure, manage, and monitor AWS services and resources. 78

AWS Service A cloud service provided by Amazon Web Services that delivers a specific functionality, such as compute, storage, monitoring, or data processing. Individual AWS services can be combined to build scalable cloud architectures and applications. 74

BizDevOps BizDevOps extends DevOps by including the business side. In this constellation, the business becomes part of the team. This setup improves communication between developers and the people requesting the work. 3, 8, 10, 11, 15

Business analyst A business analyst (BA) is a person who processes, interprets and documents business processes, products, services and software through analysis of data. The role of a business analyst is to ensure business efficiency increases through their knowledge of both IT and business function. 11

Business intelligence Business intelligence (BI) consists of strategies, methodologies, and technologies used by enterprises for data analysis and management of business information to inform business strategies and business operations. Common functions of BI technologies include reporting, online analytical processing, analytics, dashboard development, data mining, process mining, complex event processing, business performance management, benchmarking, text mining, predictive analytics, and prescriptive analytics. 11

Curated database layer The curated database layer is the first layer in the Swiss Post DWH setup in which data is cleaned and standardised and duplicates are removed. This occurs after the data is received in a raw format. 46

D & F, Delivery and Finance The Delivery and Finance team is a BizDevOps team in DWH LS OPS handling requirements related to delivery and finance, including regional controlling and process controlling. 11

Data Availability (DA) Data availability, in the context of this document, refers to whether the data is accessible e.g. whether a report can be retrieved and used when needed. 1, 7, 35, 81

Data Quality (DQ) Data quality refers to the state of qualitative or quantitative pieces of information. There are many definitions of data quality, but data is generally considered high quality if it is "fit for [its] intended uses in operations, decision making and planning". Data is deemed of high quality if it correctly represents the real-world construct to which it refers. Apart from these definitions, as the number of data sources increases, the question of internal data consistency becomes significant, regardless of fitness for use for any particular external purpose. 1, 7, 35, 81

Data Quality Management (DQM) Data quality management is the management of data quality topics. It describes how data quality is handled in a company. However, in this document, most references to DQM refer to the IT team working at Swiss Post on the DQM topic. This team is implementing defined software that is intended to promote improvements in data quality. 10, 12, 13, 16, 39, 41, 42, 56, 70–74, 81, 82, 87

DAX Data Analysis Expressions (DAX) is a formula language used in Microsoft Power BI, Analysis Services, and Power Pivot to define calculations, measures, calculated columns, and tables for data analysis and reporting. 63, 64

DevOps DevOps is the integration and automation of software development and information technology operations. DevOps encompasses the tasks necessary for software development and can lead to both shortening development time and improving the development life cycle. 54

DSP, SAP Datasphere DSP (SAP Datasphere) enables the creation of data models that integrate SAP and non-SAP data sources. These sources can include databases as well as Excel files. The data sources can then be further joined and aggregated. 48, 50–52

DWH In computing, a data warehouse (DW or DWH), also known as an enterprise data warehouse (EDW), is a system used for reporting and data analysis and is a core component of business intelligence. Data warehouses are central repositories of data integrated from disparate sources. 10, 46, 71

DWH LS OPS A Swiss post abreviation, DWH -> "Data Warehouse", LS -> "Logistic Services", OPS -> "Operations". Together, this refers to the Data Warehouse that supports and manages the operational processes of the postal logistics services. 3, 8, 10, 15, 35, 36, 39, 52, 61–63, 73, 74, 81, 83, 87

Epic Epics are large bodies of work in Agile that are broken down into smaller user stories for incremental delivery. They help teams organize, prioritize, and track progress toward bigger goals while maintaining agility. 19

ETL Extract, transform, load (ETL) is a three-phase computing process where data is extracted from an input source, transformed (including cleaning), and loaded into an output data container. 8, 12

Greenfield A greenfield refers to a system landscape where no prior system exists. In IT, a greenfield project means that a new system is built from scratch, without having to consider or integrate with any existing legacy systems. Because there is no predecessor system, development can proceed more freely, without compatibility constraints — which often makes the implementation simpler and more flexible compared to a migration or redesign of an existing system (a brownfield). 7

High Level High level refers to viewing a use case from a general or abstract perspective, rather than analyzing all details. It focuses on the overall picture instead of individual elements. Instead of examining every single event during a day, a high-level analysis would simply check whether most events fall within the expected range during that day. 8

IaC Infrastructure as Code (IaC) is the process of managing and provisioning computer data center resources through machine-readable definition files, rather than via physical hardware configuration or interactive configuration tools. 55

Infoblock Infoblock (at Swiss Post in the context of AWS data storage) An Infoblock is a standardized data product that is provided within the data architecture of Swiss Post for reporting and analysis purposes. It serves as a logical unit of processed and quality-assured data that is extracted from different source systems, transformed, and stored in a consistent structure in the AWS cloud. This concept is similar to data products as defined in modern data management practices. 50, 70

Jour Fixe A Jour Fixe is a weekly meeting used in project management to provide updates on progress, issues encountered during work, and problems that remain unresolved and require assistance. It serves to maintain communication and alignment with stakeholders. 17

Kafka Topics A Kafka topic is a core component of Apache Kafka. It is a named storage structure where producers write messages or data streams. The content of the topic can then be read and processed by consumers. 8

Kanban Kanban (Japanese: meaning signboard) is a scheduling system for lean manufacturing (also called just-in-time manufacturing). Taiichi Ohno, an industrial engineer at Toyota, developed kanban to improve manufacturing efficiency. The system takes its name from the cards that track production within a factory. 16

KPI, key performance indicator A performance indicator or key performance indicator (KPI) is a type of performance measurement. KPIs evaluate the success of an organization or of a particular activity (such as projects, programs, products and other initiatives) in which it engages. KPIs provide a focus for strategic and operational improvement, create an analytical basis for decision making and help focus attention on what matters most. 12

LS, Logistics Services Logistics Services (LS) is responsible for efficient and secure logistics of letters, printed matter and newspapers, as well as parcels, goods and merchandise in Switzerland and internationally. Logistics Services is the largest business unit of Swiss Post and makes a substantial contribution to the company's success with over 21 000 employees. 10

MIAR MIAR is the internal reporting and project effort tracking system used at Swiss Post. It is integrated with applications for working time recording, such as check-in and check-out. It allows employees to allocate worked hours to projects, which in turn enables monitoring of project efforts. 28

Microsoft Copilot AI-assisted productivity tool by Microsoft that supports users with code generation, text generation, and problem-solving based on large language models. 63

Microsoft Fabric A cloud-based analytics platform by Microsoft that integrates data engineering, data warehousing, data science, real-time analytics, and business intelligence in a unified environment. It is used to manage, analyse, and visualise data across different workloads. 63

Microsoft Power BI (PBI) Microsoft Power BI (PBI) is an interactive data visualization software product developed by Microsoft with a primary focus on business intelligence (BI). It is part of the Microsoft Power Platform. Power BI is a collection of software services, apps, and connectors that work together to turn various sources of data into static and interactive data visualizations. 61

Microsoft Power BI (PBI) Microsoft Power BI (PBI) is an interactive data visualization, database management, and data modeling software product developed by Microsoft with a primary focus on business intelligence (BI). 52

MPP Mengenplanung und Prognose Pakete MPP Parcel Volume Planning and Forecasting is a team within Swiss Post. It is responsible for improving current planning and forecasting methods. Based on this information, important decisions are made, such as determining the personnel required for peak periods like the Christmas season with increased parcel volumes. The team analyses current data sources and aims to provide more precise methods and data sources for the process. 11

MVP A Minimal Viable Product (MVP) is the simplest version of a system that successfully fulfils a specific use case. It is developed with just the core functionality needed to provide value to users. An MVP can serve two main purposes: to demonstrate that a concept or method is feasible (proof of concept), or as an initial implementation that will later be improved through further development and updates. 7, 39

OLAP In computing, online analytical processing (OLAP) , is an approach to quickly answer multi-dimensional analytical (MDA) queries. The term OLAP was created as a slight modification of the traditional database term online transaction processing (OLTP). OLAP is part of the broader category of business intelligence, which also encompasses relational databases, report writing and data mining. 52

One to One Migrations In the context of reporting, a one-to-one migration refers to rebuilding an existing report exactly as it is, but on a different software or hardware platform. The structure, logic, and content of the report remain unchanged only the underlying technology is replaced. 7

Parcel center A parcel center is a large sorting center for parcels at Swiss Post. Each region has one parcel center, which serves as the main hub for parcel sorting. Every parcel is first gathered at a parcel center and later distributed to smaller regional parcel centers if its delivery is not scheduled for the immediate vicinity of the center. 42

Post day The post day starts at 06:00 AM and lasts until 05:59 AM the next day. This is the essential date format used by Swiss Post for reporting. 50, 52

Power Query M A functional, case-sensitive programming language used by Microsoft Power Query to import, transform, combine, and prepare data from multiple data sources for analytics and reporting. 63, 64

Product Owner (PO) Each scrum team has one product owner. The product owner focuses on the business side of product development and spends the majority of time liaising with stakeholders and the team. The role is intended to primarily represent the product's stakeholders, the voice of the customer, or the desires of a committee, and bears responsibility for the delivery of business results. 36

Proof of concept (POC) Proof of concept (POC), also known as proof of principle, is a realization of a certain method or idea in order to demonstrate its feasibility, or a demonstration in principle with the aim of verifying that some concept or theory has practical potential. A proof of concept is usually small and may or may not be complete. 8, 58, 73

PyDeequ PyDeequ is a Python API for Deequ, a library built on top of Apache Spark for defining "unit tests for data", which measure data quality in large datasets. PyDeequ is written to support usage of Deequ in Python. 41

Region Swiss Post organises its postal services into four regions covering the whole of Switzerland. These regions are West, Middle, East, and South. The regions are not based on cantonal borders and may split cantons between regions. 42

Regional parcel center A regional parcel center is a smaller sorting center for parcels at Swiss Post. Multiple regional parcel centers are located within one region and handle parcels for smaller geographic areas. 42

Risk Appetite Risk appetite is a term used in risk management. It describes an individual's perception and tolerance of risk. Even when two people are facing the same risk, they may perceive its severity and probability differently, based on their personal risk appetite. 29, 31

Rollbox (RX) The rollbox (RX) is used to temporarily transport parcels throughout the Swiss Post workflow. It can hold many mail pieces and is used to move them between different processing areas, making it an essential component of parcel handling at Swiss Post. 43

SAC, SAP Analytics Cloud SAC, SAP Analytics Cloud is a plattform for online analytical reporting. It allows to show data in tables and dashboards and allows Reporting and predictive analysis. 48, 50–53, 64

Scrum Scrum is an agile team collaboration framework commonly used in software development and other industries. Scrum prescribes for teams to break work into goals to be completed within timeboxed iterations, called sprints. 15

Smart Goals S.M.A.R.T. (or SMART) is an acronym used as a mnemonic device to establish criteria for effective goal-setting and objective development. The S.M.A.R.T. criteria he proposes are as follows: Specific: Targeting a particular area for improvement, Measurable: Quantifying, or at least suggesting, an indicator of progress, Assignable: Defining responsibility clearly, Realistic: Outlining attainable results with available resources, Time-related: Including a timeline for expected results. 36

Soft Criteria Non mandatory evaluation criteria that do not determine whether a solution is technically suitable, but improve usability, maintainability, or the overall user experience. Examples include intuitive user interfaces, simplified administration, or easy management of user roles and permissions. 57

Sorting Center In the postal services of Swiss Post, sorting centers are an essential part of the logistics process for handling letters and parcels. Items are first collected and sent to regional centralized facilities, where they are either: routed to sorting centers in other regions, or processed locally and dispatched to the distribution network of the current region. These centers enable efficient processing and routing of mail within Switzerland and across regions. 3

Star Schema A dimensional data modelling approach following the Kimball methodology, consisting of a central fact table linked to multiple denormalised dimension tables. The denormalisation is performed to improve the performance of analytical queries. It is widely used in data warehouses and business intelligence systems. 61

Terraform An open-source infrastructure as code (IaC) tool developed by HashiCorp that enables infrastructure to be defined and managed using declarative configuration files. Terraform automates the provisioning, modification, and removal of cloud resources while maintaining the desired infrastructure state. It supports multiple cloud providers and promotes consistent, version-controlled infrastructure deployments. 59, 76

Wide table A wide table is a database table structure characterized by a large number of columns, often dozens or hundreds, storing multiple attributes about each record in a single row. In time-series databases, wide tables are commonly used for storing sensor data, financial market data, and other scenarios where many metrics need to be captured at each timestamp. 53

WYSIWYG (what you see is what you get) In computing, WYSIWYG (what you see is what you get) is software that allows content to be edited in a form that resembles its appearance when printed or displayed as a finished product, such as a printed document, web page, or slide presentation. WYSIWYG implies a user interface that allows the user to view something very similar to the result while the document is being created. 26

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A. Amazon Athena Queries

This appendix provides the Amazon Athena queries. They are stored here since putting them into the document would disrupt normal flow.

A.1. Full DQM view query

The full query which creates the view beeing main data source of the DQM report.

```
CREATE OR REPLACE VIEW "sendungsverarbeitung_dqm_v" AS
WITH
key_to_plz AS (
    SELECT 'VET' AS locationKey, '196000' AS plz
    UNION ALL SELECT 'HRK', '462150'
    UNION ALL SELECT 'CAD', '659100'
    UNION ALL SELECT 'WAL', '830000'
    UNION ALL SELECT 'HAE', '462000'
    UNION ALL SELECT 'UTZ', '720000'
    UNION ALL SELECT 'PRA', '413400'
    UNION ALL SELECT 'FRA', '852000'
    UNION ALL SELECT 'DAI', '131000'
    UNION ALL SELECT 'URD', '892000'
    UNION ALL SELECT 'BCH', '503000'
    UNION ALL SELECT 'RUE', '815000'
    UNION ALL SELECT 'OMU', '307100'
),

plz_to_key AS (
    SELECT '196000' AS plz, 'VET' AS locationKey
    UNION ALL SELECT '462150', 'HRK'
    UNION ALL SELECT '659100', 'CAD'
    UNION ALL SELECT '830470', 'WAL'
    UNION ALL SELECT '830000', 'WAL'
    UNION ALL SELECT '462000', 'HAE'
    UNION ALL SELECT '720000', 'UTZ'
    UNION ALL SELECT '413400', 'PRA'
    UNION ALL SELECT '852000', 'FRA'
    UNION ALL SELECT '131000', 'DAI'
    UNION ALL SELECT '892000', 'URD'
    UNION ALL SELECT '503000', 'BCH'
    UNION ALL SELECT '815000', 'RUE'
    UNION ALL SELECT '307100', 'OMU'
),

mp_discharged_raw AS (
    SELECT
        CAST(
            DATE_ADD(
```

```
                'hour', -6,
                discharged.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        discharged.sortingunit.locationKey AS locationKey,
        discharged.sourceId AS sourceId
    FROM curated_lsops.mailpiece_discharged
),

mp_discharged AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('CAD','HRK','PRA','UTZ','VET','WAL')
                 AND sourceId NOT LIKE 'w%'
                 AND sourceId NOT LIKE 'W%'
                 AND TRIM(sourceId) <> '' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_discharged
    FROM mp_discharged_raw
),

discharged_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_discharged) AS num_discharged
    FROM mp_discharged
    GROUP BY 1, 2
),

mp_submitted_raw AS (
    SELECT
        CAST(
            DATE_ADD(
                'hour', -6,
                submitted.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        submitted.sortingunit.locationKey AS locationKey,
        submitted.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_submitted
),

mp_submitted AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
```

```
        END AS relevant_submitted
    FROM mp_submitted_raw
),

submitted_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_submitted) AS num_submitted
    FROM mp_submitted
    GROUP BY 1, 2
),


mp_coded_raw AS (
    SELECT
        coded.sortingCycleId AS sortingCycleId,
        CAST(
            DATE_ADD(
                'hour', -6,
                coded.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        coded.sortingunit.locationKey AS locationKey,
        coded.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_coded
),

mp_coded AS (
    SELECT DISTINCT
        sortingCycleId,
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_coded
    FROM mp_coded_raw
),

coded_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_coded) AS num_coded
    FROM mp_coded
    GROUP BY 1, 2
),

mp_destinationAssigned_raw AS (
    SELECT
        destinationAssigned.sortingCycleId AS sortingCycleId,
        CAST(
            DATE_ADD(
```

```
                'hour', -6,
                destinationAssigned.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        destinationAssigned.sortingunit.locationKey AS locationKey,
        destinationAssigned.sortingParameters.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_destinationassigned
),

mp_destinationAssigned AS (
    SELECT DISTINCT
        sortingCycleId,
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_destinationAssigned
    FROM mp_destinationAssigned_raw
),

destinationassigned_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_destinationAssigned) AS num_destinationassigned
    FROM mp_destinationAssigned
    GROUP BY 1, 2
),

sendungsverarbeitung_raw AS (
    SELECT
        sva.datum_sort.datum AS Posttag,
        plz_to_key.locationKey AS locationKey,
        sva.sortierung.sortierprozess_typ AS sortierprozess_typ
    FROM infoblock_lsops.sendungsverarbeitung sva
    LEFT JOIN plz_to_key
        ON plz_to_key.plz = sva.zentrum.zentrum_plz
),

sendungsverarbeitung AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                 AND sortierprozess_typ = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_sva
    FROM sendungsverarbeitung_raw
),
```

```
sva_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_sva) AS num_sva
    FROM sendungsverarbeitung
    GROUP BY 1, 2
),

device_stats AS (
    SELECT
        Posttag,
        sorting_unit_location_key AS locationKey,
        SUM(content_value) AS amount
    FROM "reporting_lsops"."anlagestatistik_curated_v"
    WHERE model = 'BEUMER'
      AND content_type = 'INDUCTION'
      AND content_value_index = 1
    GROUP BY 1, 2
)

SELECT
    submitted.Posttag,
    submitted.locationKey,
    key_to_plz.plz,
    discharged.num_discharged,
    submitted.num_submitted,
    coded.num_coded,
    destinationassigned.num_destinationassigned,
    sva.num_sva,
    dvs.amount AS num_stats
FROM submitted_agg submitted
INNER JOIN key_to_plz
    ON submitted.locationKey = key_to_plz.locationKey
INNER JOIN discharged_agg discharged
    ON submitted.Posttag = discharged.Posttag
   AND submitted.locationKey = discharged.locationKey
INNER JOIN coded_agg coded
    ON submitted.Posttag = coded.Posttag
   AND submitted.locationKey = coded.locationKey
INNER JOIN destinationassigned_agg destinationassigned
    ON submitted.Posttag = destinationassigned.Posttag
   AND submitted.locationKey = destinationassigned.locationKey
INNER JOIN sva_agg sva
    ON submitted.Posttag = sva.Posttag
   AND submitted.locationKey = sva.locationKey
LEFT JOIN device_stats dvs
    ON submitted.Posttag = dvs.Posttag
   AND submitted.locationKey = dvs.locationKey
WHERE submitted.locationKey IN (
    'HRK','VET','HAE','CAD','WAL','UTZ',
    'PRA','FRA','DAI','URD','BCH','RUE','OMU'
);
```

A.2. Compact DQM view query

The query after removing the parts which are not needed for the MVP.

```
WITH
key_to_plz AS (
    SELECT 'VET' AS locationKey, '196000' AS plz
    UNION ALL SELECT 'HRK', '462150'
    UNION ALL SELECT 'CAD', '659100'
    UNION ALL SELECT 'WAL', '830000'
    UNION ALL SELECT 'HAE', '462000'
    UNION ALL SELECT 'UTZ', '720000'
    UNION ALL SELECT 'PRA', '413400'
    UNION ALL SELECT 'FRA', '852000'
    UNION ALL SELECT 'DAI', '131000'
    UNION ALL SELECT 'URD', '892000'
    UNION ALL SELECT 'BCH', '503000'
    UNION ALL SELECT 'RUE', '815000'
    UNION ALL SELECT 'OMU', '307100'
),

plz_to_key AS (
    SELECT '196000' AS plz, 'VET' AS locationKey
    UNION ALL SELECT '462150', 'HRK'
    UNION ALL SELECT '659100', 'CAD'
    UNION ALL SELECT '830470', 'WAL'
    UNION ALL SELECT '830000', 'WAL'
    UNION ALL SELECT '462000', 'HAE'
    UNION ALL SELECT '720000', 'UTZ'
    UNION ALL SELECT '413400', 'PRA'
    UNION ALL SELECT '852000', 'FRA'
    UNION ALL SELECT '131000', 'DAI'
    UNION ALL SELECT '892000', 'URD'
    UNION ALL SELECT '503000', 'BCH'
    UNION ALL SELECT '815000', 'RUE'
    UNION ALL SELECT '307100', 'OMU'
),

mp_discharged_raw AS (
    SELECT
        CAST(
            DATE_ADD(
                'hour', -6,
                discharged.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        discharged.sortingunit.locationKey AS locationKey,
        discharged.sourceId AS sourceId
    FROM curated_lsops.mailpiece_discharged
),

mp_discharged AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('CAD','HRK','PRA','UTZ','VET','WAL')
```

```
                AND sourceId NOT LIKE 'w%'
                AND sourceId NOT LIKE 'W%'
                AND TRIM(sourceId) <> '' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_discharged
    FROM mp_discharged_raw
),

discharged_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_discharged) AS num_discharged
    FROM mp_discharged
    GROUP BY 1, 2
),

mp_submitted_raw AS (
    SELECT
        CAST(
            DATE_ADD(
                'hour', -6,
                submitted.timestamp AT TIME ZONE 'Europe/Zurich'
            ) AS DATE
        ) AS Posttag,
        submitted.sortingunit.locationKey AS locationKey,
        submitted.sortingProcess AS sortingProcess
    FROM curated_lsops.mailpiece_submitted
),

mp_submitted AS (
    SELECT
        Posttag,
        locationKey,
        CASE
            WHEN locationKey IN ('HRK','VET','CAD','WAL','UTZ','PRA')
                AND sortingProcess = 'AUTOMATIC' THEN 1
            WHEN locationKey IN ('BCH','RUE','OMU') THEN 1
            WHEN locationKey IN ('HAE','FRA','DAI','URD') THEN 1
            ELSE 0
        END AS relevant_submitted
    FROM mp_submitted_raw
),

submitted_agg AS (
    SELECT
        Posttag,
        locationKey,
        SUM(relevant_submitted) AS num_submitted
    FROM mp_submitted
    GROUP BY 1, 2
)

SELECT
    submitted.Posttag,
```

```
    submitted.locationKey,
    key_to_plz.plz,
    discharged.num_discharged,
    submitted.num_submitted
FROM submitted_agg submitted
INNER JOIN key_to_plz
    ON submitted.locationKey = key_to_plz.locationKey
INNER JOIN discharged_agg discharged
    ON submitted.Posttag = discharged.Posttag
   AND submitted.locationKey = discharged.locationKey
WHERE submitted.locationKey IN (
    'HRK','VET','HAE','CAD','WAL','UTZ',
    'PRA','FRA','DAI','URD','BCH','RUE','OMU'
);
```

A.3. New Star Shema DQM view query

The for star shema DQM checks. Similar structure to the previous full query.

```
CREATE OR REPLACE VIEW "sendungsverarbeitung_dqm_starschema_v" AS
WITH
  key_to_plz AS (
   SELECT
     'VET' locationKey
   , '196000' plz

UNION ALL    SELECT
     'HRK' locationKey
   , '462150' plz

UNION ALL    SELECT
     'CAD' locationKey
   , '659100' plz

UNION ALL    SELECT
     'WAL' locationKey
   , '830000' plz

UNION ALL    SELECT
     'HAE' locationKey
   , '462000' plz

UNION ALL    SELECT
     'UTZ' locationKey
   , '720000' plz

UNION ALL    SELECT
     'PRA' locationKey
   , '413400' plz

UNION ALL    SELECT
     'FRA' locationKey
   , '852000' plz
```

```
UNION ALL    SELECT
      'DAI' locationKey
    , '131000' plz

UNION ALL    SELECT
      'URD' locationKey
    , '892000' plz

UNION ALL    SELECT
      'BCH' locationKey
    , '503000' plz

UNION ALL    SELECT
      'RUE' locationKey
    , '815000' plz

UNION ALL    SELECT
      'OMU' locationKey
    , '307100' plz

)
, plz_to_key AS (
    SELECT
      '196000' plz
    , 'VET' locationKey

UNION ALL    SELECT
      '462150' plz
    , 'HRK' locationKey

UNION ALL    SELECT
      '659100' plz
    , 'CAD' locationKey

UNION ALL    SELECT
      '830470' plz
    , 'WAL' locationKey

UNION ALL    SELECT
      '830000' plz
    , 'WAL' locationKey

UNION ALL    SELECT
      '462000' plz
    , 'HAE' locationKey

UNION ALL    SELECT
      '720000' plz
    , 'UTZ' locationKey

UNION ALL    SELECT
      '413400' plz
    , 'PRA' locationKey

UNION ALL    SELECT
      '852000' plz
    , 'FRA' locationKey
```

```
UNION ALL    SELECT
     '131000' plz
   , 'DAI' locationKey

UNION ALL    SELECT
     '892000' plz
   , 'URD' locationKey

UNION ALL    SELECT
     '503000' plz
   , 'BCH' locationKey

UNION ALL    SELECT
     '815000' plz
   , 'RUE' locationKey

UNION ALL    SELECT
     '307100' plz
   , 'OMU' locationKey

)
, mp_discharged_raw AS (
   SELECT
     CAST(DATE_ADD('hour', -6, discharged.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
     ↪ Posttag
   , discharged.sortingunit.locationKey locationKey
   , discharged.sourceId sourceId
   FROM
     curated_lsops.mailpiece_discharged_v2
   WHERE (1 = 1)
)
, mp_discharged AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , (CASE WHEN ((locationKey IN ('CAD', 'HRK', 'PRA', 'UTZ', 'VET', 'WAL')) AND (NOT (↩
     ↪ sourceId LIKE 'w%')) AND (NOT (sourceId LIKE 'W%')) AND (NOT (trim(BOTH FROM sourceId) ↩
     ↪ = ''))) THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ↩
     ↪ ('HAE', 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_discharged
   FROM
     mp_discharged_raw
)
, discharged_agg AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , SUM(relevant_discharged) num_discharged
   FROM
     mp_discharged
   GROUP BY 1, 2
)
, mp_submitted_raw AS (
   SELECT
     CAST(DATE_ADD('hour', -6, submitted.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
     ↪ Posttag
   , submitted.sortingunit.locationKey locationKey
```

```
    , submitted.sortingProcess sortingProcess
    FROM
      curated_lsops.mailpiece_submitted_v2
)
, mp_submitted AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((↩
    ↪ locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = '↩
    ↪ CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND (↩
    ↪ sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'UTZ') AND (sortingProcess =↩
    ↪  'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) ↩
    ↪ THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ('HAE', ↩
    ↪ 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_submitted
    FROM
      mp_submitted_raw
)
, submitted_agg AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , SUM(relevant_submitted) num_submitted
    FROM
      mp_submitted
    GROUP BY 1, 2
)
, mp_coded_raw AS (
    SELECT
      coded.sortingCycleId sortingCycleId
    , CAST(DATE_ADD('hour', -6, coded.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) Posttag
    , coded.sortingunit.locationKey locationKey
    , coded.sortingProcess sortingProcess
    FROM
      curated_lsops.mailpiece_coded_v2
)
, mp_coded AS (
    SELECT DISTINCT
      sortingCycleId sortingCycleId
    , Posttag Posttag
    , locationKey locationKey
    , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((↩
    ↪ locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = '↩
    ↪ CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND (↩
    ↪ sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'UTZ') AND (sortingProcess =↩
    ↪  'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) ↩
    ↪ THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ('HAE', ↩
    ↪ 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_coded
    FROM
      mp_coded_raw
)
, coded_agg AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , SUM(relevant_coded) num_coded
    FROM
```

```
      mp_coded
    GROUP BY 1, 2
)
, mp_destinationAssigned_raw AS (
    SELECT
      destinationAssigned.sortingCycleId sortingCycleId
    , CAST(DATE_ADD('hour', -6, destinationAssigned.timestamp AT TIME ZONE 'Europe/Zurich') AS ↩
    ↪ DATE) Posttag
    , destinationAssigned.sortingunit.locationKey locationKey
    , destinationAssigned.sortingParameters.sortingProcess sortingProcess
    FROM
      curated_lsops.mailpiece_destinationassigned_v2
)
, mp_destinationAssigned AS (
    SELECT DISTINCT
      sortingCycleId sortingCycleId
    , Posttag Posttag
    , locationKey locationKey
    , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((↩
    ↪ locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = '↩
    ↪ CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND (↩
    ↪ sortingProcess = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'UTZ') AND (sortingProcess =↩
    ↪  'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) ↩
    ↪ THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1 WHEN (locationKey IN ('HAE', ↩
    ↪ 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_destinationAssigned
    FROM
      mp_destinationAssigned_raw
)
, destinationassigned_agg AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , SUM(relevant_destinationAssigned) num_destinationassigned
    FROM
      mp_destinationAssigned
    GROUP BY 1, 2
)
, sendungsverarbeitung_raw AS (
    SELECT
      post_day posttag
    , plz_to_key.locationKey locationKey
    , dim_sor.sortierprozess_typ sortierprozess_typ
    FROM
      (((reporting_lsops.fact_sendungsverarbeitung_v sndv
    LEFT JOIN reporting_lsops.dim_sortierung_v dim_sor ON (sndv.sortierung_hash_key = dim_sor.↩
    ↪ sortierung_hash_key))
    LEFT JOIN reporting_lsops.dim_zentrum_v dim_zent ON (sndv.zentrum_hash_key = dim_zent.↩
    ↪ zentrum_hash_key))
    LEFT JOIN plz_to_key ON (plz_to_key.plz = dim_zent.zentrum_plz))
)
, sendungsverarbeitung AS (
    SELECT
      Posttag Posttag
    , locationKey locationKey
    , (CASE WHEN ((locationKey = 'HRK') AND (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((↩
    ↪ locationKey = 'VET') AND (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((locationKey ↩
    ↪ = 'CAD') AND (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'WAL') AND↩
```

```
    ↪  (sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((lo-cationKey = 'UTZ') AND (↩
    ↪ sortierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN ((locationKey = 'PRA') AND (sor-↩
    ↪ tierprozess_typ = 'AUTOMATIC')) THEN 1 WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN↩
    ↪  1 WHEN (locationKey IN ('HAE', 'FRA', 'DAI', 'URD')) THEN 1 ELSE 0 END) relevant_sva
   FROM
     sendungsverarbeitung_raw
)
, sva_agg AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , SUM(relevant_sva) num_sva
   FROM
     sendungsverarbeitung
   GROUP BY 1, 2
)
, device_stats AS (
   SELECT
     Posttag Posttag
   , sorting_unit_location_key locationKey
   , sum(content_value) amount
   FROM
     "reporting_lsops"."anlagestatistik_curated_v"
   WHERE ((1 = 1) AND ((1 = 0) OR ((model = 'BEUMER') AND (content_type = 'INDUCTION') AND (↩
    ↪ con-tent_value_index = 1))))
   GROUP BY 1, 2
)
SELECT
  submitted.Posttag Posttag
, submitted.locationKey locationKey
, key_to_plz.plz plz
, discharged.num_discharged num_discharged
, submitted.num_submitted num_submitted
, coded.num_coded num_coded
, destinationassigned.num_destinationassigned num_destinationassigned
, sva.num_sva num_sva
, dvs.amount num_stats
FROM
  (((((submitted_agg submitted
INNER JOIN key_to_plz ON ((1 = 1) AND (submitted.locationKey = key_to_plz.locationKey)))
INNER JOIN discharged_agg discharged ON ((1 = 1) AND (submitted.Posttag = discharged.Posttag) ↩
    ↪ AND (submit-ted.locationKey = discharged.locationKey)))
INNER JOIN coded_agg coded ON ((1 = 1) AND (submitted.Posttag = coded.Posttag) AND (submitted.↩
    ↪ locationKey = coded.locationKey)))
INNER JOIN destinationassigned_agg destinationassigned ON ((1 = 1) AND (submitted.Posttag = ↩
    ↪ destina-tionassigned.Posttag) AND (submitted.locationKey = destinationassigned.↩
    ↪ locationKey)))
INNER JOIN sva_agg sva ON ((1 = 1) AND (submitted.Posttag = sva.Posttag) AND (submitted.↩
    ↪ locationKey = sva.locationKey)))
LEFT JOIN device_stats dvs ON ((1 = 1) AND (submitted.Posttag = dvs.Posttag) AND (submitted.↩
    ↪ locationKey = dvs.locationKey)))
WHERE ((1 = 1) AND (submitted.locationKey IN ('HRK', 'VET', 'HAE', 'CAD', 'WAL', 'UTZ', 'PRA',↩
    ↪  'FRA', 'DAI', 'URD', 'BCH', 'RUE', 'OMU')))
```

A.4. Testing Monitoring Query

To enable testing of the monitoring data counts, the DQM query had to be adapted. The following shows the final version of this query.

```
CREATE OR REPLACE VIEW "sendungsverarbeitung_dqm_starschema_no_aggregation_small_v" AS
WITH
  key_to_plz AS (
   SELECT
     'VET' locationKey
   , '196000' plz

UNION ALL    SELECT
     'HRK' locationKey
   , '462150' plz

UNION ALL    SELECT
     'CAD' locationKey
   , '659100' plz

UNION ALL    SELECT
     'WAL' locationKey
   , '830000' plz

UNION ALL    SELECT
     'HAE' locationKey
   , '462000' plz

UNION ALL    SELECT
     'UTZ' locationKey
   , '720000' plz

UNION ALL    SELECT
     'PRA' locationKey
   , '413400' plz

UNION ALL    SELECT
     'FRA' locationKey
   , '852000' plz

UNION ALL    SELECT
     'DAI' locationKey
   , '131000' plz

UNION ALL    SELECT
     'URD' locationKey
   , '892000' plz

UNION ALL    SELECT
     'BCH' locationKey
   , '503000' plz

UNION ALL    SELECT
     'RUE' locationKey
```

```
    , '815000' plz

UNION ALL    SELECT
      'OMU' locationKey
    , '307100' plz

)
, plz_to_key AS (
   SELECT
      '196000' plz
    , 'VET' locationKey

UNION ALL    SELECT
      '462150' plz
    , 'HRK' locationKey

UNION ALL    SELECT
      '659100' plz
    , 'CAD' locationKey

UNION ALL    SELECT
      '830470' plz
    , 'WAL' locationKey

UNION ALL    SELECT
      '830000' plz
    , 'WAL' locationKey

UNION ALL    SELECT
      '462000' plz
    , 'HAE' locationKey

UNION ALL    SELECT
      '720000' plz
    , 'UTZ' locationKey

UNION ALL    SELECT
      '413400' plz
    , 'PRA' locationKey

UNION ALL    SELECT
      '852000' plz
    , 'FRA' locationKey

UNION ALL    SELECT
      '131000' plz
    , 'DAI' locationKey

UNION ALL    SELECT
      '892000' plz
    , 'URD' locationKey

UNION ALL    SELECT
      '503000' plz
    , 'BCH' locationKey

UNION ALL    SELECT
```

```
     '815000' plz
   , 'RUE' locationKey

UNION ALL    SELECT
     '307100' plz
   , 'OMU' locationKey

)

, mp_submitted_raw AS (
   SELECT
     CAST(DATE_ADD('hour', -6, submitted.timestamp AT TIME ZONE 'Europe/Zurich') AS DATE) ↩
    ↪ Posttag
   , submitted.sortingunit.locationKey locationKey
   , submitted.sortingProcess sortingProcess
   FROM
     curated_lsops.mailpiece_submitted_v2
)
, mp_submitted AS (
   SELECT
     Posttag Posttag
   , locationKey locationKey
   , (CASE WHEN ((locationKey = 'HRK') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'VET') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'CAD') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'WAL') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'UTZ') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN ((locationKey = 'PRA') AND (sortingProcess = 'AUTOMATIC')) THEN 1
   WHEN (locationKey IN ('BCH', 'RUE', 'OMU')) THEN 1
   WHEN (locationKey IN ('HAE', 'FRA', 'DAI', 'URD')) THEN 1
   ELSE 0 END) relevant_submitted
   FROM
     mp_submitted_raw
)

SELECT
  submitted.Posttag Posttag
, submitted.locationKey locationKey
, key_to_plz.plz plz
, submitted.relevant_submitted relevant_submitted

FROM
  (mp_submitted submitted
INNER JOIN key_to_plz ON ((1 = 1) AND (submitted.locationKey = key_to_plz.locationKey)))
WHERE ((1 = 1) AND (submitted.locationKey IN ('HRK', 'VET', 'HAE', 'CAD', 'WAL', 'UTZ', 'PRA',↩
    ↪  'FRA', 'DAI', 'URD', 'BCH', 'RUE', 'OMU')))
```

B. Miro Usage

As a reference, three screenshots of the Miro board at different stages of this project are provided. They show the increase in information as well as the restructuring of information blocks. Three observations can be made. First, the zoom level required to display the entire board on one screen had to be increased multiple times due to the growing number of meetings, Kanban cards, and other elements. Second, the implementation and document writing section in the bottom right grew considerably compared to the first screenshot. Third, new information was added continuously, which can be seen mainly in the bottom right. Finally, after these reference screenshots, some sections are highlighted to show readable parts of the work carried out on the board.

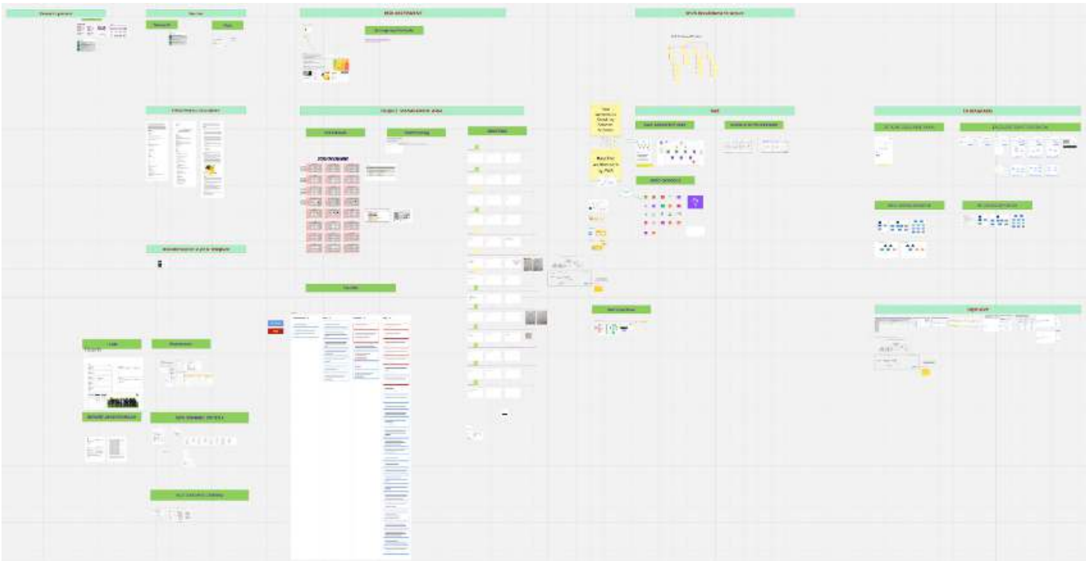

Figure B.1.: Miro board documentation, view of 10.11.2025

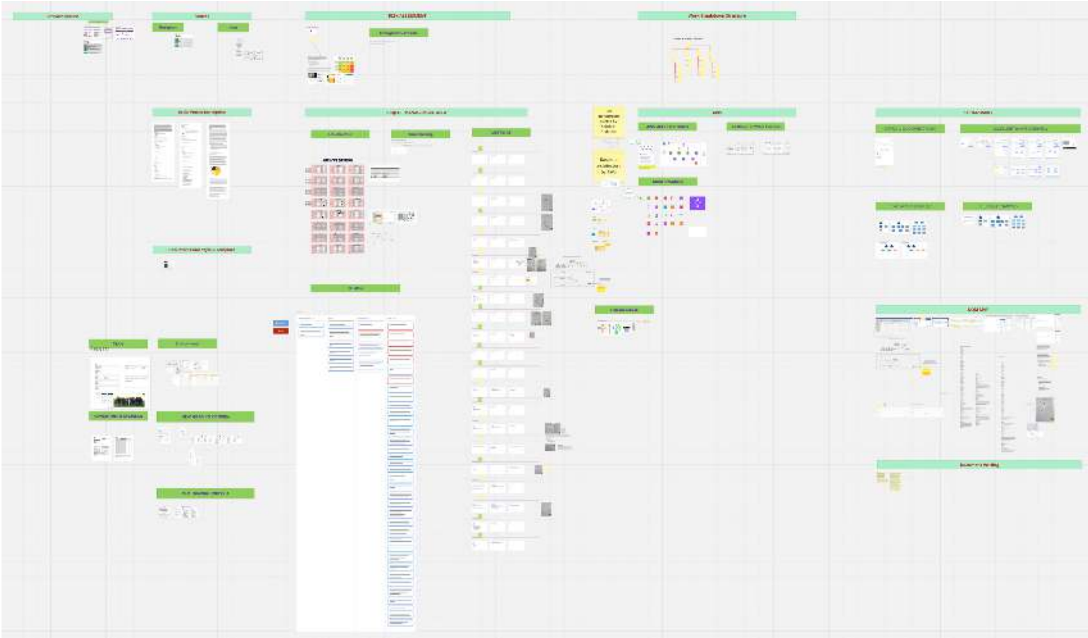

Figure B.2.: Miro board documentation, view of 25.11.2025

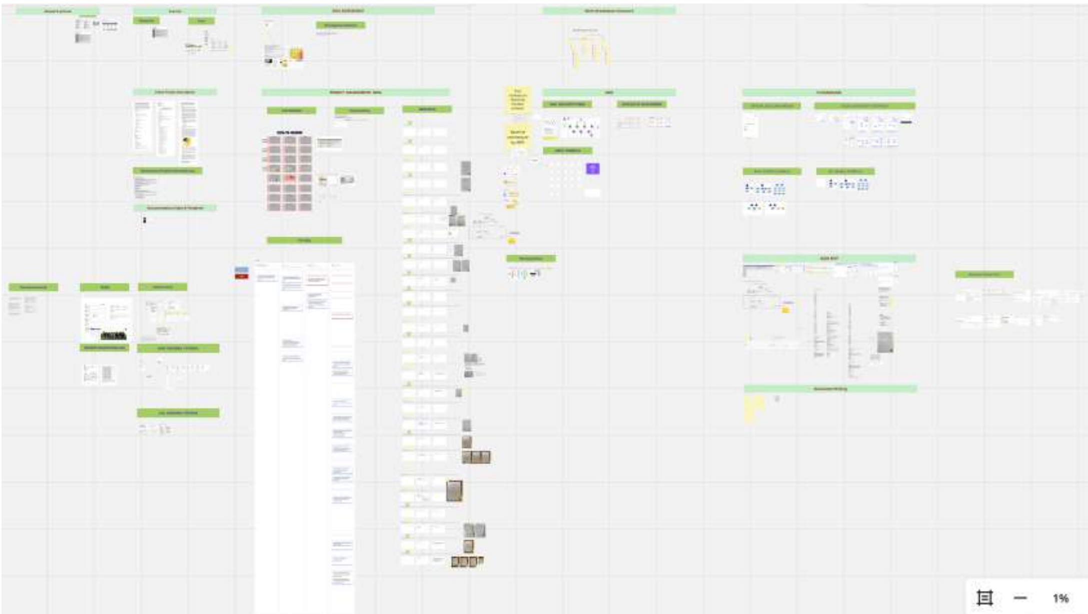

Figure B.3.: Miro board documentation, view of 23.12.2025

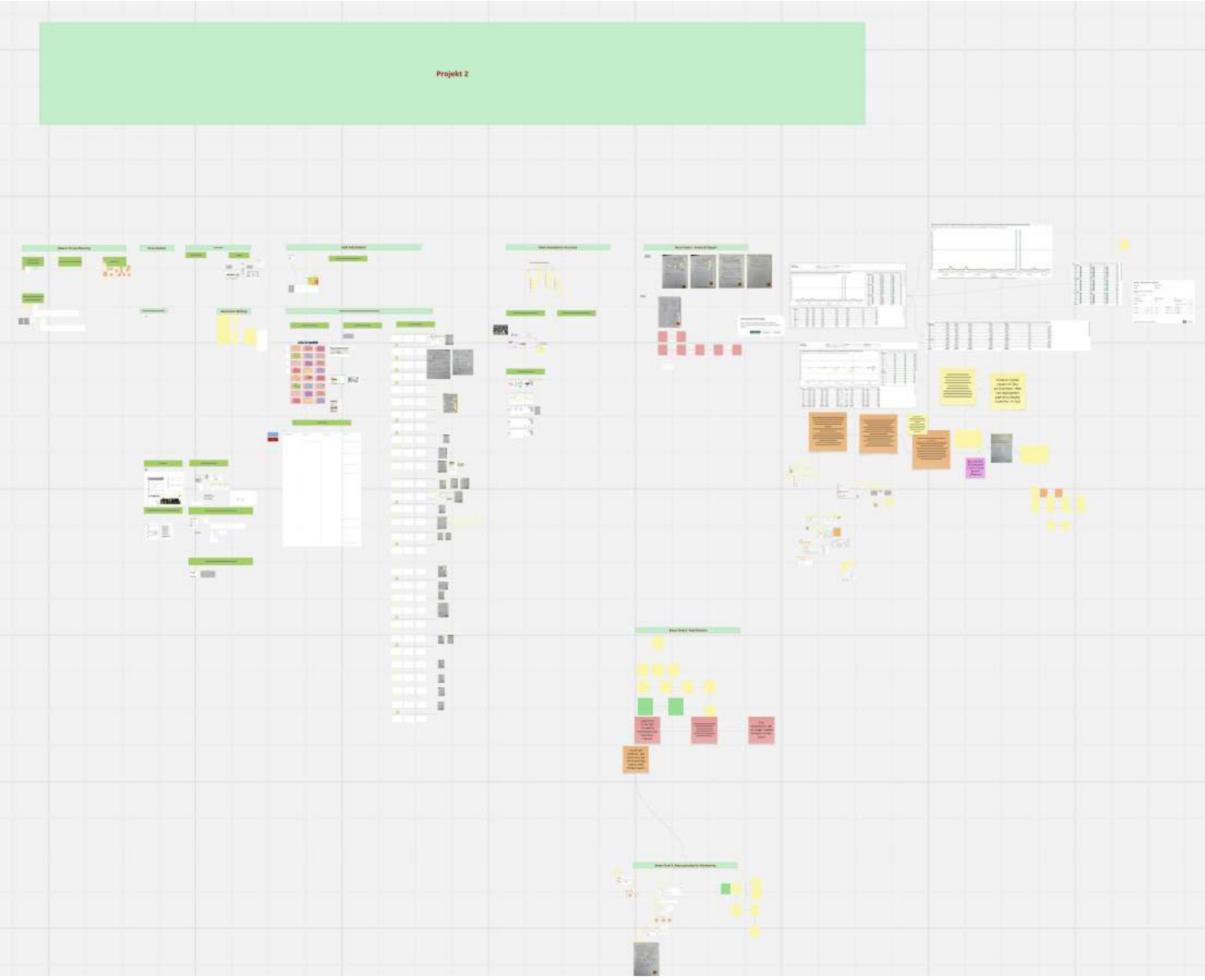

Figure B.4.: Complete Miro board documenting the evolution of the project. The board contains the complete collection of notes, diagrams, screenshots, milestones, implementation artefacts and planning information accumulated during the project. Screenshot taken on 24 July 2026.

B.1. Miro Details of SMART Goals

The following screenshots provide a high-level overview of the work carried out for the individual SMART Goals. These screenshots are intended only as a reference and are not meant to be readable in detail, but rather to provide a brief overview of the methods used.

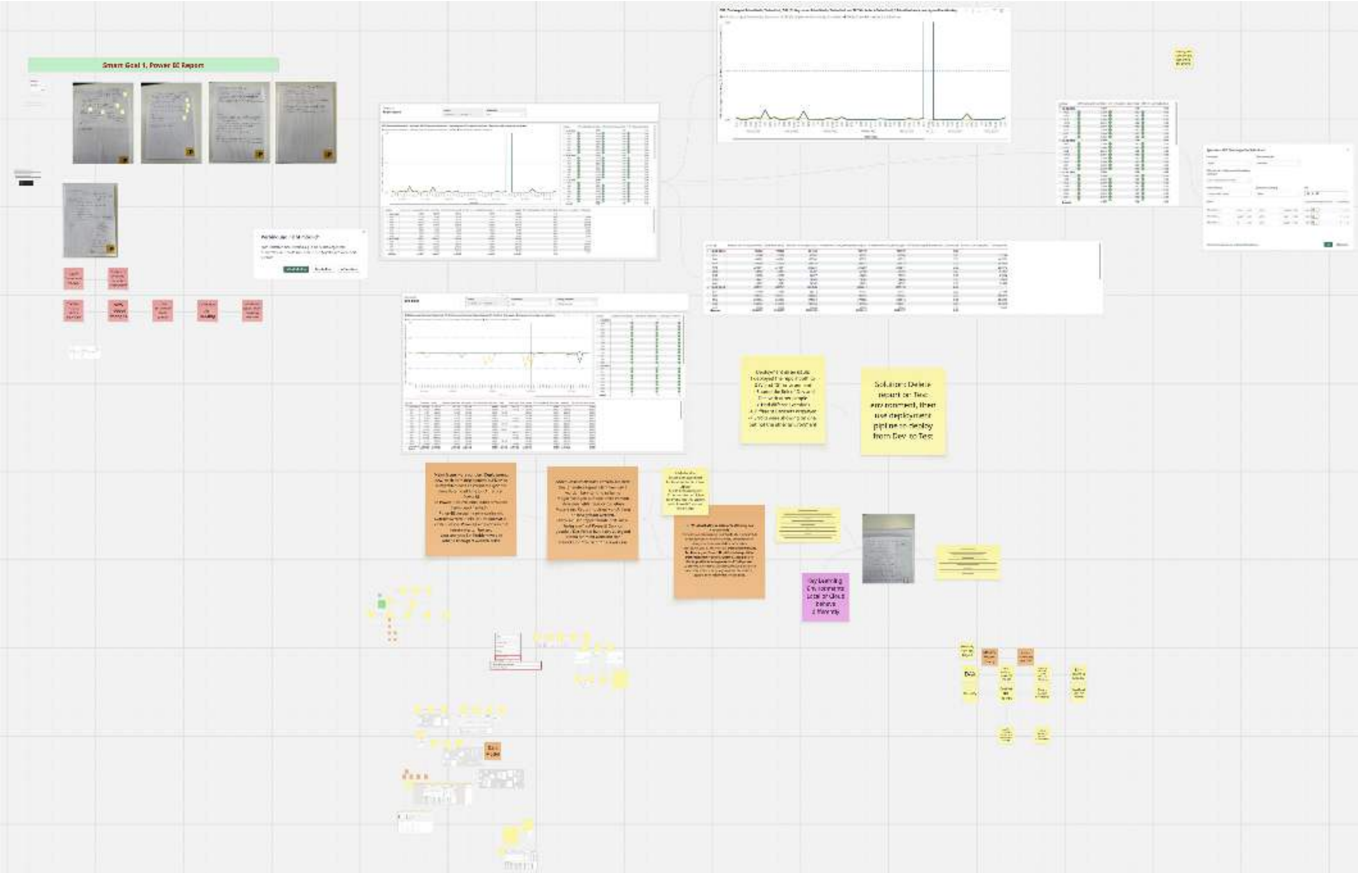

Figure B.5.: Extract of the Miro board related to Smart Goal 1. The board illustrates the planning and documentation of the reporting solution, including design concepts, implementation notes and supporting artefacts collected throughout the project.

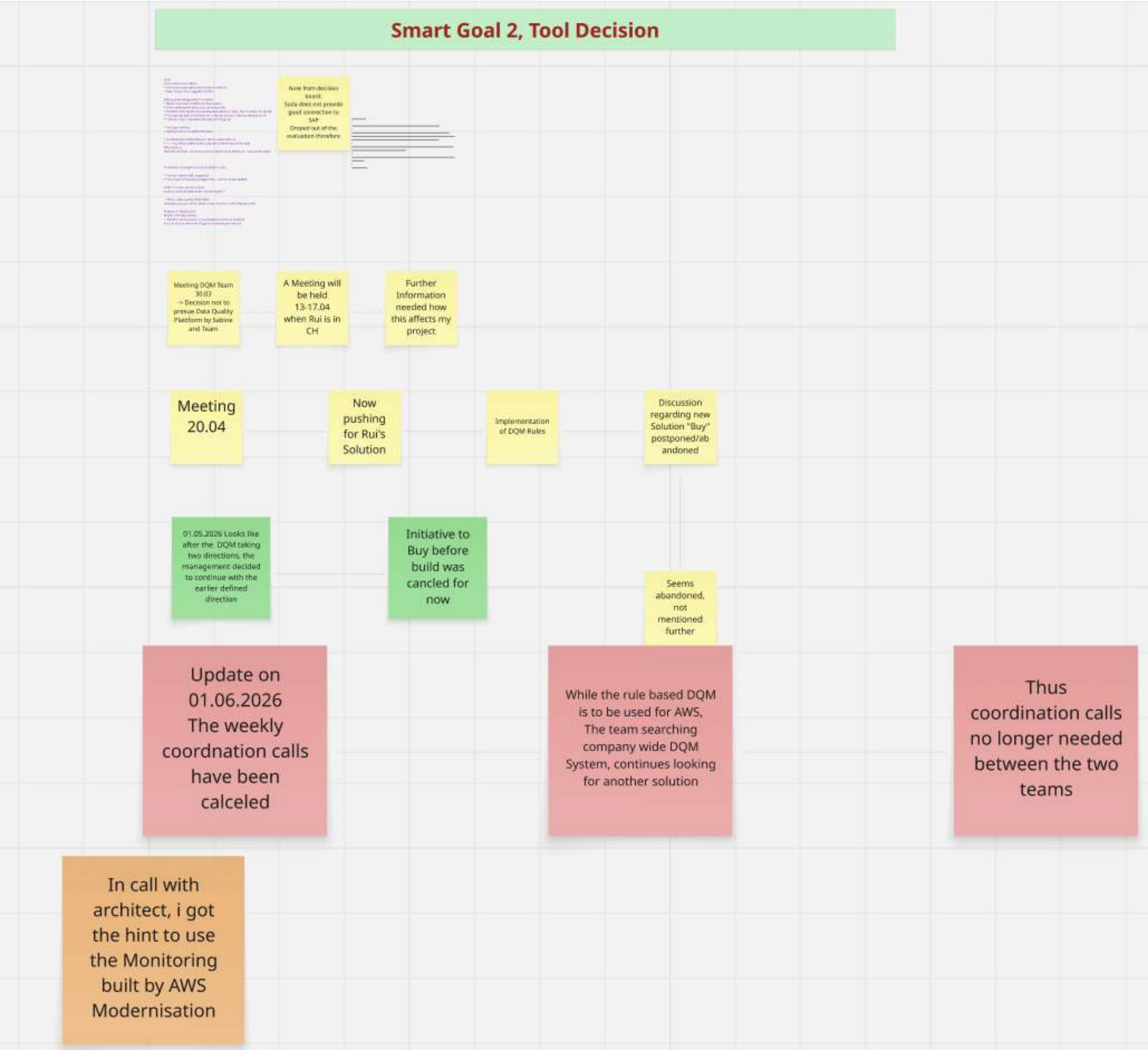

Figure B.6.: Extract of the Miro board related to Smart Goal 2. The material summarises the evaluation activities, comparisons, vendor discussions and supporting documentation used during the assessment of data quality management solutions.

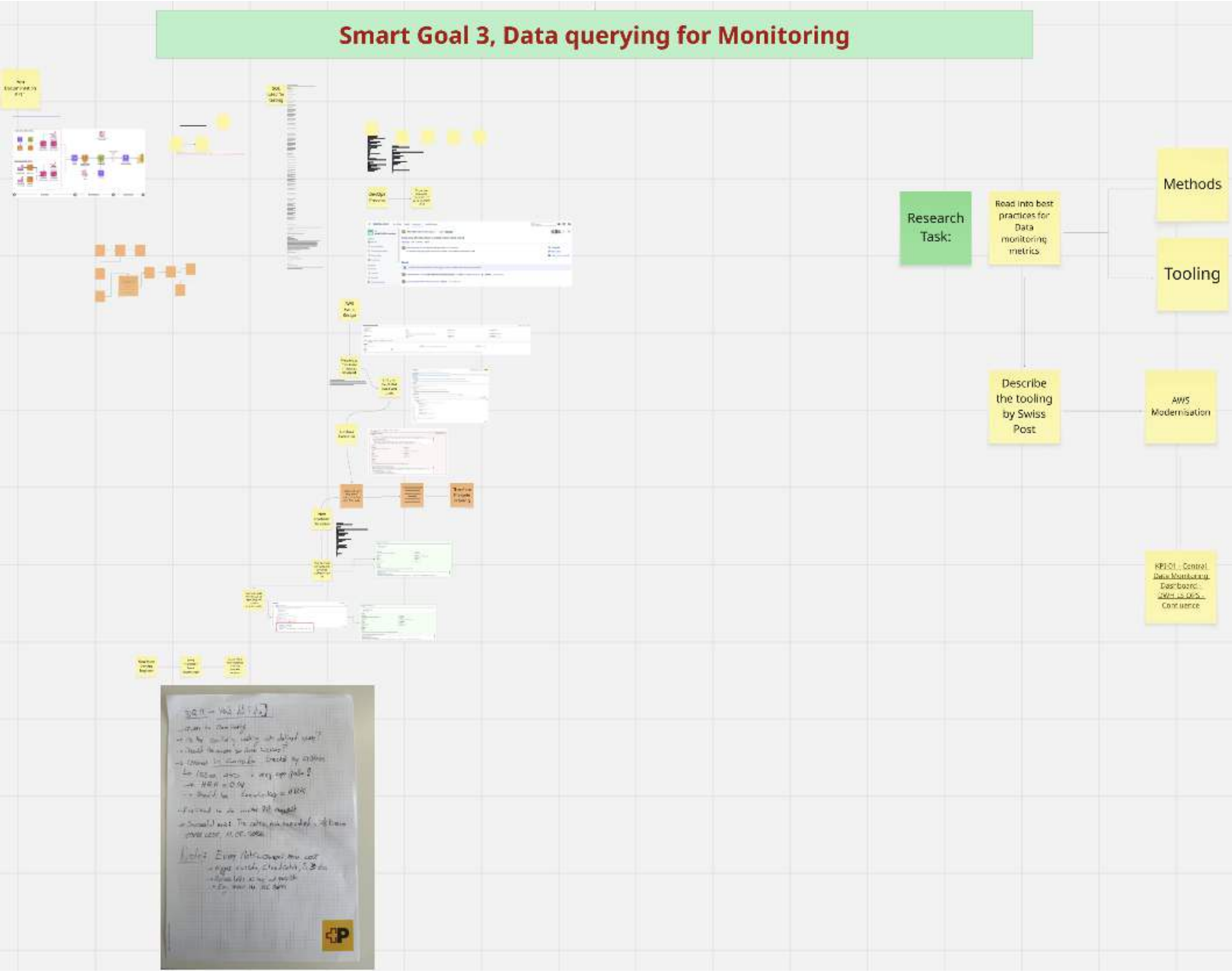

Figure B.7.: Extract of the Miro board related to Smart Goal 3. The board documents the collection of implementation notes, design ideas, intermediate results and decisions supporting the validation of the selected proof-of-concept.

C. Risk Monitoring

The risk Monitoring files are linked in the following tables.

July 2026

A final monitoring performed on 24.07.2026, concluded the writing phase on project 2.

Number	Risk	Occured	Detail	Why	Measures taken	Measure date
1	Unclear question	No			Measures taken	
2	Unclear goals	No		Team Members, PO in team		
3	Insufficient expertise	Yes	Latex usage to change document structure	Some parts of the used BFH template had to be changed due to amendments with tables and additional content	Used ChatGPT for review and changes	24.07.2026
4	Underestimated time/-effort	Yes	Efforts for writing were not committable before holiday	Took too much time in relation to needed cool down times	Moved writing partially to holiday	24.07.2026
5	Other study projects	Yes	No time invested	Focus on this project		24.07.2026
6	Work projects	No	In December much more work on the project was possible, no incidents	Full focus on new topics		

Number	Risk	Occured	Detail	Why	Measures taken	Measure date
7	Procrastination	No		Now time to finish	Working hard during beach holiday	24.07.2026
8	Unavailability of main project staff (e.g., illness)	No				
9	Unavailability of advisor (e.g., illness, accident)	No				
10	Software Version / Library / Dependency	No				
11	Hardware is damaged	No				
12	Data related	No				
13	Not reproducible work	No				
14	Plagiarism or wrong citation	No				
15	Copyright	No				
16	Bad communication with stakeholder or advisor	No				
17	Insufficient exchange with external partners	No				
18	Unforeseen familiar crisis	No				
19	General force majeure	No				
20	Unknown process	Yes	Finally after a lot of back and forth the invoice was handled through sending it via post mail	BFH could not sign the new Terms of Swiss Post	Next invoice to be sent by mail directly	01.07.2026